

Foundations of Econometrics

Estimands, identification, and inference: what a regression can tell you, and how sure to be

ELI5 — the short version

In one sentence. A regression tells you how two things move together; turning that into “this causes that” needs an assumption the data can never check for you, and this document is about which assumption, how to argue for it, and how sure you are then allowed to be.

1. A coefficient is not an effect (§1, §3). What a regression computes is a ratio: how much two things vary together, divided by how much one of them varies. That number always exists and always means something. Calling it “the effect of X on Y” is a separate claim about *why* X differs from row to row. If the firms that advertise more are also better run, the advertising coefficient is carrying both.

2. Three questions, strictly in order (§1). What exactly am I trying to measure? Can it be recovered at all from this data? How precisely has it been estimated? Errors flow downward and never upward: flawless statistics cannot rescue a quantity the data cannot identify, and no amount of data fixes the wrong target. Nearly all real disputes are about the first two questions, and nearly all the effort goes into the third.

3. Your residuals cannot warn you (§3). This is the fact that surprises people most. The leftovers from a regression are *mathematically guaranteed* to be uncorrelated with the variables you regressed on — that is how the fitting works. So no residual plot and no diagnostic computed from the output can detect the problem that matters most. The evidence for it has to come from outside the regression entirely.

4. Two unrelated ways to be wrong (§1, §4). Being wrong about *why* the variation exists gives you a wrong answer. Being wrong about *how dependent* your observations are gives you a right answer with a fake error bar. Every standard-error correction you have heard of addresses the second, and none of them touches the first.

5. Rows are not information (§2, §4). Precision depends on the number of genuinely independent pieces of information. A highly persistent series of 1,000 observations can be worth about 53 of them; forty years of overlapping twelve-month returns can be worth about forty. Grouped data are worse than they look: with fifty groups of a hundred and a within-group correlation of just 0.05, the naive standard error is 2.4 times too small.

6. An underpowered study does not merely miss things — it misleads (§4). If your design can only detect effects much larger than the truth, then the only findings that clear the significance bar are the overestimates. Work out the smallest effect you *could* detect — roughly 2.8 standard errors — before you look at the result.

7. Controls create bias as easily as they remove it (§6). Control for a genuine common cause and you remove bias. Control for something sitting *between* cause and effect and you subtract away part of what you were measuring. Control for something that both variables influence and you can reverse the sign outright. Survivorship, database inclusion and any filter based on what happened later belong to that third category, and they are everywhere in financial data.

8. Two traps specific to time (§8). Regress one aimless wandering series on a completely unrelated one and you will usually find a “significant” relationship — and collecting more data makes this worse rather than better. And “Granger causality” is not causality: it says one series helps predict another, which is a statement about precedence.

9. What machine learning does and does not change (§10). It is very good at “what happens next” and it is not a research design — it estimates things, it does not identify them, and feature importances describe the model rather than the world. Ordinary cross-validation also leaks badly here: walk forward, drop the overlapping labels, group by date, and treat a hyperparameter search as the pile of multiple tests it is.

10. Where results actually go wrong (§9, §11). Mostly in the data: joins, units, timing, corporate actions, delistings. Plot everything before estimating anything. Published return predictors lose about a quarter of their edge out of sample and more than half after publication, and published test statistics bunch suspiciously just above the significance threshold — most of all in the designs that give the researcher the most freedom. The remedies are procedural: keep a log of everything you tried, and report the range of estimates across all the reasonable choices instead of a curated table of the flattering ones.

If you do only three things: state what you are trying to measure before estimating anything, argue for identification from outside the regression rather than from its output, and count independent observations rather than rows.

What this is. A first-principles tutorial on econometrics, meaning the methods for learning about economic and financial relationships from data that nobody designed as an experiment. It covers the foundations (what a regression actually estimates, and what probability theory contributes), the methods (the estimator family, the inference machinery, the research designs that turn correlations into causal claims, and time-series econometrics), and the practice (how empirical work goes wrong, and the working habits that stop it). The emphasis throughout is on intuition and use. Every formula is preceded by what it means and followed by a number, and every method is described by how it breaks as well as by how it works.

Who it is for. A reader who is mathematically comfortable and technically strong but has not been trained as an econometrician, and who wants to *do* something with this material: run a regression and know what it means, evaluate an empirical claim, build a forecasting or signal-research pipeline that does not fool its builder, or read the empirical finance literature critically. Nothing is assumed beyond calculus, linear algebra, and the vocabulary of probability. Where a concept is used before it is fully explained, Appendix A builds it from the ground up.

How to read it. The document has six parts.

- **Part I (§1–§3): what econometrics is for.** §1 states the idea that organises everything else, and nobody should skip it. §2 is the minimum probability you need; skim it if you know the material, but read §2.2 and §2.5. §3 builds regression from the conditional expectation function, and its §3.4 and §3.5 are the two results the rest of the document leans on most.
- **Part II (§4–§5): estimation and inference.** §4 is about standard errors, tests, and the ways they mislead. It is the most practically important section for someone who already runs regressions. §5 shows that OLS, instrumental variables, GMM, and maximum likelihood are one estimator with different inputs.
- **Part III (§6–§7): identification.** §6 is the anatomy of endogeneity, the reason a regression coefficient differs from a causal effect. §7 surveys the research designs that address it: experiments, instruments, panels, difference-in-differences, regression discontinuity, synthetic control, and structural models.
- **Part IV (§8): time series.** What changes when observations are ordered in time, from autoregressions to unit roots, cointegration, VARs, volatility models, breaks, and forecast evaluation.
- **Part V (§9–§11): practice.** §9 covers the econometrics specific to finance: predictive regressions, event studies, asset-pricing tests, panel standard errors, and the replication debate. §10 is prediction, model selection, and machine learning. §11 is workflow, reporting, and a catalogue of failure modes.
- **Part VI (§12–§15): synthesis.** §12 collects the taxonomy and the equivalences, §13 is the history, §14 the synthesis and a decision tree, and §15 the references.

If you read four things, read §1.5–§1.6 (the master form and the sandwich), §3.4–§3.5 (what “controlling for” does and what leaving something out costs), §4.4 (why 5,000 observations can carry the information of 840), and §6.5 (the controls that create bias rather than remove it). If you work on return prediction, add §2.5, §8.3, §9.2, and §10.6.

Relationship to the other notes. Simple and Log Returns §6 treats the time-series econometrics of returns (unit roots, volatility models, cointegration, long-horizon predictive regressions) from the side of the return convention. This document supplies the general machinery behind it. Market Regimes and Machine Learning goes deep on structural breaks, regime-switching models, and time-series validation, which appear here only in outline (§8.7, §10.6). Portfolio Construction and the Covariance Matrix covers covariance estimation, shrinkage, and the statistics of Sharpe ratios. The momentum and trend-following notes apply the tools here to specific signals. Each note stands alone.

A warning about scope. [Practice] Nothing here is investment advice. The numerical illustrations are simulations built to isolate one mechanism at a time, not evidence about any market. Where a number comes from my own simulation rather than a published study, I say so.

Epistemic tags. Claims are flagged by status where the status changes what you should do:

- **[Fact]:** a mathematical result, or an empirical finding that has been replicated or independently reanalysed, with broad agreement among people who have looked.
- **[Contested]:** documented, but with live disagreement about magnitude, robustness, or cause.
- **[Hypothesis]:** a proposed mechanism, not decisively tested.
- **[Practice]:** practitioner or professional convention. It may well be right, but the evidence for it is informal or absent.

Untagged sentences are definitions, derivations, or arithmetic, true by construction rather than by evidence. Results from simulations I ran while writing are labelled **[Simulated]**. The generating code is committed alongside this document in `figures/em_*.py`, and each script prints the numbers quoted in the text, so you can change the parameters and rerun.

Notation. Units of observation are indexed by $i = 1, \dots, n$ in a cross-section and by $t = 1, \dots, T$ in a time series. A panel has N units observed over T periods. Data can be grouped into G **clusters** indexed by g , with n_g observations in cluster g .

Y_i is an outcome. $X_i \in \mathbb{R}^k$ is a column vector of k regressors, usually including a constant, so $X_i'\beta$ is a scalar (a prime denotes transpose). Bold capitals stack observations: $\mathbf{X}$ is the $n \times k$ matrix whose rows are X_i' , and $\mathbf{Y}$ is the n -vector of outcomes. The one exception is §8.5, where bold lowercase letters are vectors of several time series at one date and plain capitals are coefficient matrices, as is standard for vector autoregressions. $Z_i \in \mathbb{R}^\ell$ is a vector of $\ell \geq k$ **instruments**, and C_i is a control variable. A hat denotes an estimate ($\hat{\beta}$), a bar a sample average ($\bar{Y}$), and a subscript zero the true value of a generic parameter vector θ (θ_0).

Two error terms are kept strictly apart, because the distinction between them carries much of §1 and §3. e_i is the **projection error**, defined as whatever is left of Y_i after the best linear prediction from X_i . It is uncorrelated with X_i *by construction* (§3.2). u_i is a **structural error**, the unobserved part of an equation that is meant to describe how Y_i is actually generated, such as a causal or behavioural relationship. Whether u_i is uncorrelated with X_i is an *assumption*. $\hat{e}_i$ is a regression residual. ε_t is a white-noise innovation in a time-series model.

$m(x) = \mathbb{E}[Y_i | X_i = x]$ is the **conditional expectation function** (CEF), and $\epsilon_i = Y_i - m(X_i)$ its **CEF error**: mean-independent of X_i by construction (§3.1), a stronger property than e_i 's mere uncorrelatedness above, per the hierarchy of §2.2. In causal settings $D_i \in \{0, 1\}$ is a treatment indicator, $Y_i(1)$ and $Y_i(0)$ are **potential outcomes** with and without treatment, and $\tau_i = Y_i(1) - Y_i(0)$ is the unit-level treatment effect (§6.1).

The master form of §1.5 writes an estimator as the solution of a **moment condition** $\mathbb{E}[\psi_i(\theta_0)] = 0$, where $\psi_i(\theta)$ is a function of unit i 's data and the parameter. $Q = \mathbb{E}[\partial\psi_i(\theta_0)/\partial\theta']$ is its **Jacobian** (the “bread”), and Ω is the long-run variance of the scaled moment sum $n^{-1/2} \sum_i \psi_i(\theta_0)$ (the “meat”). W is a GMM weighting matrix. $Q_{XX} = \mathbb{E}[X_i X_i']$.

In regressions, α is an intercept, α_i a unit fixed effect, and λ_t a time fixed effect. In the omitted-variable formula of §3.5, γ is the effect of the omitted variable and δ its projection coefficient on the included regressor. π is a first-stage coefficient in instrumental variables and F the first-stage F -statistic. κ is a regularisation penalty (§5.6, §10.4).

In time series, ϕ is an autoregressive coefficient, ρ_j the autocorrelation at lag j , and h a forecast or return horizon. ρ_u and ρ_x , with letter subscripts, are **intra-cluster** correlations of the error and the regressor (§4.4). σ^2 is an error variance and σ_t^2 a conditional variance. The GARCH parameters of §8.6 are written ω, a, b rather than the literature's ω, α, β , so that α and β keep their regression meanings. n_{eff} is an effective sample size.

Φ and φ are the standard normal distribution and density functions. $\mathbb{1}\{\cdot\}$ is an indicator. $\xrightarrow{p}$ and $\xrightarrow{d}$ denote convergence in probability and in distribution, plim a probability limit, and Avar an asymptotic variance. $\text{se}(\cdot)$ is a standard error and SR a Sharpe ratio. Significance levels are written as percentages and have no symbol.

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Part I — What econometrics is for

1. What econometrics is

1.1 The wrong intuition

Here is the picture most people arrive with:

Econometrics is statistics applied to economic data. You regress an outcome on the variables you think matter, check which coefficients are significant, and read each significant coefficient as the effect of its variable.

Every clause is defensible on its own, and the sentence as a whole is badly wrong in three specific ways. Each one corresponds to a part of this document.

A regression coefficient is not an effect. A coefficient summarises how the average outcome differs between units whose regressor differs, after a linear adjustment for the other regressors. It becomes an *effect*, meaning what would happen if you changed the regressor, only under an assumption about *why* the regressor differs across units. Take firms that buy back their own shares. A regression of subsequent returns on a buyback indicator compares firms that chose to repurchase with firms that did not. Managers tend to repurchase when they believe their stock is cheap, so the coefficient mixes any effect of the buyback itself with the private information that prompted it. Both components are economically interesting, and they call for opposite conclusions. Separating them is the subject of Part III.

Significance is neither truth nor importance. A *t*-statistic is an estimate divided by its standard error, and the standard error is computed under an assumed model of how the sample could have come out differently. Get that model wrong, for instance by ignoring that observations share shocks, that returns overlap, or that twenty specifications were tried before this one, and the *t*-statistic is off by *factors*, not decimals (§4). Separately, a precisely estimated effect can be economically trivial, and an important effect can be imprecisely estimated.

The hard part is not the fitting. A regression takes microseconds to compute. The intellectual content of econometrics is the argument connecting the number the computer produces to the question being asked, and that argument is about the process that *generated* the data, not about the data themselves. This is why two competent econometricians can run the same regression on the same data and disagree completely about what it means. They are not disagreeing about arithmetic. They disagree about the data-generating process.

The profession's own reckoning with the naive picture came from a clean test. In the 1970s the National Supported Work Demonstration randomly assigned disadvantaged workers to a job-training programme, so its true effect on earnings was known from the experiment. LaLonde (1986) set aside the experimental control group, substituted comparison samples drawn from standard household surveys, and applied the non-experimental estimators that economists were then using to evaluate training programmes. **[Fact]** The resulting estimates were scattered widely around the experimental benchmark, many of them far off and some with the wrong sign, and the specification tests of the day did not reliably separate the good estimates from the bad. That paper is a large part of why the field now talks about research design before it talks about estimators (§13).

The better mental model:

Econometrics is the discipline of stating, and defending, the conditions under which a number computed from data answers a question about the world. The computation is easy and is almost always a valid answer to *some* question. The work is making it an answer to yours.

1.2 The problem: data that nobody randomised

A physicist who wants to know how a spring responds to force can apply the force. An economist who wants to know how earnings respond to schooling cannot assign schooling, and a financial economist who wants to know how prices respond to index inclusion cannot assign stocks to indices. Almost all economic and financial data are **observational**: they record the outcomes of decisions made by people and firms pursuing their own ends.

That single structural fact is what makes econometrics a discipline rather than a branch of statistics. **The variation in the data was generated by agents optimising with information the econometrician does not see.** People choose schooling partly on ability, which also affects earnings. Firms choose leverage partly on the stability of their cash flows, which also affects their risk. Investors buy funds after good performance, and the resulting inflows affect subsequent performance. Central banks raise rates when they expect inflation, so a naive regression can make rate rises appear to cause the inflation they were responding to.

Trygve Haavelmo gave the field its framing in 1944. Treat observed economic data as a single draw from a joint probability distribution. That distribution is generated by a system of behavioural relationships, such as demand curves, decision rules, and pricing equations. The relationships are what you want to learn. The distribution is all you can see. Whether the former can be recovered from the latter is the question of **identification**, and it has to be answered before any estimation starts.

Three consequences recur throughout the document:

1. **Correlation mixes effects with selection.** When a regressor is a choice, it is correlated with the reasons for the choice, and those reasons usually affect the outcome too (§6).
2. **Jointly determined variables cannot be regressed on each other and interpreted.** Price and quantity are set together by supply and demand, so a scatter of the two traces out neither curve (Working, 1927). The same holds for returns and order flow, volume and volatility, and fund flows and fund performance (§6.4).
3. **Relationships estimated under one regime can break when the regime changes.** Agents re-optimize when policy or market structure changes, so a relationship that was stable in the past need not survive an intervention (Lucas, 1976). This is the econometric root of models that work in backtests and fail when deployed (§7.8, §8.7).

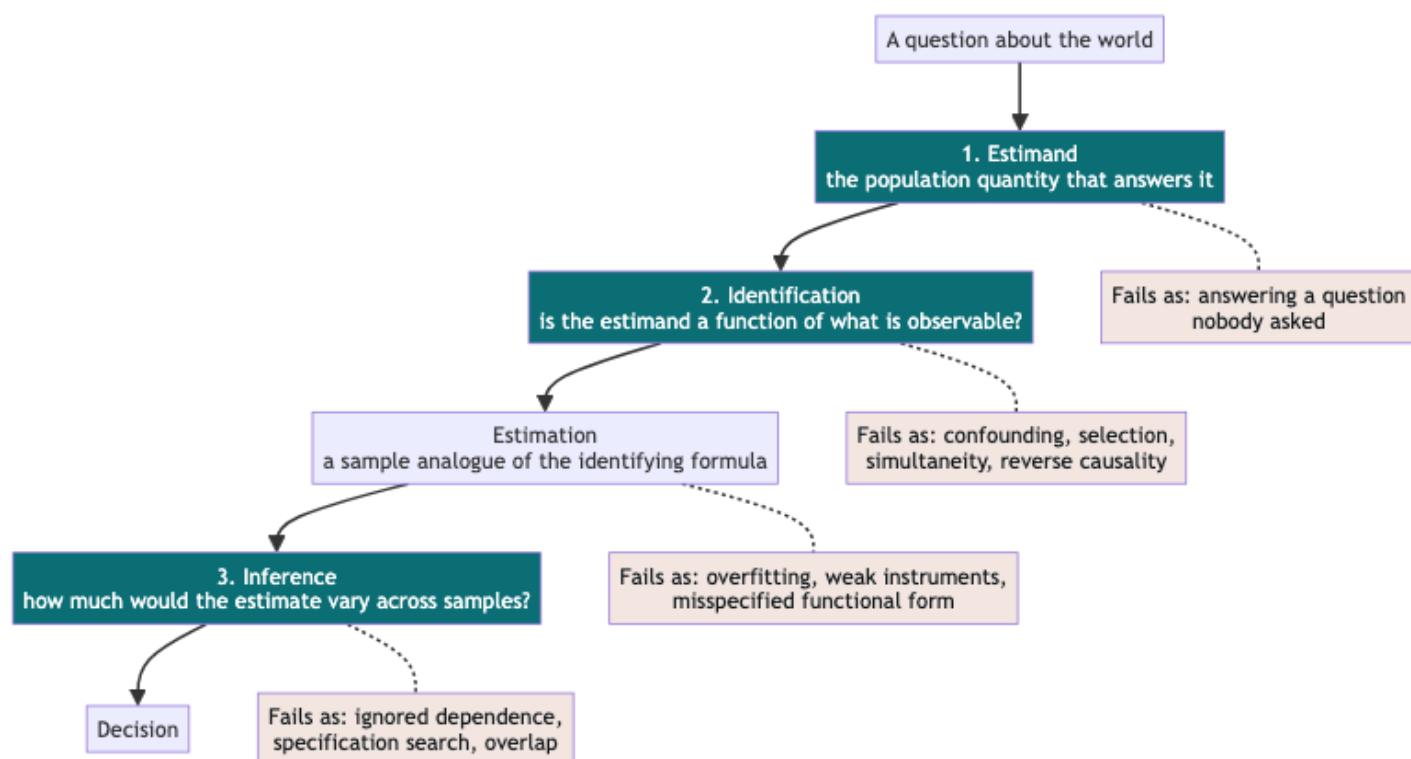
Finance deserves a remark here, because much of it looks like it escapes these problems. A trader who wants tomorrow's expected return is asking a *prediction* question, not a causal one, and prediction does not need identification (§1.4). But two things keep finance inside the tent. First, a great deal of finance *is* causal: whether index inclusion moves prices, whether buybacks create value, whether a regulation changes liquidity. Second, prediction has its own version of the problem. A predictive relationship is only useful if it holds in data the model has not seen, and the forces that make it unstable are exactly the re-optimisation and selection above. The inference problems of §4 (dependent data, overlapping horizons, many tested hypotheses) apply everywhere with full force.

1.3 Three questions, in order

Every empirical claim, whether a published paper, a backtest, or a regression in a notebook, is an answer to three questions. They have to be answered in order.

1. **What is the estimand?** The estimand is the precise population quantity you want to know. "The effect of index inclusion on prices" is not an estimand. "The average change in a stock's price over the month after it is added to the S&P 500, relative to what its price would have been had it not been added, among stocks of the kind that get added" is one. The estimand is defined by a thought experiment, not by the data. The data will never tell you which question to ask.
2. **Is it identified?** Could the estimand be recovered from the population distribution of the observable variables, that is, from an infinite amount of data? Identification is a property of the assumptions and the data-generating process, not of the sample. If the estimand is not identified, more data only give you a more precise estimate of the wrong thing. This is where the economics lives. Every identifying assumption is a claim about behaviour, such as "the lottery number affected earnings only through military service" or "firms just above and just below the index cutoff are otherwise comparable."
3. **How precisely is it estimated?** Given identification and an estimator, how much would the estimate vary across the samples you could have drawn instead? This is **inference**, and its central object is the standard error (§4).

The ordering matters because errors propagate downwards and never upwards. A perfect standard error around an unidentified parameter is worthless, and an elegant estimator cannot rescue a badly chosen estimand. Yet practice routinely inverts the order. People choose the estimator first (because the software offers it), worry about standard errors second, and state the identifying assumption last or not at all.



1.4 Four kinds of question

“What does the data say about X and Y ?” hides four different questions. They need different assumptions and different evaluation, and confusing them is one of the most common errors in empirical work.

Question	Estimand	What must be true	How it is checked	Typical tools
Description	A conditional mean or linear projection	Only sampling assumptions	Inference alone	Regression read as a projection; portfolio sorts
Prediction	$\mathbb{E}[Y_{t+h} \mid \mathcal{F}_t]$, or a forecast that minimises a loss	The relationship is stable out of sample	Out-of-sample evaluation	Forecasting models; machine learning (§8.8, §10)
Causal effect	An average effect of a well-defined intervention	An identifying assumption about why treatment varies	Design diagnostics, placebo tests, sensitivity analysis	Experiments, IV, panels, DiD, RD (§7)
Structural parameter	A deep parameter of preferences or technology	The economic model is correctly specified	Fit to features of the data the model was not tuned to	GMM, maximum likelihood, simulated moments (§5, §7.8)

$\mathcal{F}_t$ in the table is the information available at time t .

The same regression can be read as any of the four, and the reading is determined by what you are willing to assume, not by the output. A regression of next month’s return on this month’s book-to-market ratio is a perfectly valid *description* and a possibly useful *prediction*. It is not an estimate of what would happen to a firm’s return if you changed its book value, and nothing in the output warns you against reading it that way. §10.1 returns to this as the distinction between problems about $\hat{Y}$ and problems about $\hat{\beta}$.

Most machine learning in finance is prediction. Most empirical corporate finance is causal. Asset-pricing tests sit closer to the structural end, because the moment conditions come from an economic model of prices. Knowing which one you are doing tells you which of the later sections are your problem.

1.5 The master form: one equation behind every estimator

Here is the idea that organises the rest of the document. Almost every estimator in econometrics is defined by a **moment condition**: a function of the data and the parameters whose population expectation is zero at the true parameter value,

$$\mathbb{E}[\psi_i(\theta_0)] = 0.$$

The estimator is the **sample analogue**. Choose $\hat{\theta}$ so that the sample average of ψ_i is zero, or as close to zero as possible when there are more equations than unknowns:

$$\frac{1}{n} \sum_{i=1}^n \psi_i(\hat{\theta}) = 0.$$

That is all an estimator is. What distinguishes one method from another is only the choice of ψ .

Estimator	Moment function $\psi_i(\theta)$	The assumption hidden in “= 0”
Sample mean	$Y_i - \mu$	None; this is the definition of μ
OLS	$X_i(Y_i - X_i'\beta)$	The error is uncorrelated with X_i . This is automatic for a projection and an assumption for a causal β
Instrumental variables	$Z_i(Y_i - X_i'\beta)$	The instruments are uncorrelated with the structural error
Maximum likelihood	$\partial \log f(Y_i X_i; \theta) / \partial \theta$	The model's density is correct, or at least its score has mean zero
Fixed effects	$(X_{it} - \bar{X}_i)(Y_{it} - \bar{Y}_i - (X_{it} - \bar{X}_i)'\beta)$	Regressors are uncorrelated with the error at every date: strict exogeneity
Quantile regression	$X_i(q - \mathbb{1}\{Y_i < X_i'\beta\})$	$X_i'\beta$ is the conditional q -quantile
Asset-pricing Euler equation	$Z_t(M_{t+1}(\theta) R_{t+1} - 1)$	Investors' first-order condition holds and Z_t is known at t

In the last two rows, $q \in (0, 1)$ is the quantile of interest, R_{t+1} a gross asset return, and $M_{t+1}(\theta)$ a **stochastic discount factor**, the model's valuation of a unit payoff at $t + 1$ (§5.3). In the fixed-effects row, $\bar{X}_i$ and $\bar{Y}_i$ are unit i 's averages over time (§7.4).

Three facts follow from writing estimators this way, and each one is used repeatedly later on.

Identification is uniqueness of the solution. A parameter is identified when the population moment condition holds at θ_0 and nowhere else. For linear moment conditions this reduces to a rank condition. Instrumental variables, for example, needs $\mathbb{E}[Z_i X_i']$ to have full column rank, which is the formal version of “the instruments must actually move the regressors” (§7.3).

The substantive assumption is always an orthogonality condition. Some observed variable is uncorrelated with some unobserved one. And in the just-identified case, where there are exactly as many equations as unknowns, *the data cannot check it*, because the estimator forces the sample analogue to hold exactly. OLS residuals are orthogonal to the regressors by construction, so no plot of residuals against X can ever reveal that X is correlated with the structural error. **[Fact]** Endogeneity is invisible in the residuals of the regression it contaminates. Over-identified models permit a partial check, and only of whether the instruments agree with each other (§5.3).

A new method changes only what is required to be orthogonal to the error. Adding controls asks for orthogonality conditional on the controls. Fixed effects ask for orthogonality with within-unit variation only. Instrumental variables ask for orthogonality with the instrument rather than the regressor. Regression discontinuity asks for it only in a small window around a cutoff. When you meet a new design, the first question to ask is *which orthogonality condition it buys, and at what price*.

1.6 The sandwich: one formula behind every standard error

The master form also gives every standard error in econometrics, through one derivation. Start from the sample moment condition at the estimate and expand it around the truth. To first order,

$$0 = \frac{1}{n} \sum_{i=1}^n \psi_i(\hat{\theta}) \approx \frac{1}{n} \sum_{i=1}^n \psi_i(\theta_0) + \underbrace{\left[\frac{1}{n} \sum_{i=1}^n \frac{\partial \psi_i(\theta_0)}{\partial \theta'} \right]}_{\rightarrow Q} (\hat{\theta} - \theta_0).$$

Solve for the estimation error and scale by $\sqrt{n}$:

$$\sqrt{n}(\hat{\theta} - \theta_0) \approx -Q^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_i(\theta_0).$$

The estimation error is a fixed matrix times a scaled sum of the moment contributions. A central limit theorem (§2.4) makes that sum approximately normal, with variance Ω . So

$$\text{Avar}(\sqrt{n}(\hat{\theta} - \theta_0)) = \underbrace{Q^{-1}}_{\text{bread}} \underbrace{\Omega}_{\text{meat}} \underbrace{Q^{-1'}}_{\text{bread}}, \quad \Omega = \lim_{n \rightarrow \infty} \text{Var}\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_i(\theta_0)\right).$$

Dividing by n gives the approximate variance of $\hat{\theta}$ itself, and the square roots of its diagonal are the standard errors.

The two ingredients do different jobs, and it is worth holding them apart.

- **The bread, Q , is set by the model.** It measures how sharply the moment condition responds when the parameter moves. For OLS, $\psi_i = X_i(Y_i - X_i'\beta)$, so $Q = -\mathbb{E}[X_i X_i'] = -Q_{XX}$. More variation in the regressors gives a larger bread and a smaller variance, which is why an experiment with a wide spread of treatment doses is more informative than one with a narrow spread.
- **The meat, Ω , is set by the dependence in the data.** It is the variance of a sum. If the moment contributions are independent across observations, it is the sum of their variances. If they are correlated, the covariances enter too, and with many positively correlated pairs they come to dominate.

Every standard-error “correction” in applied work is a different estimate of the meat. Each makes a different assumption about which observations are independent of which.

Name	Estimate of the meat Ω	Independence it assumes
Classical	$\hat{\sigma}^2 \hat{Q}_{XX}$	All observations independent, with equal error variance
Heteroskedasticity-robust (White)	$\frac{1}{n} \sum_i \hat{e}_i^2 X_i X_i'$	All observations independent, variances free
Cluster-robust	$\frac{1}{n} \sum_g \left(\sum_{i \in g} X_i \hat{e}_i \right) \left(\sum_{i \in g} X_i \hat{e}_i \right)'$	Independent across clusters; anything goes within
Newey–West (HAC)	Weighted sum of sample autocovariances of $X_t \hat{e}_t$	Serial dependence that dies out with the lag
Driscoll–Kraay	Newey–West applied to cross-sectional sums	Arbitrary cross-sectional dependence, fading serial dependence
Fama–MacBeth	Variance of the period-by-period estimates	Independence over time; arbitrary within a period
Two-way clustered	Firm-clustered + time-clustered – heteroskedasticity-robust	Independence across both firms and dates, except through those two groupings

All of them use the same bread. §4 derives the table’s entries and says when each is right, and the equivalence between the rows is exact in a few cases (§12.2).

The table implies the most important practical fact about inference, and the second idea that organises this document:

The precision of an estimate is set by the number of independent pieces of information in the meat, not by the number of rows in the data.

Two numbers make the point. Take 5,000 observations in 50 groups of 100, where the regressor is set at the group level and 5% of the error variance is shared within a group. The effective sample size is about 840, and classical standard errors reject a true null hypothesis 42% of the time at a nominal 5% (§4.4). **[Simulated]** Or take 40 years of monthly data on 12-month returns. It has 480 rows, but when estimating an average 12-month return it carries roughly 40 non-overlapping observations' worth of information (§4.5).

The spine, stated once. Every estimate can go wrong in two ways, and they come from different places:

Bias comes from a false orthogonality condition. False precision comes from a wrong meat.

Almost every method in the rest of this document does one of three things. It makes an orthogonality condition more credible (Part III), it estimates the meat honestly (§4, §8, §9), or it trades a little bias for a large reduction in variance (§5.6, §10). When a new technique appears, placing it in one of those three bins tells you most of what you need to know about it.

1.7 What econometrics is not

Several distinctions come up so often that they are worth fixing at the start.

Often conflated	The difference
Econometrics and statistics	The same mathematics. Econometrics adds data generated by optimising agents, which makes identification the central problem, and economic models that supply moment conditions. Other fields have converged on the same problems under other names: epidemiology's "confounding" is econometrics' "omitted variable bias."
Econometrics and machine learning	Machine learning optimises out-of-sample prediction of Y . Much of econometrics estimates a parameter with a causal or structural meaning and an honest measure of its uncertainty. The tools now cross over (§10); the goals do not.
A good fit and a good model	A high R^2 says nothing about identification, and a low one is compatible with a precisely estimated causal effect (§3.8).
Statistically significant and important	Significance measures signal relative to noise. Importance is about magnitude (§4.10).
Unbiased and consistent	Unbiased means right on average at a fixed sample size. Consistent means converging to the truth as the sample grows. Neither implies the other (§2.3).
Granger causality and causality	Granger causality means that the past of X improves forecasts of Y . It is predictive precedence and says nothing about what an intervention on X would do (§8.5).
The error term and the residual	The error (u_i or e_i) is a population object. The residual $\hat{e}_i$ is computed from a sample, and is orthogonal to the regressors by construction (§3.3).
A backtest and an econometric study	They are the same thing. A backtest estimates a mean from dependent data after a search over specifications, and inherits every problem in §4 and §9 whether or not anyone acknowledges it (§9.7).

§1 Key takeaways

1. A regression coefficient is a ratio of covariance to variance, not an effect. It becomes an effect only under an assumption about why the regressor varies across units.
2. Econometrics exists because economic data are generated by agents who choose their regressors with information the analyst does not have.
3. Answer three questions in order: what is the estimand, is it identified, and how precisely is it estimated. Errors propagate downwards, never upwards.
4. Description, prediction, causal effects, and structural parameters are four different questions. The same regression can serve any of them, and the output never tells you which.

5. Almost every estimator solves a sample moment condition. The substantive assumption is always an orthogonality condition, and in the just-identified case the data cannot test it.
6. Every standard error is a sandwich: bread from the model, meat from the dependence in the data. Every standard-error correction is a different estimate of the meat.
7. Bias comes from a false orthogonality condition; false precision comes from a wrong meat. Precision is governed by the number of independent pieces of information, not the number of rows.

2. The probability you actually need

This section is the minimum probability theory that the rest of the document leans on. It is written to be read quickly by someone who has seen the definitions before. The two subsections that are not standard textbook fare, and that nobody should skip, are §2.2 (three different strengths of “unrelated”) and §2.5 (how dependence spends observations).

2.1 Populations, samples, and the data-generating process

A random variable is a model of what the data *could have been*. The **data-generating process** (DGP) is the joint probability distribution from which the observed data are one draw. In a cross-section it is the distribution of $(Y_i, X_i, Z_i, \dots)$ for a randomly drawn unit. In a time series it is the stochastic process that produced the whole path.

An estimate is uncertain because a different draw would have produced a different estimate, and a standard error quantifies that variation. It is therefore only meaningful once you have said *what is being redrawn*. There are two standard answers.

- **Sampling-based uncertainty.** The data are a random sample from a larger population, and the uncertainty is about which units happened to be sampled. This is the textbook picture, and it fits survey data.
- **Design-based uncertainty.** The data cover the entire population of interest, such as all fifty US states or every stock in an index, so no sampling variation exists in the usual sense. What is random is which units received the treatment, and so which potential outcome of each unit you get to see. Abadie, Athey, Imbens & Wooldridge (2020) show that conventional standard errors remain sensible for causal questions under this reading, but can be conservative, and that for purely descriptive questions about a fully observed population they answer a question nobody asked.

Finance has a third answer, and it matters for everything that follows. When you study every listed stock, the cross-section is not a sample. **The sample is the history.** The relevant thought experiment is that the same economy could have produced a different sequence of shocks, and the uncertainty is over which history occurred. That is why the number of independent time periods, not the number of stocks, usually governs precision in empirical finance (§2.5, §9.1).

2.2 Conditional expectation, and three strengths of “unrelated”

Conditional expectation. $\mathbb{E}[Y | X]$ is the average of Y among units with a given value of X , viewed as a function of X . Three properties make it the central object of the field.

1. **It is the best predictor.** Among all functions $g(X)$, the one that minimises the mean squared error $\mathbb{E}[(Y - g(X))^2]$ is $g(X) = \mathbb{E}[Y | X]$. To see why, add and subtract the conditional mean:

$$\mathbb{E}[(Y - g(X))^2] = \mathbb{E}[(Y - \mathbb{E}[Y | X])^2] + \mathbb{E}[(\mathbb{E}[Y | X] - g(X))^2].$$

The cross term vanishes because the prediction error $Y - \mathbb{E}[Y | X]$ has mean zero at every value of X . The first term does not depend on g , and the second is zero only when g equals the conditional mean.

2. **The law of iterated expectations.** $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y | X]]$: the overall average is the average of the group averages, weighted by group size.
3. **The law of total variance.** $\text{Var}(Y) = \text{Var}(\mathbb{E}[Y | X]) + \mathbb{E}[\text{Var}(Y | X)]$. Total variation splits into the part explained by X and the part left over. R^2 is a sample version of the first share (§3.8).

Three strengths of “unrelated.” Econometric assumptions constantly require an unobserved U (with $\mathbb{E}[U] = 0$) to be “unrelated” to an observed X . There are three distinct ways to say this, in increasing strength:

Condition	Statement	What it rules out	What needs it
Uncorrelated	$\mathbb{E}[XU] = 0$	A <i>linear</i> relationship between X and U	Consistency of OLS and IV
Mean-independent	$\mathbb{E}[U X] = 0$	Any relationship between X and the average of U	Unbiasedness; treating any function of X as a valid regressor or instrument
Independent	$U \perp X$	Any relationship between X and the whole distribution of U	Quantile and distributional claims; randomisation

Each implies the one above it, and neither converse holds. The standard counterexample takes $X \sim N(0, 1)$ and $U = X^2 - 1$. Then $\mathbb{E}[XU] = \mathbb{E}[X^3] - \mathbb{E}[X] = 0$, so the two are uncorrelated and a regression of U on X has a slope of exactly zero, even though U is a deterministic function of X .

Financial returns are the practical version of the same gap. **[Fact]** Daily returns on broad equity indices are close to uncorrelated over time, but they are far from independent, because large moves cluster: squared returns are strongly autocorrelated. Between the two sits the **martingale difference** property, $\mathbb{E}[r_{t+1} \mid \mathcal{F}_t] = 0$ for a demeaned return r_{t+1} , which says the past cannot predict the *level* of future returns while leaving their variance free to be predictable. The distinction has consequences. A test of “no autocorrelation” whose standard errors assume independence is mis-sized on returns, because independence is false even when the hypothesis being tested is true (§8.1).

2.3 Estimators are random variables

An estimator $\hat{\theta}$ is a function of the sample, so it has a distribution: the **sampling distribution**, meaning the distribution of $\hat{\theta}$ across all the samples the DGP could have produced. Three summaries of it matter.

- **Bias:** $\mathbb{E}[\hat{\theta}] - \theta_0$, whether the estimator is right on average.
- **Variance:** how much it moves from sample to sample. The **mean squared error** combines the two: $\text{MSE} = \text{bias}^2 + \text{variance}$.
- **Consistency:** whether $\hat{\theta} \xrightarrow{p} \theta_0$ as the sample grows, that is, whether the estimator converges to the truth.

Unbiasedness and consistency are logically independent. Using only the first observation, $\hat{\mu} = Y_1$, is an unbiased estimator of a mean, but its variance never shrinks, so it is inconsistent. The OLS estimate of an autoregressive coefficient ϕ is biased downward in finite samples, by roughly $-(1 + 3\phi)/T$ (Kendall, 1954), but it is consistent. The bias matters when T is small and ϕ is near one, as §9.2 shows.

Econometrics mostly cares about consistency rather than unbiasedness, for a practical reason: almost no estimator used in practice is unbiased. Instrumental variables, maximum likelihood, and every nonlinear estimator are biased in finite samples. The working questions are whether the estimator converges to the right thing, and whether the sample is large enough that the remaining bias is small relative to the standard error.

The MSE decomposition also foreshadows a theme of §5.6 and §10. **An estimator with some bias can have a lower mean squared error than an unbiased one**, if the bias buys a large enough reduction in variance. Shrinkage estimators exploit this deliberately.

2.4 The two limit theorems, and what they quietly require

The law of large numbers (LLN) says that sample averages converge to population means. It needs a finite mean and either independence or dependence that fades with distance. For time series the relevant condition is **ergodicity**, which says that averaging along one long path is equivalent to averaging across many independent paths.

The central limit theorem (CLT) says that sample averages are approximately normal, with a spread that shrinks at rate $\sqrt{n}$:

$$\sqrt{n}(\bar{Y} - \mu) \xrightarrow{d} N(0, \sigma^2).$$

It needs a finite variance and, again, independence or fading dependence. Under dependence, σ^2 becomes the *long-run* variance of §2.5.

Three consequences for practice:

1. **The $\sqrt{n}$ rule.** Standard errors fall like $1/\sqrt{n}$, so halving a standard error takes four times the data. Equivalently, detecting an effect half as large requires a sample four times as big. Precision is expensive, and it gets more expensive quickly.
2. **The approximation is to the average, not to the data.** The CLT says nothing about whether individual observations are normal. It justifies *t*-tests on regression coefficients, which are weighted averages, with non-normal data. It does not justify treating tail probabilities of the data themselves as normal.
3. **Heavy tails slow the approximation down.** When the variance exists but higher moments are large or infinite, the CLT still holds eventually, but the sample size needed for it can be very large. **[Fact]** Daily equity returns have tails heavy enough that the fourth moment may not exist (Cont, 2001). Their means and variances behave tolerably, but any statistic that depends on fourth moments, such as sample kurtosis, the standard error of a variance estimate, or the non-normal standard error of a Sharpe ratio, is far less precise than its formula suggests.

2.5 Effective sample size: dependence spends observations

The CLT's variance for the mean of a stationary series is not σ^2 but the **long-run variance**, which includes all the autocovariances. The variance of a sample mean of T observations is approximately

$$\text{Var}(\bar{Y}) \approx \frac{\sigma^2}{T} \left(1 + 2 \sum_{j=1}^{\infty} \rho_j \right),$$

where ρ_j is the autocorrelation at lag j . Positive autocorrelation inflates the variance, because neighbouring observations repeat part of the same information. Comparing this with the independent case defines the **effective sample size**:

$$n_{\text{eff}} = \frac{T}{1 + 2 \sum_{j \geq 1} \rho_j}.$$

For a first-order autoregression, $\rho_j = \phi^j$ and the sum is geometric, which gives

$$n_{\text{eff}} = T \frac{1 - \phi}{1 + \phi}.$$

Persistence ϕ	n_{eff}/T	1,000 observations are worth
0	1.000	1,000
0.5	0.333	333
0.9	0.053	53
0.99	0.005	5

The same arithmetic holds across clusters. If n observations fall into groups of size $\bar{n}_g$ and every pair within a group has correlation ρ , the effective sample size for a mean is $n/(1 + (\bar{n}_g - 1)\rho)$. For a regression coefficient the formula picks up the within-group correlation of the regressor as well, which gives the Moulton factor of §4.4.

Overlapping observations are the case finance meets most. A series of overlapping h -period returns built from T one-period returns carries roughly the information of T/h non-overlapping ones when estimating a mean. Forty years of monthly data on 12-month returns have 480 rows and about 40 independent observations.

Merton's asymmetry. The precision of an estimated *mean* return depends on the calendar span of the data, not on how finely it is sampled. The mean return over forty years is the total log price change divided by forty, and sampling daily instead of monthly does not change the endpoints. With annual volatility of 16%, the standard error of a mean annual return estimated from 40 years is $16\%/\sqrt{40} = 2.5\%$, so a 6% equity premium is known only to within about ± 5 percentage points at 95% confidence. *Variances* behave oppositely: summing squared high-frequency returns estimates variance more and more precisely as the sampling interval shrinks (Merton, 1980). This single asymmetry explains a great deal of empirical finance: why risk models work far better than return forecasts, why the covariance matrix is estimable and the mean vector is not, and why the evidence for any return anomaly is always weaker than its t -statistic makes it look.

2.6 The delta method, and other tools for approximations

Three results let you manipulate estimates the way you would manipulate the numbers they estimate.

- **The continuous mapping theorem.** If $\hat{\theta} \xrightarrow{p} \theta_0$ and g is continuous, then $g(\hat{\theta}) \xrightarrow{p} g(\theta_0)$.
- **Slutsky's theorem.** A statistic that converges in distribution, multiplied by something that converges in probability, converges to the obvious product. This is what licenses replacing an unknown variance with a consistent estimate inside a t -statistic.
- **The delta method.** If $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, V)$ and g is differentiable, then

$$\sqrt{n}(g(\hat{\theta}) - g(\theta_0)) \xrightarrow{d} N(0, g'(\theta_0)^2 V).$$

The standard error of a smooth function of an estimate is the standard error of the estimate times the slope of the function.

A worked example, and its failure. A shock to an AR(1) process decays like ϕ^h , so its **half-life**, the horizon at which half of it is gone, is $\ln(0.5)/\ln(\phi)$. Suppose $\hat{\phi} = 0.95$ with a standard error of 0.02. The point estimate of the half-life is $0.693/0.0513 = 13.5$ periods. Its derivative with respect to ϕ is $\ln(2)/(\phi(\ln \phi)^2) = 0.693/(0.95 \times 0.00263) = 277$, so the delta-method standard error is $277 \times 0.02 = 5.5$ periods, giving an interval of roughly 13.5 ± 11 .

That interval is badly wrong, and the way it is wrong generalises. Two standard errors above the estimate, $\phi = 0.99$ implies a half-life of 69 periods, not $13.5 + 11 = 24.5$. The half-life is so nonlinear near $\phi = 1$ that a linear approximation across the confidence region is meaningless. Whenever a parameter sits near a boundary, whether a unit root, a variance near zero, or a correlation near one, delta-method intervals mislead. Transform the endpoints of an interval for ϕ instead, or bootstrap (§4.9).

A second example is the one quants use most. For independent, normally distributed returns, the delta method gives the standard error of an estimated per-period Sharpe ratio as approximately $\sqrt{(1 + \text{SR}^2/2)/T}$ (Lo, 2002). In annual units, with a true Sharpe ratio of 0.5, that is about $1.06/\sqrt{\text{years}}$. It takes around 18 years of data for the estimate to sit two standard errors from zero.

2.7 When asymptotics lie

Asymptotic approximations are statements about what happens as some quantity goes to infinity. Trouble starts when the quantity that actually governs the approximation is not the one you think is large.

Situation	What is really small	Symptom	Where
Few clusters, or few <i>treated</i> clusters	The number of clusters, not observations	Tests reject true nulls far too often	§4.4
Weak instruments	The first-stage signal	IV biased toward OLS; oversized <i>t</i> -tests	§7.3
Near-unit-root regressors	The distance of ϕ from one, times T	Spurious significance; Stambaugh bias	§8.3, §9.2
Long overlapping horizons	The number of non-overlapping periods	Oversized Newey–West tests	§4.5, §9.2
Heavy tails	The effective number of tail observations	Unstable higher-moment estimates	§2.4
Many parameters	Observations per parameter	In-sample fit far above out-of-sample fit	§10.2
Estimation after model selection	Nothing: the approximation is invalid	Confidence intervals too narrow	§10.3
Maxima over many tests	Nothing: not a smooth function of means	Naive bootstrap fails	§4.8, §4.9

§2 Key takeaways

1. A standard error is meaningless until you say what is being redrawn. In finance the sample is usually the history, not the cross-section.
2. “Uncorrelated,” “mean-independent,” and “independent” are different assumptions. Returns are close to uncorrelated and far from independent, and tests that confuse the two are mis-sized.
3. Consistency matters more than unbiasedness in practice, because almost no estimator in use is unbiased. A little bias can buy a large reduction in variance.
4. Precision grows like $\sqrt{n}$: four times the data halves the standard error.
5. Dependence spends observations. An AR(1) with $\phi = 0.9$ turns 1,000 observations into about 53, and 40 years of overlapping 12-month returns into about 40.
6. The precision of a mean return depends on calendar span, not sampling frequency; the precision of a variance improves with frequency. This asymmetry shapes most of empirical finance.
7. The delta method fails near boundaries such as unit roots. When a parameter is near a boundary, transform interval endpoints or bootstrap.
8. Asymptotic approximations fail when the quantity that governs them (clusters, instrument strength, distance from a unit root, independent periods) is small, however many rows the data have.

3. Regression from first principles

Regression is usually taught as a model: assume $Y = X'\beta + u$ with some list of properties for u , then estimate β . This section builds it the other way round, starting from objects that exist whether or not any model is true. The payoff is a clean answer to the question the rest of the document keeps asking: when does a regression coefficient mean what you want it to mean?

3.1 The conditional expectation function

The **conditional expectation function** (CEF) $m(x) = \mathbb{E}[Y_i \mid X_i = x]$ gives the average outcome at each value of the regressors. Define the CEF error as whatever is left over:

$$Y_i = m(X_i) + \epsilon_i, \quad \epsilon_i = Y_i - \mathbb{E}[Y_i \mid X_i].$$

By the law of iterated expectations, $\mathbb{E}[\epsilon_i \mid X_i] = 0$ automatically. **This is a definition, not an assumption.** Every outcome can be split into its conditional mean and a mean-independent remainder, whatever the data are and whatever caused them. So a statement like “the error has mean zero given X ” has no content until the error is given some meaning *beyond* “whatever is left over.” That meaning is exactly what §3.2 separates out.

By §2.2, the CEF is the best predictor of Y given X . The average log wage at each level of schooling, or the average next-month return at each level of the dividend yield, is a CEF. It describes the data. It does not, by itself, say what would happen to a person’s wage if they stayed in school longer, or to a market’s return if its dividend yield were changed.

3.2 Linear projection: the regression that always exists

The CEF can be any shape. The **linear projection** of Y on X is the best *linear* predictor, the vector β that minimises the mean squared error of $X'\beta$:

$$\beta = \arg \min_b \mathbb{E}[(Y_i - X_i'b)^2] = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i Y_i].$$

It exists whenever second moments are finite and no regressor is an exact linear combination of the others. Its error $e_i = Y_i - X_i'\beta$ satisfies the first-order condition

$$\mathbb{E}[X_i e_i] = 0$$

by construction. With a single regressor and an intercept, the slope reduces to the familiar $\text{Cov}(X, Y) / \text{Var}(X)$.

Two facts connect the projection to the CEF. If the CEF happens to be linear, the two coincide. If it is not, the projection is still the best linear *approximation* to the CEF, in the sense of minimising $\mathbb{E}[(m(X_i) - X_i'\beta)^2]$. So a regression is always a valid summary of how the average outcome varies with the regressors, to the extent a straight line can summarise it.

That is the first half of the central point of this section:

A regression is never wrong. It always estimates the linear projection, a well-defined population object. What can be wrong is the claim about what the projection measures.

Now bring in the second half. Suppose that Y is actually generated by a **structural equation**, one meant to describe cause and effect:

$$Y_i = X_i'\beta_s + u_i,$$

where β_s is defined by a thought experiment (change X , hold everything else fixed, watch Y) and u_i collects everything else that affects Y . Nothing guarantees $\mathbb{E}[X_i u_i] = 0$. Substitute the structural equation into the projection formula:

$$\beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i (X_i'\beta_s + u_i)] = \beta_s + \underbrace{\mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i u_i]}_{\text{endogeneity bias}}.$$

This one line is the whole problem of Part III. **The regression estimates β . You want β_s . They differ by the projection of the structural error on the regressors**, and that term is zero exactly when the regressors are uncorrelated with everything

else that moves the outcome. No amount of data shrinks it, because it is a difference between two population quantities. The projection error e_i is uncorrelated with X_i by definition. The structural error u_i is uncorrelated with X_i only if the world cooperates.

3.3 OLS: the sample moment condition, and its geometry

Ordinary least squares replaces population moments with sample moments:

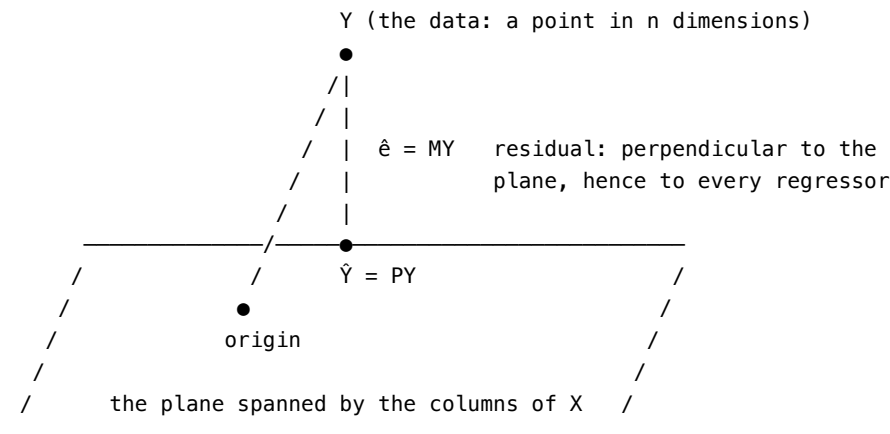
$$\hat{\beta} = \left(\sum_{i=1}^n X_i X_i' \right)^{-1} \sum_{i=1}^n X_i Y_i = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}.$$

Its defining property is the **normal equations** $\mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\beta}) = 0$, the sample analogue of $\mathbb{E}[X_i e_i] = 0$. This is the master form of §1.5 with $\psi_i(\beta) = X_i(Y_i - X_i'\beta)$.

The geometry. Think of $\mathbf{Y}$ as a point in n -dimensional space and the k columns of $\mathbf{X}$ as spanning a k -dimensional flat subspace. OLS finds the point of that subspace closest to $\mathbf{Y}$, which is its orthogonal projection:

$$\hat{\mathbf{Y}} = \mathbf{P}\mathbf{Y}, \quad \mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \quad \hat{\mathbf{e}} = \mathbf{M}\mathbf{Y}, \quad \mathbf{M} = \mathbf{I} - \mathbf{P}.$$

The residual vector is perpendicular to every column of $\mathbf{X}$, and therefore to the fitted values.



Two consequences of the picture are worth fixing.

- **Pythagoras gives R^2 .** With an intercept, the total sum of squares of Y around its mean splits exactly into the explained part, $\|\hat{\mathbf{Y}} - \bar{Y}\|^2$, and the residual part, $\|\hat{\mathbf{e}}\|^2$.
- **Residuals cannot reveal endogeneity.** $\mathbf{X}'\hat{\mathbf{e}} = 0$ holds in every sample by construction, whatever the true relationship between X and u (§1.5).

Leverage and influence. The diagonal element $\mathbf{P}_{ii}$ is observation i 's **leverage**, a number between 0 and 1 that measures how far its regressors lie from the bulk of the data. The leverages sum to k . High-leverage observations pull the fitted plane toward themselves, so their residuals are *shrunk*: under homoskedasticity $\text{Var}(\hat{e}_i) = \sigma^2(1 - \mathbf{P}_{ii})$. Dropping observation i changes the estimate by

$$\hat{\beta} - \hat{\beta}_{(-i)} = (\mathbf{X}'\mathbf{X})^{-1} X_i \frac{\hat{e}_i}{1 - \mathbf{P}_{ii}},$$

so an observation is influential when it has both high leverage and a large residual. Financial data are full of such points: crash months, microcap stocks, and firms with extreme accounting ratios. **[Practice]** Before believing a coefficient, compute how much it moves when the most influential 1% of observations are dropped. If the answer is “most of the way to zero,” the finding belongs to those observations, not to the sample.

3.4 Frisch–Waugh–Lovell: what “controlling for” means

Split the regressors into two groups, $Y = X_1\beta_1 + X_2\beta_2 + e$. The **Frisch–Waugh–Lovell (FWL) theorem** says that the multiple-regression coefficient $\hat{\beta}_1$ can be computed in two steps:

1. Regress X_1 on X_2 and keep the residuals $\tilde{X}_1 = M_2X_1$, the part of X_1 that X_2 cannot linearly predict. Here $M_2 = I - X_2(X_2'X_2)^{-1}X_2'$.
2. Regress Y on $\tilde{X}_1$: $\hat{\beta}_1 = (\tilde{X}_1'\tilde{X}_1)^{-1}\tilde{X}_1'Y$.

Residualising Y on X_2 as well gives the same coefficient. The theorem follows from the geometry of §3.3: the component of X_1 lying inside the space of X_2 is already accounted for by X_2 's coefficients, so only the orthogonal component can identify β_1 .

The interpretation is the most useful single idea in applied regression:

“Controlling for X_2 ” means using only the variation in X_1 that X_2 cannot predict. The coefficient is estimated entirely from that residual variation.

Four consequences follow, and each one comes back later.

1. **Controls consume identifying variation.** The precision of $\hat{\beta}_1$ depends on how much residual variation remains. If $R_{1,2}^2$ is the R^2 from regressing X_1 on the controls, the variance of $\hat{\beta}_1$ is inflated by the **variance inflation factor** $1/(1 - R_{1,2}^2)$ relative to a regression with no controls. With $R_{1,2}^2 = 0.9$, the standard error is $\sqrt{10} = 3.2$ times larger, as if the sample were ten times smaller. “Multicollinearity” is not a violation of any assumption. It is a shortage of independent variation, which is why Goldberger (1991) suggested renaming it “micronumerosity.”
2. **The estimate is local to the residual variation.** Suppose you regress stock returns on analyst coverage, controlling for firm size. Size explains most of the cross-sectional variation in coverage, so the coefficient is identified from firms with *unusually* high or low coverage for their size. Those firms are unusual in other ways too, such as recent listings, controversy, or index membership. The estimate describes them, and extrapolating it to a typical firm is an additional assumption.
3. **Regression weights observations by the conditional variance of the regressor.** With a binary treatment and a set of discrete controls, the OLS coefficient is a weighted average of the treatment–control differences within each cell of the controls, with each cell weighted by its size times $p(1 - p)$, where p is the share treated in that cell (Angrist, 1998). Cells where almost everyone or almost no one is treated get almost no weight. If the effect differs across cells, the regression coefficient is neither the average effect in the population nor the average effect on the treated. It is a variance-weighted average that can be dominated by an unrepresentative subset (Aronow & Samii, 2016; Słoczyński, 2022).
4. **Fixed effects and “partialling out” are FWL.** Adding a dummy for every firm is algebraically identical to subtracting each firm's mean from both sides (§7.4). The double machine learning estimator of §10.5 is FWL with the linear first-step regressions replaced by flexible predictions.

3.5 The omitted variable bias formula

FWL says what including a control does. The **omitted variable bias (OVB) formula** says what leaving one out costs. Suppose the regression you would like to run is

$$Y_i = \alpha + \beta X_i + \gamma C_i + u_i,$$

and the one you can run omits C_i . Write the projection of the omitted variable on the included one as $C_i = \delta_0 + \delta X_i + v_i$. Substituting,

$$Y_i = (\alpha + \gamma\delta_0) + (\beta + \gamma\delta) X_i + (\gamma v_i + u_i),$$

and because v_i is uncorrelated with X_i by construction, the short regression's slope converges to

$$\text{plim } \hat{\beta}_{\text{short}} = \beta + \gamma\delta.$$

In words: **the bias equals the effect of the omitted variable on the outcome, times the coefficient of the omitted variable on the included one.** Both links must be present. An omitted variable that affects Y but is uncorrelated with X costs precision, not bias. One that is correlated with X but does not affect Y costs nothing.

	Omitted variable correlated positively with X ($\delta > 0$)	Correlated negatively ($\delta < 0$)
Omitted variable raises Y ($\gamma > 0$)	Upward bias	Downward bias
Omitted variable lowers Y ($\gamma < 0$)	Downward bias	Upward bias

A worked example, with illustrative numbers. Suppose a regression of log earnings on years of schooling gives $\hat{\beta}_{\text{short}} = 0.10$, a 10% earnings gain per year. Ability is unobserved. Suppose one standard deviation of ability raises log earnings by $\gamma = 0.05$ holding schooling fixed, and each year of schooling is associated with $\delta = 0.4$ standard deviations more ability. The bias is $0.05 \times 0.4 = 0.02$, so the causal return would be about 0.08. A fifth of the naive estimate is selection. The formula does not tell you γ and δ , but it tells you exactly what you would need to believe about them to defend the estimate, and that turns a vague worry into a specific argument.

The formula has three further uses.

- **Signing the bias without data.** If you can argue the direction of both links, you know whether the estimate is an upper or a lower bound.
- **Coefficient stability as evidence.** If adding a set of observed controls barely moves the coefficient, and those controls are informative about the unobserved ones, the remaining bias may be small. §6.7 makes this argument formal, and shows its limits.
- **Noisy controls control only partly.** A control measured with error is only partially correlated with the true confounder, so including it removes only part of the bias. In the simulation of §6.5, controlling for a proxy with a reliability of 0.5 removes under 40% of the bias. **[Simulated]** “We controlled for it” often means “we controlled for a noisy version of it.”

The multivariate generalisation is the endogeneity-bias formula at the end of §3.2. OVB is its most common special case.

3.6 Reading coefficients

A coefficient is only interpretable in the units of the model it came from.

Specification	Interpretation of the slope
$Y = \alpha + \beta X$	A one-unit difference in X goes with β units of Y
$\log Y = \alpha + \beta X$	A one-unit difference in X goes with about $100\beta\%$ of Y ; exactly $100(e^\beta - 1)\%$
$Y = \alpha + \beta \log X$	A 1% difference in X goes with $\beta/100$ units of Y
$\log Y = \alpha + \beta \log X$	An elasticity : a 1% difference in X goes with $\beta\%$ of Y
$\log Y = \alpha + \beta D$, D binary	A proportional difference of exactly $e^\beta - 1$
$Y = \alpha + \beta_1 X + \beta_2 X^2$	The slope $\beta_1 + 2\beta_2 X$ varies with X
$Y = \alpha + \beta_1 X + \beta_2 Z + \beta_3 XZ$	The slope of X is $\beta_1 + \beta_3 Z$; β_1 alone is the slope where $Z = 0$

Four traps are common enough to name.

- **Log approximations fail for large coefficients.** A dummy coefficient of 0.3 in a log regression is a 35% difference, not 30%, and -0.3 is -26% , not -30% .
- **Main effects in interaction models are evaluated at zero.** If Z is firm size in dollars, β_1 is the effect of X for a firm of size zero, which does not exist. Centre Z at a meaningful value before interacting.
- **Logs of variables with zeros are not unit-free.** Transforms like $\log(1 + Y)$ or $\text{arcsinh}(Y)$, often applied to trading volume, counts, or R&D spending, give coefficients that change when Y is rescaled, so the “percentage” reading is not well-defined (Chen & Roth, 2024).
- **Predicting $\log Y$ is not predicting Y .** $\exp(\mathbb{E}[\log Y \mid X])$ is the conditional *geometric* mean, which is below $\mathbb{E}[Y \mid X]$. Retransforming a log-scale forecast needs a correction (Simple and Log Returns §11.5).

“Holding other variables constant” deserves a warning of its own. It describes the arithmetic of the linear projection, a partial derivative of $X'\beta$. It is *not* a description of a feasible change in the world. If analyst coverage causes firms to grow, then “the effect of coverage holding size constant” asks what coverage does through every channel *except* size, which may not be the question anyone wanted answered (§6.5).

3.7 The classical assumptions, ranked by what breaks

Textbooks list the assumptions of the linear model as if they were equally important. They are not. What matters is what goes wrong when each one fails.

Assumption	What it buys	If it fails	The modern response
Exogeneity: $\mathbb{E}[u_i X_i] = 0$	$\hat{\beta}$ estimates the structural β_s	Wrong answer , and more data do not help	Research design (Part III)
Independence across observations	The simple meat	Wrong precision , often by large factors	Cluster-robust or HAC standard errors (§4)
Homoskedasticity	Classical standard errors; OLS is efficient	Wrong precision, usually too small; some efficiency lost	Heteroskedasticity-robust standard errors (§4.3)
Linearity of the interpreted model	The coefficient has the meaning assigned to it	You estimate the best linear approximation, which may mislead	Flexible specifications; binned scatter plots (§11.4)
No perfect collinearity	A unique $\hat{\beta}$	Not identified	Reparametrise. Near-collinearity costs precision, not bias
Normal errors	Exact finite-sample t and F distributions	Almost nothing in moderate samples, by the CLT	Asymptotic inference; the bootstrap in small samples

The ranking is the point. **[Fact]** A failure of exogeneity gives the wrong answer. A failure of independence or homoskedasticity gives the wrong confidence in an answer. A failure of normality in a sample of reasonable size gives almost nothing.

The **Gauss–Markov theorem** says that under the first five assumptions OLS has the smallest variance among linear unbiased estimators. It gets more attention than it earns. The class of “linear unbiased” estimators is narrow, efficiency is a second-order concern next to bias, and since White (1980) valid inference has not needed homoskedasticity at all. Its main modern use is as a reminder of what is lost by *not* weighting when the error variance is known.

3.8 R-squared, and what it is not

$R^2 = 1 - \sum_i \hat{e}_i^2 / \sum_i (Y_i - \bar{Y})^2$ is the share of the sample variance of Y captured by the linear projection. It also equals the squared correlation between Y and the fitted values $\hat{Y}$. It measures fit, and nothing else.

- **It is not a test of causal validity.** A randomised experiment with a noisy outcome can estimate a treatment effect precisely with an R^2 of 0.02. Two unrelated trending series can produce an R^2 above 0.3 about a third of the time, from nothing but shared drift (§8.3).
- **It is not a measure of out-of-sample performance.** In-sample R^2 never falls when a regressor is added, whatever the regressor is (§10.2).
- **A small R^2 can be economically large.** This is the version that matters in finance.

The last point deserves its derivation, because it recalibrates intuition about return predictability. Consider an investor with mean-variance preferences and risk aversion γ_r who holds a risky asset with excess return mean μ , volatility σ , and Sharpe ratio $S = \mu/\sigma$. Without a forecast, the optimal weight is $\mu/(\gamma_r \sigma^2)$ and the expected excess return on the investor’s portfolio is S^2/γ_r . Now give the investor a predictor that explains a fraction R^2 of the variance of the asset’s return. The conditional mean μ_t then has variance $R^2 \sigma^2$ around μ , and the unexplained variance is $(1 - R^2) \sigma^2$. The optimal weight becomes $\mu_t/(\gamma_r (1 - R^2) \sigma^2)$, and the expected portfolio excess return is this weight times μ_t , averaged over its distribution: $\mathbb{E}[\mu_t^2]/(\gamma_r (1 - R^2) \sigma^2)$. Since μ_t has mean μ and variance $R^2 \sigma^2$, $\mathbb{E}[\mu_t^2] = \mu^2 + R^2 \sigma^2$, so the expected portfolio excess return becomes

$$\frac{\mathbb{E}[\mu_t^2]}{\gamma_r (1 - R^2) \sigma^2} = \frac{\mu^2 + R^2 \sigma^2}{\gamma_r (1 - R^2) \sigma^2} = \frac{S^2 + R^2}{\gamma_r (1 - R^2)}.$$

Relative to no forecast, expected return rises by a factor of $(1 + R^2/S^2)/(1 - R^2)$, which is approximately $1 + R^2/S^2$. **The right benchmark for a predictive R^2 is the squared Sharpe ratio, not 1.** A monthly Sharpe ratio of 0.12 (about 0.42 annualised) gives $S^2 = 0.0144$. A monthly R^2 of 0.5% would then raise expected return by about 35%. This is the logic of Campbell & Thompson (2008), and it is why an out-of-sample R^2 that looks negligible can be worth money.

The same logic cuts the other way. Because the benchmark is so small, an in-sample R^2 of that size is also easy to produce by accident, through a search over predictors, overlapping observations, or a persistent regressor (§9.2). The relevant statistic for prediction is the **out-of-sample** R^2 ,

$$R_{\text{os}}^2 = 1 - \frac{\sum_t (r_t - \hat{r}_t)^2}{\sum_t (r_t - \bar{r}_t)^2},$$

where $\hat{r}_t$ is the forecast made with information available before t and $\bar{r}_t$ is the historical average at that date. It can be, and often is, negative (§8.8).

§3 Key takeaways

1. The error of the conditional expectation function is mean-independent of the regressors by definition. “The error has mean zero” says nothing until the error is given a structural meaning.
2. A regression always estimates the linear projection, a valid population object. It equals the causal coefficient only when the structural error is uncorrelated with the regressors, and the gap does not shrink with sample size.
3. Residuals are orthogonal to regressors by construction, so no residual diagnostic can detect endogeneity.
4. Controlling for a variable means using only the variation it cannot predict. Controls consume identifying variation and make the estimate local to the observations that retain it.
5. Omitted variable bias is the omitted variable’s effect times its coefficient on the included regressor. Use the formula to sign biases, and remember that a noisy control removes only part of one.
6. With heterogeneous effects, regression reports a variance-weighted average that can be dominated by an unrepresentative subset of the data.
7. Rank the classical assumptions by consequence: exogeneity failures give wrong answers, dependence and heteroskedasticity give wrong precision, and non-normality rarely matters.
8. Judge a predictive R^2 against the squared Sharpe ratio. A monthly R^2 of 0.5% can be economically large, and is also easy to manufacture in-sample.

Part II — Estimation and inference

4. Inference: how sure should you be?

Identification decides whether an estimate is aimed at the right target. Inference decides how far from the target it is likely to land. This section is about the second question, and its central message is the one from §1.6: every standard error rests on an assumption about which observations are independent of which, and when that assumption is wrong the error is measured in factors, not decimals.

4.1 What a standard error measures

A **standard error** is an estimate of the standard deviation of an estimator's sampling distribution: how much the estimate would vary across the samples the data-generating process could have produced. A ***t*-statistic** divides the distance between the estimate and a hypothesised value by the standard error. A **95% confidence interval**, roughly the estimate plus or minus two standard errors, is a procedure that covers the true value in 95% of repeated samples.

That last sentence is easy to misread. A computed interval either contains the true value or it does not, so it is not “95% likely” to contain it in any frequentist sense. The reading that is both correct and useful is this: **the confidence interval is the set of parameter values that the data do not reject at the 5% level**. Values inside it are compatible with the data, and values outside it are not. Read that way, an interval of $[-0.1, 3.0]$ and one of $[-0.01, 0.02]$ are both “not significant,” and they say completely different things.

Every standard error is conditional on the model of how the sample could have come out differently. §4.2–§4.5 are four versions of that model.

4.2 The sandwich, derived for OLS

Substitute $Y_i = X_i' \beta + e_i$ into the OLS formula and rearrange:

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i e_i.$$

The first factor converges to Q_{XX}^{-1} . The second is a scaled sum of the moment contributions $X_i e_i$, and a central limit theorem makes it approximately normal with variance $\Omega = \text{Var}(n^{-1/2} \sum_i X_i e_i)$. So

$$\text{Avar}(\sqrt{n}(\hat{\beta} - \beta)) = Q_{XX}^{-1} \Omega Q_{XX}^{-1}.$$

This is §1.6 with the OLS bread. Everything in the rest of this section is a statement about Ω .

- **If the observations are independent**, the cross-products between different observations have expectation zero and $\Omega = \mathbb{E}[e_i^2 X_i X_i']$.
- **If, in addition, the error variance does not depend on X** ($\mathbb{E}[e_i^2 | X_i] = \sigma^2$), then $\Omega = \sigma^2 Q_{XX}$, and the sandwich collapses to the classical formula $\sigma^2 Q_{XX}^{-1}$.

The classical standard error is therefore the *most* restrictive special case, not the neutral default. It assumes both independence and constant variance, and it is correct only when both hold.

4.3 Heteroskedasticity

Replace the population meat with its sample analogue and you get the **heteroskedasticity-robust** or **White** variance estimator (White, 1980):

$$\hat{V}_{\text{HC}} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n \hat{e}_i^2 X_i X_i' \right) (\mathbf{X}'\mathbf{X})^{-1}.$$

It is consistent whatever form the heteroskedasticity takes. Several small-sample variants exist. HC1 rescales by $n/(n-k)$. HC2 and HC3 divide each squared residual by $(1 - \mathbf{P}_{ii})$ and $(1 - \mathbf{P}_{ii})^2$ respectively, which undoes the shrinkage of high-leverage residuals noted in §3.3. HC3 is close to a jackknife and performs best in small samples with influential points (MacKinnon & White, 1985).

Three practical points:

- **Heteroskedasticity does not bias $\hat{\beta}$.** It only corrupts the classical standard error, and usually makes it too small. That happens whenever the error variance is larger where the regressor is far from its mean, which is typical: small firms have both extreme characteristics and volatile returns.
- **[Practice] Use robust standard errors by default.** In samples of reasonable size they cost almost nothing when the errors happen to be homoskedastic, and they protect you when they are not.
- **Weighting recovers efficiency when the variance is predictable.** Weighted least squares divides each observation by its error standard deviation. Scaling returns by a volatility forecast before regressing is exactly this, and it is often a larger improvement than any change of estimator. A large gap between classical and robust standard errors is also a useful signal that the model is misspecified in some other way (King & Roberts, 2015).

4.4 Clustering

Now drop independence. Suppose observations come in G groups (states, industries, trading days, firms), and errors are correlated within a group but independent across groups. How wrong is the classical standard error?

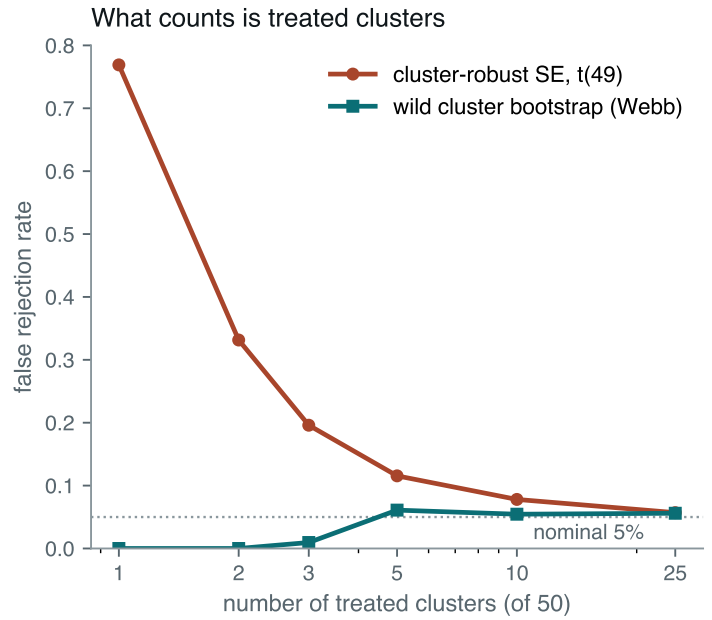
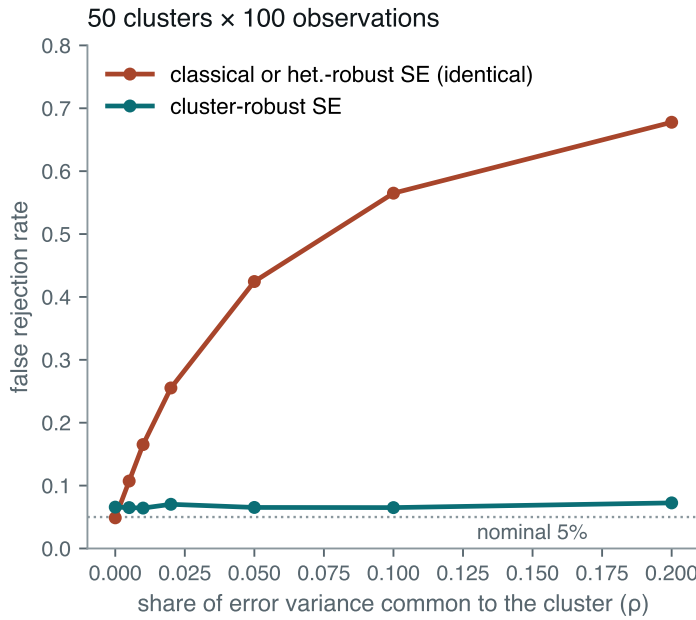
The Moulton factor. With equal group sizes $\bar{n}_g$, an intra-cluster correlation of the error ρ_u , and an intra-cluster correlation of the regressor ρ_x , the true variance of the OLS slope exceeds the classical formula by approximately (Moulton, 1990)

$$\frac{\text{Var}_{\text{true}}(\hat{\beta})}{\text{Var}_{\text{classical}}(\hat{\beta})} \approx 1 + (\bar{n}_g - 1) \rho_x \rho_u.$$

The formula is worth reading term by term.

- **Both correlations must be present.** If the regressor varies independently within groups ($\rho_x = 0$), correlated errors cost nothing. Clustering matters most when the regressor is *constant* within groups ($\rho_x = 1$), as with a state law, an industry shock, or a date-level event.
- **Large groups amplify small correlations.** The factor multiplies ρ_u by $\bar{n}_g - 1$. An error correlation of 0.05 is negligible with groups of 3 and enormous with groups of 1,000.

Take 50 groups of 100 observations, a group-level regressor, and $\rho_u = 0.05$. The factor is $1 + 99 \times 0.05 = 5.95$, so the true standard error is $\sqrt{5.95} = 2.44$ times the classical one, and the 5,000 observations carry the information of about 840 independent ones. A classical t -statistic of 2.4 is really about 1.0. With groups of 1,000 instead, the same correlation gives a factor of $1 + 999 \times 0.05 = 51$: standard errors 7.1 times too small. The simulation in the left panel below confirms it: classical standard errors reject a true null 42% of the time at a nominal 5%, and 57% of the time when $\rho_u = 0.1$. Heteroskedasticity-robust errors are no better, because the problem is not heteroskedasticity. **[Simulated]**



The **cluster-robust estimator** lets the errors within each group be correlated arbitrarily and sums the moment contributions group by group before squaring:

$$\hat{V}_{CR} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{X}'_g \hat{\mathbf{e}}_g \hat{\mathbf{e}}'_g \mathbf{X}_g \right) (\mathbf{X}'\mathbf{X})^{-1},$$

where $\mathbf{X}_g$ and $\hat{\mathbf{e}}_g$ stack the observations in group g . The usual small-sample adjustment multiplies by $\frac{G}{G-1} \cdot \frac{n-1}{n-k}$, and tests use the t distribution with $G - 1$ degrees of freedom. The meat is now estimated from G group sums. **The effective number of observations for inference is the number of clusters, not the number of rows.**

Where to cluster. Two rules are in use, and they usually agree.

- **The design rule.** Cluster at the level at which the treatment or regressor was assigned, or at which the sample was drawn. If a policy varies by state, cluster by state. Abadie, Athey, Imbens & Wooldridge (2023) show that this, rather than the presence of within-group residual correlation, is what justifies clustering, and that clustering at a coarser level than the design requires can be unnecessarily conservative.
- **The conservative rule.** Cluster at the broadest level at which errors might be correlated, provided there are enough clusters (Cameron & Miller, 2015). Coarser clusters remove more dependence, and leave fewer clusters to estimate the meat from.

Few clusters, and few treated clusters. Cluster-robust inference is an asymptotic approximation in G , and it degrades when clusters are few (a common rule of thumb is fewer than 40 to 50), unequal in size, or, worst of all, when only a few of them are treated. The right panel above holds $G = 50$ and varies the number of treated clusters. With one treated cluster the cluster-robust t -test rejects a true null 77% of the time, with two 33%, with five 12%, and with ten 8%. **[Simulated]** The reason is mechanical: a treated cluster's own dummy-like variation lets OLS fit that cluster's mean almost exactly, so its residual sum, which is precisely the piece of the meat that measures the treatment's noise, is forced toward zero. The **wild cluster bootstrap**, which flips the signs of whole clusters' residuals at random and re-estimates, is the standard remedy (Cameron, Gelbach & Miller, 2008). As the panel shows, it errs the other way with one or two treated clusters, where it almost never rejects, and becomes reliable from about five (MacKinnon & Webb, 2017; MacKinnon, Nielsen & Webb, 2023). With a single treated unit, cluster-based inference is not available at all, and the design has to change: randomisation inference (§7.2), synthetic control (§7.7), or the approach of Conley & Taber (2011).

Two dimensions at once. Finance panels are often correlated both within firms over time and across firms at a date. **Two-way clustering** adds the firm-clustered and date-clustered meats and subtracts the heteroskedasticity-robust one that both double-count (Cameron, Gelbach & Miller, 2011; Thompson, 2011). It needs many clusters in *both* dimensions, which is why §9.5 recommends it for panels of thousands of firms over decades and not for ten years of annual data.

The canonical warning. **[Fact]** Bertrand, Duflo & Mullainathan (2004) took about 20 years of state-level data on women's wages from the Current Population Survey, assigned fake "laws" to randomly chosen states and years, and ran the standard difference-in-differences regression with conventional standard errors. The placebo laws were significant at 5% in up to 45%

of simulations. Outcomes that are serially correlated within a state make the effective sample the number of states, not the number of state-years. Clustering by state repairs it, when there are enough states.

4.5 Serial correlation and HAC standard errors

Time series are one cluster, ordered. The moment contributions $\psi_t = X_t e_t$ are correlated with their neighbours, and the meat is the **long-run variance**

$$\Omega = \sum_{j=-\infty}^{\infty} \Gamma_j, \quad \Gamma_j = \mathbb{E}[\psi_t \psi_{t-j}'],$$

the sum of all the autocovariances of the moment contributions. It is the multivariate version of the variance inflation in §2.5.

The **Newey–West estimator** truncates the sum at a lag L and down-weights distant lags linearly:

$$\hat{\Omega}_{\text{NW}} = \hat{\Gamma}_0 + \sum_{j=1}^L \left(1 - \frac{j}{L+1}\right) (\hat{\Gamma}_j + \hat{\Gamma}_j').$$

The linear **Bartlett weights** are not decoration. They guarantee that the estimate is positive semi-definite, so that no variance comes out negative, which the unweighted sum does not (Newey & West, 1987). Such estimators are called **HAC**, for heteroskedasticity and autocorrelation consistent.

The bandwidth is a bias–variance choice. Too small an L misses autocorrelation and understates the variance. Too large an L makes the variance estimate noisy, which fattens the tails of the t -statistic. The textbook rule $L = 0.75 T^{1/3}$ errs toward the first failure. Lazarus, Lewis, Stock & Watson (2018) simulate a regression with $T = 200$ and moderately persistent regressor and error, and find that this rule with normal critical values rejects a true null 9% to 18% of the time at a nominal 5%, depending on the persistence. **[Fact]** Their recommendation is a much larger bandwidth, $L = 1.3 T^{1/2}$, combined with **fixed- b** critical values that account for the noise in the variance estimate, or an equal-weighted cosine estimator with $\nu = 0.4 T^{2/3}$ terms and Student- t critical values with ν degrees of freedom. Both cut the distortions roughly in half.

Overlapping observations are where HAC inference fails most often in finance. Regress h -month returns on a predictor using monthly data, and consecutive dependent variables share $h - 1$ months. Under the null of no predictability the error is a moving average of order $h - 1$, so Newey–West with at least $h - 1$ lags is consistent in principle. In a sample of realistic length it is not close. With 40 years of monthly data and an exogenous, persistent predictor, Newey–West with h lags rejects a true null 15% of the time at $h = 12$ and 28% at $h = 60$. Classical standard errors reject 57% and 78%. **[Simulated]** Standard errors that impose the null of no predictability when estimating the meat, due to Hodrick (1992), hold the 5% line (§9.2).

The underlying rule is the one from §2.5: **a long-horizon regression has about T/h independent observations, and no standard-error formula can manufacture information the data do not contain.** A formula can only fail to notice that it is missing.

4.6 Tests, p-values, and confidence intervals

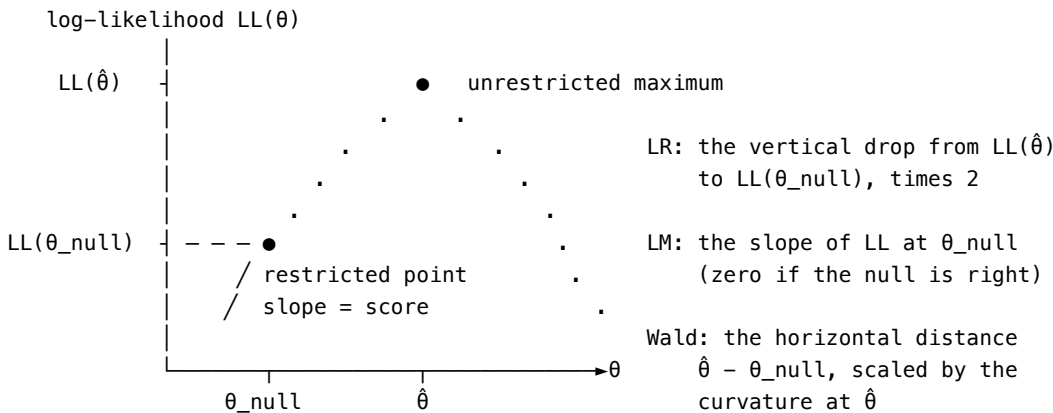
A **hypothesis test** is a rule for rejecting a null hypothesis. Its **size** is the probability of rejecting when the null is true, its **power** the probability of rejecting when a given alternative is true. Every section so far has been about keeping the actual size close to the nominal one.

The **p-value** is the probability, computed as if the null hypothesis were true, of a test statistic at least as extreme as the one observed. Implicit in that computation are the model assumptions and the premise that this was the only test that was going to be run. The American Statistical Association’s statement on p-values (Wasserstein & Lazar, 2016) lists the misreadings worth avoiding:

- A p-value is **not** the probability that the null hypothesis is true.
- It is **not** the probability that the result “is due to chance.”
- It does **not** measure the size or importance of an effect.
- A threshold such as 0.05 does **not** separate real effects from spurious ones.

Confidence intervals invert tests. The 95% interval is the set of null values that a 5% test would not reject (§4.1). Reporting the interval rather than the p-value keeps the magnitude in view.

Three ways to test a restriction. When a model is estimated by maximum likelihood, there are three classical ways to test a restriction such as $\theta = \theta_{\text{null}}$. They measure the same discrepancy along different axes of the log-likelihood curve, written $LL(\theta)$ in the diagram.



- The **Wald** test estimates only the unrestricted model and asks whether $\hat{\theta}$ is far from θ_{null} in standard-error units. Every t -test in a regression table is a Wald test.
- The **likelihood ratio (LR)** test estimates both models and compares their fit.
- The **Lagrange multiplier (LM)** or **score** test estimates only the restricted model and asks whether the log-likelihood is still rising there.

Under the null and a correctly specified likelihood, the three are asymptotically equivalent. In finite samples they differ, and the Wald test has one known weakness: it is not invariant to how the restriction is written, so testing $\beta = 1$ and testing $1/\beta = 1$ can give different answers. The LM form is convenient when the unrestricted model is hard to estimate, which is why many specification tests are LM tests. With one restriction, the F -statistic is the square of the t -statistic.

4.7 Power, and the winner's curse

A test with low power does not just miss real effects. **When it does find one, it exaggerates it.** The estimates that clear a significance bar are disproportionately the ones that overshoot.

Two numbers make power concrete. For a two-sided 5% test to have 80% power, the true effect must be about $1.96 + 0.84 = 2.8$ standard errors. That is the **minimum detectable effect**, and it should be computed *before* an analysis rather than after. The right panel of the figure in §4.8 plots the exaggeration of statistically significant estimates against power. At 80% power, significant estimates overstate the true effect by 12% on average. At 50% power they overstate it by 41%, and at 18% power by a factor of 2.4. At very low power a meaningful share of significant estimates even have the wrong sign: about one in five at 6% power. **[Simulated]** The framework, “Type M” (magnitude) and “Type S” (sign) errors, is from Gelman & Carlin (2014).

This matters because low power is normal. Ioannidis, Stanley & Doucouliagos (2017) surveyed 159 empirical economics literatures containing about 64,000 estimates. They put median statistical power at 18% or less, and estimate that nearly 80% of reported effects are exaggerated, typically by a factor of two. **[Contested]** The magnitudes depend on their meta-analytic estimate of each literature’s true effect, which is itself uncertain, but the direction of the conclusion is not in serious dispute.

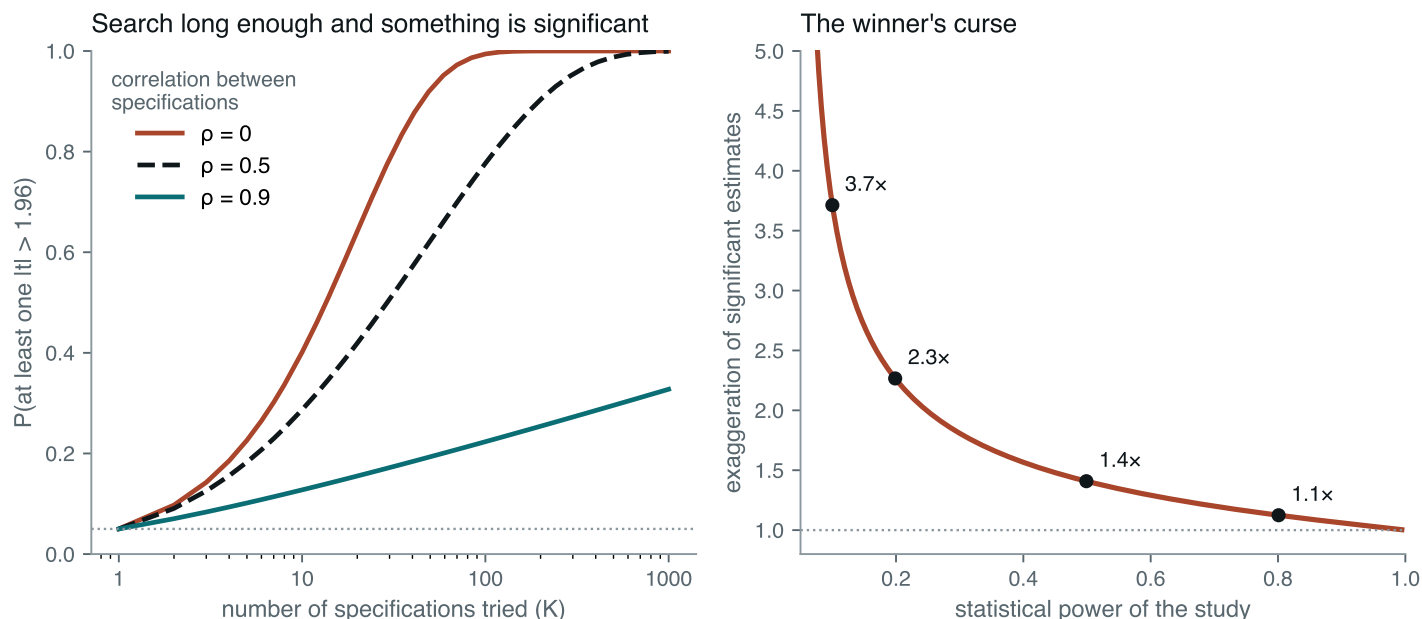
The finance version. Suppose a strategy has a true annual Sharpe ratio of 0.3 and you backtest it over ten years. Its expected t -statistic is $0.3 \times \sqrt{10} = 0.95$, and the power of a 5% test is about 16%. If the backtest nonetheless comes out significant, the Sharpe ratio it reports is about 0.8 on average, roughly 2.6 times the truth. The strategies that look good enough to deploy are systematically the ones whose backtests got lucky, which is one reason live performance disappoints even when nothing is wrong with the strategy.

4.8 Multiple testing and the garden of forking paths

A 5% test rejects a true null 5% of the time. Run 20 independent tests of true nulls and the chance that at least one rejects is $1 - 0.95^{20} = 64\%$.

Real specification searches are not independent. Specifications that share data and differ in a control or a sample period produce correlated test statistics. Correlation slows the problem down but does not stop it. With a correlation of 0.5 between

every pair of specifications, 20 tries find “significance” 42% of the time and 100 tries 78% of the time. With a correlation of 0.9, the figures are 16% and 22%. The expected largest $|t|$ among K independent null statistics is about 1.9 for $K = 10$, 2.8 for $K = 100$, and 3.4 for $K = 1,000$. [Simulated]



Explicit multiplicity has standard corrections. They control different error rates.

Target	What it controls	Standard procedure	When it fits
Family-wise error rate	The probability of <i>any</i> false rejection	Bonferroni (test at $5\%/K$), or Holm's step-down version	A few tests, where any false claim is costly
False discovery rate	The expected <i>share</i> of rejections that are false	Benjamini & Hochberg (1995)	Screening many candidates, where some false positives are tolerable
Best of many strategies	Whether the best performer beats a benchmark by more than luck	White's reality check; Hansen's SPA test; Romano–Wolf stepdown	Backtests and forecast comparisons (§9.6)

For the cross-section of stock returns, Harvey, Liu & Zhu (2016) catalogue several hundred published return predictors and argue that, given the search that produced them, a new one should clear a t -statistic of about 3.0 rather than 2.0.

Implicit multiplicity is the harder problem. Gelman & Loken (2014) call it the **garden of forking paths**. A researcher who never runs more than one regression can still face a multiple-testing problem, because the choices of sample period, outlier rule, control set, horizon, and functional form were made after seeing the data, and would have been made differently had the data been different. A p-value is only valid under the premise that the analysis would have been the same for any data set, and that premise is almost never true.

The remedies share one idea, which is to separate the choices from the data that judge them:

- **Pre-specify** the analysis before seeing the outcome data, formally in a pre-analysis plan or informally in a dated note.
- **Hold out data** that are touched once, at the end.
- **Report every specification tried**, for example as a specification curve (§11.4), so a reader can see the whole distribution rather than the chosen tail.
- **Raise the threshold** in proportion to the search, as with the $t > 3$ hurdle or the deflated Sharpe ratio (Portfolio Construction, Appendix A.46).
- **Label exploratory findings as hypotheses** to be tested on new data.

4.9 The bootstrap

The **bootstrap** approximates an estimator's sampling distribution by treating the sample as if it were the population: draw many samples from the data with replacement, recompute the estimate in each, and use the spread of the recomputed

estimates (Efron, 1979). It replaces analytical derivations with computation, and for complicated estimators it is often the only practical route to a standard error.

The one rule: resample the independent units. The bootstrap is only as good as its assumption about what is independent, which is the same assumption the meat makes.

Dependence	Resample	Name
Independent observations	Individual observations, with Y and X together	Pairs bootstrap
Clusters	Whole clusters	Cluster (block) bootstrap
Independent observations, heteroskedastic errors	Flip residual signs at random, keeping X fixed	Wild bootstrap
Few clusters	Flip the signs of whole clusters' residuals	Wild cluster bootstrap (§4.4)
Time series	Contiguous blocks of observations	Moving block (Künsch, 1989) or stationary bootstrap (Politis & Romano, 1994)

Two refinements matter in practice. Bootstrapping the t -statistic (the **percentile- t** method) is more accurate than bootstrapping the estimate itself, because the t -statistic's distribution depends less on unknown parameters. And for time series, a block has to be longer than the dependence it is meant to preserve. Volatility clustering in daily returns decays over weeks to months, so blocks of a few days destroy it.

When the bootstrap fails. It is not a universal solvent.

- **The resampling scheme destroys the relevant dependence.** An independent bootstrap of autocorrelated or volatility-clustered returns understates uncertainty in the same way classical standard errors do.
- **The statistic is an extreme.** The maximum Sharpe ratio across K strategies is not a smooth function of means, and a naive bootstrap of it is inconsistent. White's reality check exists to handle this case.
- **The parameter is at or near a boundary**, as with unit roots or variances near zero.
- **There are too few independent units:** a handful of clusters or independent periods leaves too little to resample.

4.10 Statistical significance and economic significance

Statistical significance measures how clearly an estimate is distinguished from zero relative to its noise. Economic significance measures whether the effect is large enough to matter. The two are independent. McCloskey & Ziliak (1996) found that 70% of the full-length empirical papers in the *American Economic Review* in the 1980s did not distinguish between them.

The practical consequences:

- **Report magnitudes in meaningful units:** the effect of a one-standard-deviation change, annualised returns, basis points, dollars. Put the confidence interval next to the point estimate.
- **A precisely estimated zero is a finding.** An interval of $[-0.02, 0.01]$ rules out every economically meaningful effect. An interval of $[-3, 5]$ rules out nothing, and "not significant" hides that difference.
- **With enough data everything is significant.** With ten million observations, standard errors are tiny and trivial effects clear any threshold. The question becomes whether the effect is large enough to act on, and whether it survives the biases that no sample size removes (§3.2).

§4 Key takeaways

1. A confidence interval is the set of values the data do not reject. Read it, not the p-value.
2. The classical standard error is the most restrictive special case of the sandwich. It assumes independence and constant variance, and it is correct only when both hold.
3. Heteroskedasticity does not bias coefficients; it corrupts classical standard errors. Use robust standard errors by default.
4. Clustering matters when both the regressor and the error are correlated within groups, and large groups amplify small correlations. With 50 groups of 100 and an error correlation of 0.05, classical standard errors are 2.4 times too small.

5. The effective sample size for inference is the number of clusters, and with a binary treatment the number of treated clusters. With one or two treated clusters, conventional cluster-robust inference fails outright.
6. Newey–West with textbook bandwidths is oversized in realistic samples. Use larger bandwidths with fixed- b critical values, and do not trust any HAC correction to rescue long overlapping horizons.
7. Low power does not just miss effects, it exaggerates the ones it finds. Compute the minimum detectable effect, about 2.8 standard errors, before looking at results.
8. Specification search inflates false positives even when the specifications are correlated and even when no one runs more than one regression. Separate analysis choices from the data that judge them.
9. Bootstrap the independent units. Resampling observations from dependent data reproduces the classical mistake.

5. One estimator family

Textbooks present OLS, instrumental variables, GMM, and maximum likelihood as four separate techniques, each with its own chapter. This section presents them as one technique, the master form of §1.5, with four ways of choosing the moment function. Seeing them this way makes the relationships between them obvious, gives every standard error from one formula, and makes clear which choice actually matters. That choice is the moment condition, not the estimator.

5.1 The method of moments

The oldest estimation idea is also the most general. A model implies that certain population averages take certain values. Replace population averages with sample averages and solve for the parameters.

The simplest case is a mean and a variance. The model says $\mathbb{E}[Y_i - \mu] = 0$ and $\mathbb{E}[(Y_i - \mu)^2 - \sigma^2] = 0$. Setting the sample versions to zero gives $\hat{\mu} = \bar{Y}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_i (Y_i - \bar{Y})^2$.

OLS is the same move applied to the projection's first-order condition $\mathbb{E}[X_i(Y_i - X_i'\beta)] = 0$. When there are as many moment conditions as parameters, the model is **just-identified**: the sample equations can be solved exactly, and the asymptotic variance is the sandwich $Q^{-1}\Omega Q^{-1'}$ from §1.6.

5.2 Instrumental variables as a moment condition

Suppose the structural equation is $Y_i = X_i'\beta + u_i$ and some regressors are correlated with u_i , so that OLS converges to the wrong thing (§3.2). Suppose also that a vector of **instruments** Z_i satisfies two conditions.

- **Relevance:** $\mathbb{E}[Z_i X_i']$ has full column rank. The instruments move the endogenous regressors.
- **Exogeneity:** $\mathbb{E}[Z_i u_i] = 0$. The instruments are uncorrelated with everything else that affects the outcome. This includes the **exclusion restriction**, the requirement that Z affects Y only through X .

The second condition is a moment condition, $\mathbb{E}[Z_i(Y_i - X_i'\beta)] = 0$. With as many instruments as regressors, its sample analogue solves to

$$\hat{\beta}_{IV} = (Z'X)^{-1}Z'Y.$$

With one regressor, one instrument, and an intercept, this is a ratio of two regression slopes:

$$\hat{\beta}_{IV} = \frac{\widehat{\text{Cov}}(Z, Y)}{\widehat{\text{Cov}}(Z, X)} = \frac{\widehat{\text{Cov}}(Z, Y)/\widehat{\text{Var}}(Z)}{\widehat{\text{Cov}}(Z, X)/\widehat{\text{Var}}(Z)} = \frac{\text{reduced form: effect of } Z \text{ on } Y}{\text{first stage: effect of } Z \text{ on } X}.$$

The intuition in one sentence: IV uses only the part of the variation in X that is driven by Z , and converts the effect of Z on Y into an effect of X on Y by dividing by how much Z moved X .

When Z is binary this is the **Wald estimator**, the difference in average outcomes between the $Z = 1$ and $Z = 0$ groups divided by the difference in their average treatment. The canonical example is Angrist (1990). The Vietnam-era draft lottery assigned men eligibility for conscription at random by birth date. Eligibility raised the probability of military service, and eligible men later earned less. The ratio of the earnings gap to the service gap estimates the effect of service itself: in the early 1980s, white veterans earned roughly 15% less than comparable non-veterans. Nobody chose their lottery number, which is the whole argument for exogeneity, and the lottery affected earnings, it is assumed, only by changing who served.

Two-stage least squares. With more instruments than endogenous regressors, the instruments are combined optimally under homoskedasticity by projecting X on Z and using the fitted values:

$$\hat{\beta}_{2SLS} = (X'P_Z X)^{-1}X'P_Z Y, \quad P_Z = Z(Z'Z)^{-1}Z'.$$

The name describes a computation, and the computation hides a trap. **[Practice]** Never run the second stage by hand as an OLS regression of Y on $\hat{X}$ and report its standard errors. The point estimate is right, but the residuals are computed using $\hat{X}$ instead of X , so the standard errors are wrong. Use a routine that computes 2SLS as one estimator.

IV is consistent but biased in finite samples, with the bias pointing toward OLS and growing as the first stage weakens. §7.3 covers what happens when instruments are weak or invalid, which is where most of the practical difficulty lies.

5.3 GMM: more moments than parameters

With ℓ moment conditions and $k < \ell$ parameters, the model is **over-identified**. The sample moments cannot all be set to zero at once, so the **generalised method of moments** (GMM) makes them jointly as small as possible in a weighted sense (Hansen, 1982):

$$\hat{\theta}_{\text{GMM}} = \arg \min_{\theta} \bar{\psi}(\theta)' W \bar{\psi}(\theta), \quad \bar{\psi}(\theta) = \frac{1}{n} \sum_{i=1}^n \psi_i(\theta).$$

Any positive definite weighting matrix W gives a consistent estimator. The general asymptotic variance is

$$\text{Avar}(\sqrt{n}(\hat{\theta} - \theta_0)) = (Q' W Q)^{-1} Q' W \Omega W Q (Q' W Q)^{-1},$$

the same sandwich with the weighting matrix folded into the bread. It is smallest when $W = \Omega^{-1}$, which gives $(Q' \Omega^{-1} Q)^{-1}$. **The efficient weighting matrix puts more weight on the moments that are estimated more precisely.** In practice this takes two steps: estimate with a simple W , use the residuals to estimate Ω , and re-estimate with $\hat{\Omega}^{-1}$. 2SLS is GMM with $W = (\mathbf{Z}'\mathbf{Z}/n)^{-1}$, which is the efficient choice when the errors are homoskedastic.

The J -test. Over-identification buys a test. If all ℓ moment conditions are valid, the minimised objective, scaled by n and computed with the efficient weighting matrix,

$$J = n \bar{\psi}(\hat{\theta})' \hat{\Omega}^{-1} \bar{\psi}(\hat{\theta}) \xrightarrow{d} \chi_{\ell-k}^2,$$

is chi-squared with $\ell - k$ degrees of freedom. A large J says the moment conditions cannot all hold at once. Be precise about what that does and does not establish.

- **It can detect instruments that disagree with each other.** If one instrument implies $\beta = 0.1$ and another implies $\beta = 0.5$, they cannot both be valid.
- **It cannot detect instruments that are wrong in the same way.** If every instrument is correlated with the error in the same direction, they agree on the same wrong answer and J is small.
- **In a just-identified model it is identically zero.** There is nothing to test, which is §1.5's point that the orthogonality condition is untestable when it is exactly identifying.

A passed J -test is weak evidence of validity. A failed one is strong evidence of a problem.

GMM in asset pricing. GMM was built for moment conditions that come from economic theory, and asset pricing is its native habitat. An investor's first-order condition says that for any gross return R_{t+1} ,

$$\mathbb{E}[M_{t+1}(\theta) R_{t+1} - 1 \mid \mathcal{F}_t] = 0, \quad M_{t+1}(\theta) = d \left(\frac{C_{t+1}}{C_t} \right)^{-\eta},$$

where the **stochastic discount factor** M_{t+1} in the power-utility example is built from consumption growth, a subjective discount factor d , and relative risk aversion η . Because the condition holds conditional on information at t , any variable Z_t known at t gives an unconditional moment condition $\mathbb{E}[Z_t(M_{t+1}(\theta)R_{t+1} - 1)] = 0$. Hansen & Singleton (1982) estimated d and η this way. Almost all of empirical asset pricing can be written in this form, which is the organising theme of Cochrane's *Asset Pricing* (2005).

Two practical warnings.

- **[Fact] Efficient GMM with many moments behaves badly in finite samples.** The estimated weighting matrix is noisy, and its inverse overweights moments that happen to look precise in the sample. Altonji & Segal (1996) found large small-sample biases in efficient GMM for covariance structures, with equally weighted moments performing better. Iterated and continuously updated GMM help somewhat (Hansen, Heaton & Yaron, 1996).
- **[Practice] Statistical efficiency is not economic relevance.** The efficient weighting matrix may concentrate on portfolio combinations that are estimated precisely but are economically uninteresting, such as extreme long-short positions. Asset-pricing tests often deliberately use a pre-specified weighting matrix, such as equal weights on test portfolios, so that the model is judged on the returns an investor actually cares about (Hansen & Jagannathan, 1997).

5.4 Maximum likelihood, and quasi-likelihood

If the model specifies the full conditional distribution of the data, $f(Y_i | X_i; \theta)$, the **maximum likelihood estimator** (MLE) chooses the parameter under which the observed data were most probable:

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta} \sum_{i=1}^n \log f(Y_i | X_i; \theta).$$

Its first-order condition sets the sample average of the **score** $s_i(\theta) = \partial \log f(Y_i | X_i; \theta) / \partial \theta$ to zero, so MLE is a method-of-moments estimator with $\psi_i = s_i$. The population moment condition $\mathbb{E}[s_i(\theta_0)] = 0$ holds whenever the model is correct, because differentiating $\int f(y | x; \theta) dy = 1$ with respect to θ gives $\int \partial f / \partial \theta dy = 0$, which is the expected score.

Why MLE is efficient, and when that stops being true. Under correct specification the **information matrix equality** holds: the variance of the score equals minus the expected Hessian of the log-likelihood,

$$\mathbb{E}[s_i(\theta_0) s_i(\theta_0)'] = -\mathbb{E}\left[\frac{\partial s_i(\theta_0)}{\partial \theta'}\right] \equiv \mathcal{I}(\theta_0).$$

In sandwich terms, the meat equals minus the bread, and $Q^{-1} \Omega Q^{-1'}$ collapses to $\mathcal{I}(\theta_0)^{-1}$, the inverse **Fisher information**. That is the Cramér–Rao lower bound, so no regular consistent estimator is more precise. For the linear model with normal errors, the MLE of β is exactly OLS.

The equality is a consequence of the model being right. When the density is misspecified, the MLE converges to a **pseudo-true** value, the parameter whose implied distribution is closest to the truth in Kullback–Leibler divergence, and the two sides of the equality no longer match (White, 1982). The estimator is then a **quasi-maximum likelihood estimator** (QMLE), and its standard errors must use the full sandwich: the Hessian for the bread, the outer product of the scores for the meat.

Some quasi-likelihoods are robust, and some are not. This is the practical content of QMLE theory.

- **Robust.** A Gaussian likelihood for a GARCH model gives consistent estimates of the mean and variance parameters even when returns are fat-tailed, provided the conditional mean and variance equations are right (Bollerslev & Wooldridge, 1992). The Poisson likelihood gives consistent estimates of an exponential conditional mean for any non-negative outcome, count or not, and handles zeros that a log-linear regression cannot (Santos Silva & Tenreiro, 2006).
- **Not robust.** Probit and logit coefficients are inconsistent when the latent error is heteroskedastic or has the wrong distribution, although average marginal effects are often approximately right. Censored-regression (Tobit) estimates depend heavily on normality.

When to reach for likelihood. Use it when the distribution itself is the object of interest, as with volatility, tail risk, durations, and default probabilities; when latent states must be filtered, as in the regime-switching models of Market Regimes and Machine Learning; or when efficiency in a small sample is worth the risk of misspecification.

5.5 M-estimators beyond the mean

Everything so far targets conditional *means*. The same machinery handles any estimator defined by minimising a sum, known collectively as **M-estimators**.

Quantile regression (Koenker & Bassett, 1978) estimates how the conditional q -quantile of Y depends on X , by minimising $\sum_i \rho_q(Y_i - X_i' \beta)$ with the asymmetric “check” loss $\rho_q(v) = v(q - \mathbb{1}\{v < 0\})$. The median regression $q = 0.5$ is least absolute deviations. It is the natural tool for questions about tails: the conditional 5% quantile of a portfolio’s return is a value-at-risk model (Engle & Manganelli, 2004). It is also robust to outliers in Y , though not in X . One interpretive caution: a quantile coefficient describes how a quantile of the *distribution* shifts. It is not the effect on the particular unit that happened to sit at that quantile, unless units keep their ranks.

Robust regression (Huber, 1964) replaces squared loss with a loss that grows only linearly for large residuals, so a few extreme observations cannot dominate the fit. It changes the estimand as well as the estimator. In finance the extreme observations are often the economically important ones, so “robust to the crash” is a modelling decision rather than a technical fix.

5.6 Shrinkage and the Bayesian reading

Every estimator so far is unbiased or consistent. §2.3 noted that a biased estimator can have a lower mean squared error. Shrinkage estimators exploit that trade deliberately.

Ridge regression adds a penalty on the size of the coefficients:

$$\hat{\beta}_{\text{ridge}} = \arg \min_b \sum_{i=1}^n (Y_i - X_i' b)^2 + \kappa \|b\|^2 = (\mathbf{X}'\mathbf{X} + \kappa \mathbf{I})^{-1} \mathbf{X}'\mathbf{Y}.$$

The penalty pulls the coefficients toward zero, adds bias, and reduces variance. It helps most when there are many regressors, each with a small effect, and the regressors are correlated. Replacing $\|b\|^2$ with $\sum_j |b_j|$ gives the **LASSO**, which also sets some coefficients exactly to zero (§10.4).

The Bayesian reading. With normal errors of variance σ^2 and a prior belief that each coefficient is normal around zero with variance τ^2 , the posterior mean of β is exactly the ridge estimator with $\kappa = \sigma^2/\tau^2$. The LASSO is the posterior mode under a Laplace (double-exponential) prior. **A penalty is a prior, and a prior is a penalty.** Choosing κ by cross-validation is choosing how strongly to believe that effects are small.

That shrinkage can dominate the obvious estimator is not a heuristic. **[Fact]** When estimating three or more means at once, shrinking all the sample means toward a common value gives a lower total mean squared error than the sample means themselves, whatever the true means are. This is Stein's paradox, explained accessibly by Efron & Morris (1977). Unbiasedness, in that setting, is not a virtue.

Shrinkage earns its keep in econometrics in three recurring places.

- **Many related effects.** Estimates of skill for thousands of fund managers, or of accuracy for thousands of analysts, should be shrunk toward the group average in proportion to their noise. The raw extremes are mostly luck (empirical Bayes).
- **Forecasting with many parameters.** Vector autoregressions with a “Minnesota” prior that pulls each variable toward a random walk forecast macroeconomic series far better than unrestricted ones (Litterman, 1986).
- **Covariance matrices.** The shrinkage estimators of Portfolio Construction are the same idea applied to Σ .

When the data are informative, Bayesian and frequentist answers converge: in regular models with large samples the posterior is approximately normal around the MLE with the sandwich-free variance $\mathcal{J}^{-1}/n$. They diverge exactly where the data are weak, with small samples, weak identification, or many parameters. That is where the prior is doing the work, and where it should be reported and defended like any other assumption.

5.7 Choosing among them

Estimator	Key assumption	Robust to	Typical use	Typical failure
OLS	Regressors uncorrelated with the error	Non-normality; heteroskedasticity and dependence, with the right meat	Linear conditional means	Endogeneity
IV / 2SLS	Valid, relevant instruments	Endogeneity of the regressor	Causal effects from natural experiments	Weak or invalid instruments (§7.3)
Efficient GMM	Valid moment conditions	The distributional form	Many theory-based moments	Finite-sample bias with many moments
Maximum likelihood	The full distribution is right	Nothing beyond the model	Volatility, discrete outcomes, latent states	Inconsistency under misspecification
Quasi-ML	Only the parts of the model that matter	A wrong distribution, in robust cases	GARCH; count-type outcomes	Invalid standard errors if the sandwich is skipped
Quantile regression	A linear conditional quantile	Outliers in Y	Tails, value-at-risk, heterogeneity	Misreading distributional shifts as individual effects

Estimator	Key assumption	Robust to	Typical use	Typical failure
Ridge, LASSO, Bayes	The prior or penalty is sensible	Overfitting	Prediction with many regressors	Biased coefficients; invalid naive inference (§10.3)

The table should not obscure the ordering from §1.3. **[Practice]** Among the three things you choose, the orthogonality condition, the meat, and the estimator, the estimator is usually the least consequential. A well-identified parameter estimated by plain OLS with the right standard errors beats a badly identified one estimated by an elegant method. The exact algebraic relationships among the estimators in this section are collected in §12.2.

§5 Key takeaways

1. OLS, IV, GMM, and maximum likelihood are one estimator: each sets a sample average of a moment function to zero, and each gets its standard error from the same sandwich.
2. IV is a ratio: the effect of the instrument on the outcome divided by its effect on the regressor. It uses only the variation in the regressor that the instrument drives.
3. Never compute 2SLS standard errors by running the second stage by hand.
4. Over-identification buys a J -test, which can catch instruments that disagree and cannot catch instruments that are wrong in the same way.
5. Efficient GMM is fragile in finite samples with many moments; pre-specified weights are often more reliable and more economically meaningful.
6. MLE is efficient because, under correct specification, the meat equals minus the bread. Under misspecification, use the sandwich and know which quasi-likelihoods remain consistent.
7. A penalty is a prior. Shrinkage trades bias for variance, and when many related quantities are estimated at once it provably beats the unbiased estimates.
8. The estimator is the least consequential of the three choices. Get the orthogonality condition and the meat right first.

Part III — Identification

6. Endogeneity: why a coefficient is not an effect

§3.2 reduced the central problem of the field to one line: a regression estimates β_s plus a bias term that is zero only when the regressors are uncorrelated with the structural error. This section is the anatomy of that bias term: what it means in causal language, the five mechanisms that produce it, and the one idea (randomisation) that removes it by construction. §7 then surveys the designs that try to approximate randomisation when nobody randomised.

6.1 Potential outcomes

The cleanest language for causal questions is the **potential outcomes** framework, due to Neyman and developed by Rubin (1974). For each unit i and a binary treatment D_i , imagine two outcomes: $Y_i(1)$, what would happen if the unit were treated, and $Y_i(0)$, what would happen if it were not. The unit's causal effect is the difference, $\tau_i = Y_i(1) - Y_i(0)$. The observed outcome is whichever potential outcome the treatment selected:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

The fundamental problem of causal inference is that τ_i is never observed, because no unit is both treated and untreated at the same moment (Holland, 1986). Every causal estimate is therefore a statement about *averages*, and the average has to be chosen.

Estimand	Definition	Answers
Average treatment effect (ATE)	$\mathbb{E}[\tau_i]$	What if everyone were treated, versus no one?
Average effect on the treated (ATT)	$\mathbb{E}[\tau_i \mid D_i = 1]$	What did the treatment do for those who got it?
Average effect on the untreated (ATU)	$\mathbb{E}[\tau_i \mid D_i = 0]$	What would extending it to the rest do?
Conditional average effect (CATE)	$\mathbb{E}[\tau_i \mid X_i = x]$	How does the effect vary with characteristics?
Local average effect (LATE)	The effect for units whose treatment an instrument changes	What does IV actually estimate? (§7.3)

When effects are the same for everyone, these coincide. When they differ, which is the normal case, they differ, and a method that is unbiased for one is generally biased for another. The estimand has to be stated before the design is chosen (§1.3).

Take index inclusion as an example. For a stock added to the S&P 500, $Y_i(1)$ is its return over the following month and $Y_i(0)$ is the return it would have had if it had not been added. The ATT is the average inclusion effect on stocks that were added. It is the natural estimand, and it is not the effect that adding a randomly chosen stock would have.

SUTVA. The framework assumes the **stable unit treatment value assumption**: a unit's potential outcomes do not depend on other units' treatments, and there is only one version of the treatment. The first half fails constantly in finance. The return to a trading strategy depends on how many others trade it. Adding one stock to an index moves the prices of its close substitutes. A regulation that constrains some banks pushes activity toward the others, which contaminates exactly the control group used to measure its effect. When interference is present, the "effect" a design estimates is a contrast between treated and control units *in the presence of spillovers*, which may be very different from the effect of treating everyone.

6.2 The selection bias decomposition

What does the simple comparison of treated and untreated units estimate? Add and subtract the treated units' untreated potential outcome:

$$\underbrace{\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]}_{\text{observed difference}} = \underbrace{\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0]}_{\text{selection bias}}.$$

The selection-bias term compares the *untreated* outcomes of the two groups. It asks whether the units that got treated would have done differently from the others even without treatment. It is the potential-outcomes version of the omitted-variable term of §3.5.

Back to share repurchases. If managers buy back stock when they believe it is undervalued, then repurchasing firms would have earned higher returns than other firms *even had they not repurchased*, so $\mathbb{E}[Y_i(0) \mid D_i = 1] > \mathbb{E}[Y_i(0) \mid D_i = 0]$. The selection bias is positive, and the naive return difference overstates whatever the buyback itself contributes. Selection can also run the other way. People who go to hospital are less healthy afterwards than people who do not, because the untreated health of those who seek treatment is worse, which can overwhelm any benefit of the treatment in a naive comparison.

6.3 Why randomisation works

If treatment is assigned at random, it is independent of the potential outcomes: $D_i \perp (Y_i(1), Y_i(0))$. Then the untreated potential outcome has the same distribution in both groups, the selection-bias term is exactly zero, and the simple difference in means estimates the ATT, which equals the ATE. In regression language, a randomly assigned regressor is uncorrelated with the structural error by construction, so the endogeneity term of §3.2 vanishes.

Randomisation balances *unobserved* characteristics as well as observed ones, in expectation. No other device does. That is why the randomised experiment is the reference point for every design in §7, each of which is an argument that some feature of the world assigned treatment “as good as randomly” for some units. **[Practice]** The most useful question to ask about any observational study is: *what experiment is this trying to approximate, and where does the approximation break?*

Finance has a few genuine experiments. The best known is the SEC’s **Regulation SHO pilot**. The SEC sorted the stocks in the 2004 Russell 3000 by listing market and trading volume and designated every third one a pilot stock, exempt from short-sale price tests from May 2005 to August 2007 (SEC, 2007). Dozens of papers have used it to estimate the effects of short-selling constraints on outcomes from liquidity to earnings management (Fang, Huang & Karpoff, 2016). Its heavy reuse became a lesson of its own: Heath, Ringgenberg, Samadi & Werner (2023) show that when many outcomes are tested against the same experiment, many results that were significant in isolation do not survive a correction for multiple testing (§4.8). A clean experiment does not protect against the garden of forking paths.

6.4 Five routes to a correlated error

Every failure of exogeneity comes from one of five mechanisms. Naming the mechanism tells you the likely direction of the bias and what could fix it.

1. Omitted variables (confounding). Something that affects the outcome also affects the regressor. The bias is the OVB formula of §3.5. For example, a regression of fund returns on fund size that omits manager skill is biased if skill attracts inflows (raising size) and also raises returns.

2. Measurement error. The regressor is measured with noise, $X_i = X_i^* + v_i$, where v_i is independent of everything else. Then the error in the regression on the observed X_i includes $-\beta v_i$, which is correlated with X_i , and

$$\text{plim } \hat{\beta} = \beta \lambda, \quad \lambda = \frac{\text{Var}(X^*)}{\text{Var}(X^*) + \text{Var}(v)}.$$

The **reliability ratio** λ is between 0 and 1, so classical measurement error **attenuates** the coefficient toward zero. A reliability of 0.5 halves it. Measurement error in the *outcome* adds noise without bias. With several regressors, measurement error in one biases the others too, in directions that are hard to sign.

The consequence people most often miss concerns panels. **Fixed effects and differencing remove signal and keep noise.** Suppose the true regressor is persistent, with a correlation of 0.9 between adjacent periods, and the measurement error is independent over time with a levels reliability of 0.8. First-differencing cuts the variance of the true regressor to $2(1 - 0.9) = 0.2$ of its level, but doubles the noise variance to $2 \times 0.25 = 0.5$ in the same units. The reliability of the differenced regressor is $0.2 / (0.2 + 0.5) = 0.29$, down from 0.8 (Griliches & Hausman, 1986). A “within-firm” estimate that is much smaller than the pooled one may reflect attenuation rather than confounding in the pooled regression.

Finance is full of regressors estimated with error: betas from a first-pass time-series regression (§9.4), expected returns proxied by characteristics, expectations proxied by analyst forecasts, liquidity proxied by spreads.

3. Simultaneity. The outcome also affects the regressor, because both are determined together. The founding example is supply and demand. Quantity and price are set jointly by two equations, and a regression of quantity on price traces out a mixture of the two curves, weighted by which curve’s shocks happen to be larger. If demand shocks dominate, the regression

recovers something close to the *supply* curve (Working, 1927). Instrumental variables were invented to solve exactly this problem (§7.3; Stock & Trebbi, 2003). Finance equivalents include order flow and returns (informed traders buy because they expect returns, and returns attract momentum traders), fund flows and performance, and trading volume and volatility.

4. Sample selection. The outcome is observed only for a subsample chosen in a way related to the outcome. Wages are observed only for people who work, and fund returns only for funds that survive. Heckman (1979) showed that selection is an omitted variable: within the selected sample, the expected error depends on the regressors through the probability of selection. Correcting it credibly requires a variable that affects selection but not the outcome, which is an exclusion restriction by another name.

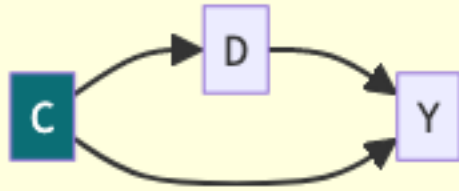
The finance case is **survivorship bias**. Brown, Goetzmann, Ibbotson & Ross (1992) showed that because poorly performing funds disappear from the data, a sample truncated by survival can display apparent persistence in performance even when none exists. **[Fact]** Its sibling is look-ahead bias in sample construction, such as backtesting on today's index constituents, which guarantees that every stock in the sample survived to today.

5. Bad controls. Conditioning on the wrong variable *creates* a correlation between the regressor and the error where none existed. This mechanism is the least intuitive of the five, because it is caused by an action normally taken to reduce bias. It gets its own subsection.

6.5 Causal graphs: confounders, mediators, colliders

A **directed acyclic graph** (DAG) draws variables as nodes and direct causal effects as arrows (Pearl, 2009). Its value for applied work is one question it answers mechanically: *which variables should I control for?* Three elementary structures cover almost every case.

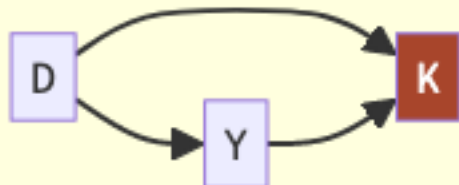
Confounder: control for it



Mediator: do not control for it when you want the total effect



Collider: never control for it

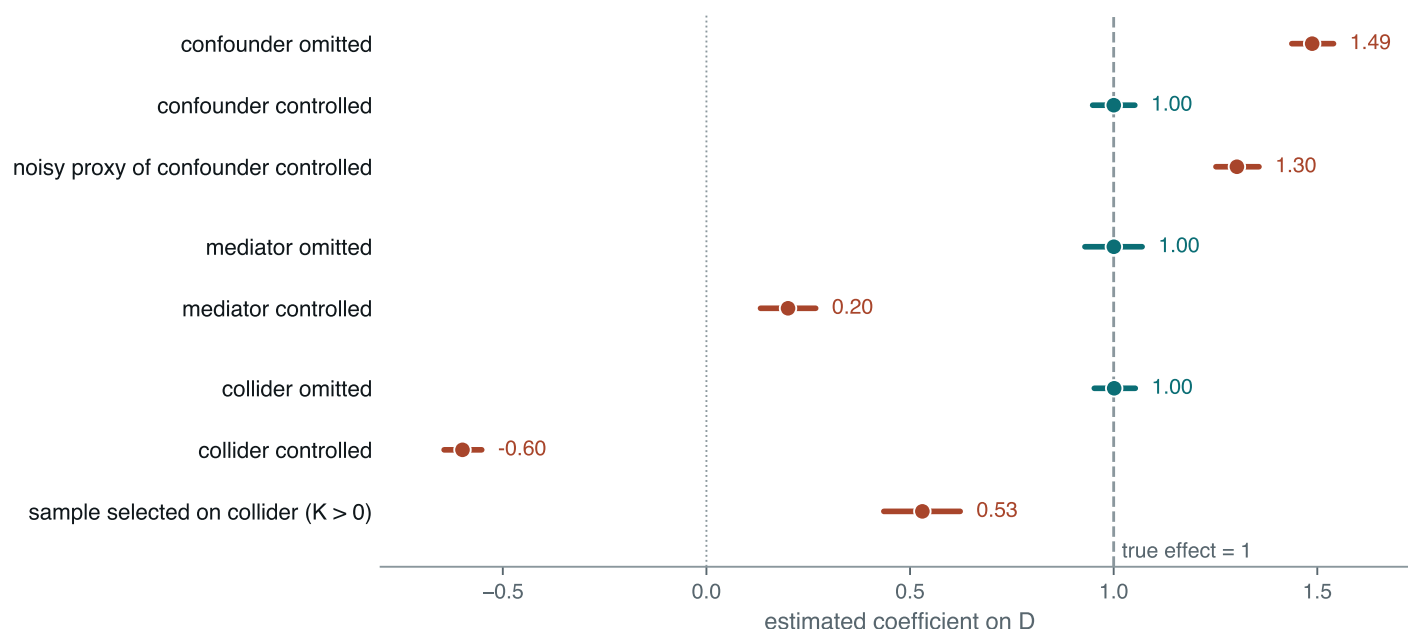


- A **confounder** ($C \rightarrow D$ and $C \rightarrow Y$) opens a non-causal “back-door” path from D to Y . Conditioning on C blocks it.
- A **mediator** ($D \rightarrow M \rightarrow Y$) carries part of the causal effect. Conditioning on M removes that part, leaving only the direct effect.
- A **collider** ($D \rightarrow K \leftarrow Y$) blocks the path between its causes *by default*. Conditioning on K opens it, manufacturing an association between D and Y that has nothing to do with the effect of D .

The formal rule is the **back-door criterion**. A set of controls identifies the effect of D on Y if it blocks every path from D to Y that begins with an arrow *into* D , and contains no variable caused by D .

The simulation below makes the three cases concrete. In every scenario the true effect of D on Y is 1. Omitting a confounder gives 1.49, and controlling for it gives 1.00, but controlling for a noisy proxy of it (reliability 0.5) gives 1.30, removing less than 40% of the bias. Omitting a mediator correctly gives the total effect of 1.00, and controlling for it gives only the direct part, 0.20. Omitting a collider correctly gives 1.00. Controlling for it gives -0.60 , flipping the sign, and merely restricting the sample to units with a positive value of the collider gives 0.53. **[Simulated]**

Controlling for a confounder removes bias; controlling for a mediator or collider creates it



The collider mechanism is easy to see with an example. Suppose that among all managers, skill and personal connections are independent, and a firm hires anyone whose skill *plus* connections clears a bar. Among hired managers, those with weak connections must have strong skill to have been hired, so skill and connections are negatively correlated in the hired sample, although they are unrelated in the population. Conditioning on having been hired, the collider, created the correlation.

Finance supplies colliders in quantity:

- **Survival and database inclusion.** A firm or fund stays in the data if its performance and its characteristics jointly keep it alive. Any study restricted to survivors conditions on a collider.
- **Post-treatment controls.** If analyst coverage raises liquidity and liquidity affects returns, “controlling for liquidity” when estimating the effect of coverage on returns removes a mediated channel. If liquidity is also affected by returns, it is a collider as well.
- **Selecting on outcomes.** Studying “stocks that hit a 52-week high” or “funds that raised a second fund” conditions on a variable that depends on past returns.

Two further lessons. First, “**only control for pre-treatment variables**” is necessary but not sufficient. A pre-treatment variable can still be a collider on a path between unobserved causes of treatment and outcome, although such cases usually produce small biases (Cinelli, Forney & Pearl, 2024). Second, graphs and potential outcomes are complements rather than rivals. Graphs are best at deciding what to control for, and potential outcomes are best at defining estimands, handling heterogeneous effects, and connecting to design-based inference (Imbens, 2020).

6.6 Selection on observables: regression, matching, and weighting

When no natural experiment is available, the fallback is to assume that the observed covariates capture everything that jointly drives treatment and outcomes. Formally, two conditions:

- **Unconfoundedness** (conditional independence): $(Y_i(1), Y_i(0)) \perp D_i \mid X_i$. Among units with the same covariates, treatment is as good as random.
- **Overlap:** $0 < p(X_i) < 1$, where $p(x) = \Pr(D_i = 1 \mid X_i = x)$ is the **propensity score**. Every kind of unit has some chance of being in either group.

Under both, the ATE is identified by comparing treated and untreated units with the same covariates and averaging: $ATE = \mathbb{E}[\mathbb{E}[Y_i \mid D_i = 1, X_i] - \mathbb{E}[Y_i \mid D_i = 0, X_i]]$. Rosenbaum & Rubin (1983) showed that if unconfoundedness holds given X , it holds given the scalar $p(X)$ alone, which turns matching on many covariates into matching on one number.

There are four families of estimator:

Estimator	What it models	Idea
Regression adjustment	The outcome, $\mathbb{E}[Y \mid D, X]$	Predict both potential outcomes for everyone and average the difference
Matching	Neither, directly	Compare each treated unit with untreated units with similar X or similar $p(X)$
Inverse probability weighting	The treatment, $p(X)$	Reweight so that both groups resemble the full population: $\mathbb{E}\left[\frac{D_i Y_i}{p(X_i)} - \frac{(1-D_i)Y_i}{1-p(X_i)}\right]$
Doubly robust (augmented IPW)	Both	Consistent if <i>either</i> the outcome model or the propensity model is right; the basis of §10.5

The point that matters most: none of these estimators relaxes unconfoundedness. They differ only in how they use X , and the choice among them is second-order compared with whether X contains the confounders. **[Practice]** Matching is not a research design. It is a way of doing covariate adjustment, and it inherits the adjustment's assumption.

Overlap is the check you can actually run. Plot the estimated propensity score separately for treated and untreated units. Where the distributions do not overlap, the effect is being extrapolated from functional form, and inverse probability weights explode as $p(X)$ approaches 0 or 1. Trimming the sample to the region of common support changes the estimand, but honestly.

The LaLonde question, forty years on. Dehejia & Wahba (1999) revisited LaLonde's data and found that propensity score methods, on a subsample with two years of pre-programme earnings, came close to the experimental benchmark. Smith & Todd (2005) showed that the result was sensitive to the sample and the specification. **[Contested]** The most recent reassessment, by Imbens & Xu (2025), finds that modern methods give robust estimates once covariate overlap is ensured, but that robustness is not credibility: whether the estimates can be read causally has to be judged by validation exercises, such as placebo tests, not by goodness of fit. My reading is that modern methods cannot recover the benchmark merely by being modern. The ingredient that separates good estimates from bad is the validation, not the estimator.

6.7 Sensitivity analysis: how much confounding would it take?

Unconfoundedness cannot be tested, because the confounders it worries about are unobserved. What can be done is to ask how strong an unobserved confounder would have to be to overturn the result, and then argue about whether one that strong is plausible.

- **Selection on observables as a guide.** Altonji, Elder & Taber (2005) proposed using the degree of selection on observed characteristics to bound selection on unobserved ones.
- **Coefficient stability, done properly.** Oster (2019) formalised the common practice of watching how a coefficient moves when controls are added. Movement is only informative relative to how much explanatory power the controls add, so the method uses the change in both the coefficient and the R^2 . It reports δ , the ratio of selection on unobservables to selection on observables that would drive the effect to zero, given an assumed maximum attainable R^2 . Her suggested defaults are a maximum R^2 of 1.3 times the R^2 with controls, and $|\delta| > 1$ as the robustness threshold. **[Contested]** Both the default and the premise that observed controls are representative of unobserved ones are debated.
- **Partial R^2 benchmarks.** Cinelli & Hazlett (2020) extend the OVB formula to report a **robustness value**: the minimum share of the residual variance of both the treatment and the outcome that a confounder would have to explain to eliminate the estimate. The number can be benchmarked against observed covariates, as in "a confounder would need to be three times as strongly related to treatment as firm size is."

[Practice] Any estimate that rests on selection on observables should be reported with a sensitivity statistic. An effect that a confounder half as strong as firm size would erase is a hypothesis. One that would need a confounder ten times as strong is evidence.

§6 Key takeaways

1. Causal effects are defined by potential outcomes, and no unit's effect is ever observed. Choose the average (ATE, ATT, LATE, CATE) before choosing the method.

2. A naive comparison equals the effect on the treated plus selection bias: the difference in what the two groups would have experienced without treatment.
3. Randomisation removes selection bias by construction, including bias from unobserved characteristics. Every observational design is an argument that something mimicked it.
4. Endogeneity has five sources: omitted variables, measurement error, simultaneity, sample selection, and bad controls. Naming the source tells you the direction of the bias.
5. Measurement error attenuates, and fixed effects make it worse: differencing a persistent regressor can cut its reliability from 0.8 to about 0.3.
6. Control for confounders; do not control for mediators when you want the total effect; never control for colliders. Controlling for a collider can reverse the sign of an estimate.
7. Survivorship, database inclusion, and selection on outcomes are colliders, and they are everywhere in financial data.
8. Regression, matching, weighting, and doubly robust estimators all rest on the same untestable assumption. What you can check is overlap, and what you should report is a sensitivity statistic.

7. Research designs

7.1 How to read this section

A **research design** is an argument that some source of variation in the regressor is as good as random, together with an estimator that uses only that variation. Each design in this section buys a particular orthogonality condition, the master form of §1.5, at a particular price.

Design	The orthogonality condition it buys	The price
Randomised experiment (§7.2)	Treatment is independent of potential outcomes, by construction	Cost, feasibility, and external validity
Instrumental variables (§7.3)	An instrument is uncorrelated with the structural error	An exclusion restriction, and a local estimand
Panel fixed effects (§7.4)	Within-unit changes in the regressor are uncorrelated with within-unit changes in the error	Strict exogeneity; only time-invariant confounders are removed
Difference-in-differences (§7.5)	Treatment timing is unrelated to untreated trends	Parallel trends
Regression discontinuity (§7.6)	Crossing a cutoff is unrelated to the error near the cutoff	An estimand that exists only at the cutoff
Synthetic control (§7.7)	A weighted donor pool tracks the treated unit's untreated path	Long pre-periods, and good pre-treatment fit
Structural estimation (§7.8)	Whatever the economic model implies	The model has to be right

Selection on observables, the fallback design, was covered in §6.6. Every design below is described with the same fields, so that failure modes can be compared directly across designs: **intuition**, **estimand**, **identifying assumption**, **estimation**, **inference**, **diagnostics**, **failure modes**, **in finance**, and **when preferred**. The failure modes carry the most practical value and get the most room.

7.2 Randomised experiments

Intuition. Assign treatment by coin flip, so that nothing about a unit can influence whether it is treated, and compare average outcomes.

Estimand. The ATE, which equals the ATT. If some units do not comply with their assignment, the comparison by *assignment* estimates the **intention-to-treat** effect, the effect of being offered treatment. The effect of treatment itself on those who comply is then an instrumental-variables problem, with assignment as the instrument (§7.3).

Identifying assumption. Random assignment, SUTVA (§6.1), and no attrition that differs by treatment arm.

Estimation. The difference in means. Adding pre-treatment covariates in a regression does not affect unbiasedness but can sharpen precision considerably. Interacting the treatment indicator with demeaned covariates guarantees that adjustment does not hurt precision in large samples (Lin, 2013). Covariates measured *after* treatment must never be added (§6.5).

Inference. Robust standard errors, clustered at the unit of randomisation. In small experiments, **randomisation inference** is exact: recompute the statistic under every reassignment that could have happened, and see where the actual one falls.

Diagnostics. A balance table of pre-treatment covariates across arms tests the randomisation *procedure*. With many covariates a few will differ by chance, which is not evidence of a problem. Also check attrition by arm and compliance rates.

Failure modes.

- **Non-compliance** dilutes the intention-to-treat effect and changes the estimand.
- **Differential attrition** turns an experiment back into an observational study. If the treated units who drop out are those for whom treatment failed, the surviving comparison is biased.
- **Spillovers** contaminate the control group. Randomising at a higher level, such as markets rather than stocks or days rather than orders, can contain them at a large cost in clusters.
- **Small samples** produce chance imbalances large enough to matter, which stratified or paired randomisation reduces.
- **Many outcomes** create multiple testing (§4.8), which is why pre-specifying the primary outcome is standard.
- **External validity.** An experiment identifies an effect for its population, its context, and its scale. General-equilibrium effects that appear only when the treatment is scaled up are invisible in it.

In finance. Genuine experiments are rare but valuable: the Regulation SHO pilot (§6.3), the SEC's Tick Size Pilot, and field experiments in household finance. The most common experiment a quant runs is an **A/B test of an execution algorithm**. Randomise at the level of the order or the day, cluster inference by day, because every order on a given day shares the same market conditions, and be alert to interference, because two algorithms trading the same names can move each other's prices.

When preferred. Whenever it is feasible. Every other design in this section is a substitute for it.

7.3 Instrumental variables

Intuition. Find something that moves the regressor for reasons unrelated to the outcome's other causes, and use only the variation it creates (§5.2).

Estimand. With a constant effect, the structural coefficient. With effects that differ across units, IV estimates a **local average treatment effect** (LATE): the average effect among **compliers**, the units whose treatment status the instrument actually changes (Imbens & Angrist, 1994; Angrist, Imbens & Rubin, 1996). Units that would be treated whatever the instrument says, or untreated whatever it says, contribute nothing. Two consequences follow. Different valid instruments can legitimately give different answers, because they shift different compliers. And a failed over-identification test may reflect heterogeneous effects rather than invalid instruments.

Identifying assumptions. Four, and the last two are untestable.

1. **Relevance.** The instrument moves the treatment. This one is testable.
2. **Independence.** The instrument is as good as randomly assigned.
3. **Exclusion.** The instrument affects the outcome *only* through the treatment.
4. **Monotonicity.** The instrument pushes everyone's treatment in the same direction, so there are no "defiers."

Estimation. 2SLS. With many instruments, limited-information maximum likelihood (LIML) is less biased.

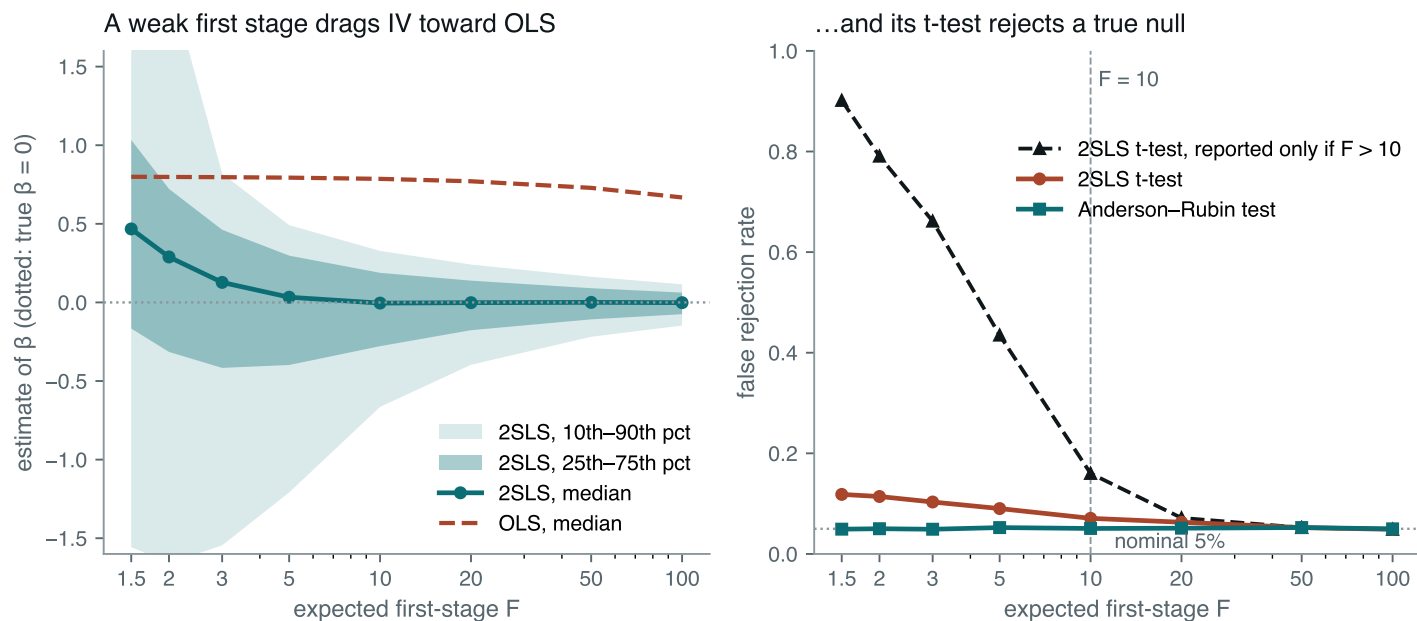
Inference. Robust or clustered standard errors when the instrument is strong. When it may be weak, use tests whose size does not depend on instrument strength. The **Anderson–Rubin test** (Anderson & Rubin, 1949) regresses $Y - X\beta_0$ on the instruments and tests whether they matter. It is valid at any strength in the just-identified case, and its confidence set is obtained by collecting the values of β_0 it does not reject. Moreira (2003) extends the idea to over-identified models.

Diagnostics.

- **The first stage.** The regression of the endogenous regressor on the instruments, $X_i = Z_i'\pi + v_i$, is the first stage. Report it, with a first-stage F -statistic for $\pi = 0$ that is robust to the dependence in the data. With heteroskedastic or clustered errors, that is the "effective F " of Montiel Olea & Pflueger (2013), not the classical one.
- **The reduced form.** Regress the outcome directly on the instrument. It is an ordinary regression, immune to weak-instrument problems, and if it is zero there is no effect to find.
- **Balance and placebos.** The instrument should not predict pre-determined characteristics, or outcomes that the treatment cannot affect.

Failure modes.

Weak instruments. When the first stage is weak, 2SLS is biased toward OLS and its t -test rejects true nulls too often. The figure below simulates a just-identified model with strong endogeneity (a correlation of 0.8 between the first-stage and structural errors) and a true effect of zero. With an expected first-stage F of 2, the median 2SLS estimate is 0.29, over a third of the way to the OLS estimate of 0.80. With an expected F of 10 the median bias has gone, but the estimate's spread is wide and the t -test still rejects at 7%. **[Simulated]**



The dashed line in the right panel is the more important result. It shows what the t -test does in only those samples whose first-stage F clears 10, which is what a literature that screens on $F > 10$ ends up reporting. When the true expected F is 5, the samples that pass the screen reject the true null 44% of the time. **[Simulated]** Passing the screen selects samples where the first stage looks strong *by chance*, and because the first-stage and structural errors are correlated, those are also the samples where the estimate is furthest from the truth.

The published thresholds tell the same story. **[Fact]** For one instrument, keeping the size of a nominal 5% test below 10% requires a first-stage F above 16.38 (Stock & Yogo, 2005). Lee, McCrary, Moreira & Porter (2022) show that a true 5% test needs F above 104.7, or equivalently a critical value of 3.43 instead of 1.96 when F is 10. Re-examining 57 IV papers in the *American Economic Review*, they find that about half of the results presumed significant are not.

Invalid exclusion. The instrument affects the outcome through another channel. Weather is a popular instrument, and Mellon (2025) catalogues close to two hundred variables that published studies have linked to weather. Each one is a potential channel that violates the exclusion restriction of every other weather-IV study.

Weak instruments amplify small violations. The large-sample bias of IV is $\text{Cov}(Z, u) / \text{Cov}(Z, X)$. A weak first stage divides a small correlation with the error by a small correlation with the treatment. **An instrument that is both slightly invalid and weak can be worse than OLS.**

Many instruments. 2SLS with many instruments over-fits the first stage and drifts toward OLS. **[Fact]** Bound, Jaeger & Baker (1995) showed that a famous estimate of the return to schooling, which used quarter of birth interacted with many other variables as instruments, could be closely reproduced with *randomly generated* instruments.

Fragile inference in practice. Young (2022) re-examined 1,309 IV regressions from 30 papers in the American Economic Association's journals. He found that non-independent errors and high leverage, results that hinge on a few clusters or observations, distort size and power, that first-stage F pretests are largely uninformative, and that IV estimates rarely reject the OLS estimate despite being substantively different from it. **[Contested]**

The wrong estimand. LATE is the effect for compliers, who may be unrepresentative. Whether that is a feature or a flaw is a live debate (Deaton, 2010; Imbens, 2010). **[Contested]**

Shift-share instruments. "Bartik" instruments combine exposure shares with aggregate shocks. The identifying variation can come from the shares or from the shocks, and the assumptions needed are different in each case (Goldsmith-Pinkham, Sorkin & Swift, 2020; Borusyak, Hull & Jaravel, 2022).

In finance. The cautionary tale is the mutual-fund flow instrument. Large outflows force funds to sell their holdings, so flow-induced selling was used as a source of price pressure unrelated to fundamentals (Coval & Stafford, 2007; Edmans, Goldstein & Jiang, 2012), and a large literature used it to study how mispricing affects corporate decisions. Wardlaw (2020) showed that the standard construction of the measure is mechanically a function of the stock's own realised return in the outflow quarter. With that removed, the price effect is small and does not reverse, and many downstream results do not hold. **[Contested]** An instrument that contains the outcome is not excluded. More constructive uses include demand-system asset pricing, which instruments for prices using features of investors' mandates (Kojen & Yogo, 2019), and monetary-policy surprises measured from futures prices (§8.5).

When preferred. When some feature of the world moves the treatment for reasons you can argue are unrelated to the outcome, and you can defend the exclusion restriction channel by channel.

7.4 Panel data and fixed effects

Intuition. Compare each unit with itself at other times, removing everything about it that does not change.

Model and estimand.

$$Y_{it} = \alpha_i + \lambda_t + X'_{it}\beta + u_{it},$$

where α_i absorbs every time-invariant characteristic of unit i , observed or not, and λ_t absorbs every shock common to all units at date t . The coefficient β is identified from **within-unit changes** in X relative to the average change at that date.

Identifying assumption. Strict exogeneity: $\mathbb{E}[u_{it} \mid X_{i1}, \dots, X_{iT}, \alpha_i] = 0$. The error at any date is unrelated to the regressors at every date. That rules out time-varying confounders, and it rules out feedback from past outcomes to current regressors.

Estimation. The **within estimator** subtracts each unit's time average from every variable and runs OLS. By Frisch–Waugh–Lovell (§3.4) this is identical to including a dummy variable for every unit. With two periods, first-differencing gives the same estimate. The two-way version also removes date means, and high-dimensional variants absorb interactions such as industry-by-year effects.

Random effects assume α_i is uncorrelated with the regressors, and in exchange use the between-unit variation too. In economics that assumption is rarely credible, because the unobserved unit characteristic is usually the reason for worrying in the first place. The **Hausman test** compares the two estimators (Hausman, 1978). Mundlak (1978) showed that adding the unit averages of the regressors to a random-effects model reproduces the fixed-effects coefficients exactly, which turns the choice into a regression specification.

Inference. Cluster by unit, which allows for serial correlation within units (§4.4). Cluster two-way if shocks are also shared across units at a date.

Diagnostics. How much of the regressor's variance is within units? How many units change their treatment, and are they typical? Does the estimate change sharply between pooled and fixed-effects specifications, and if so, is that confounding or attenuation (§6.4)?

Failure modes.

- **Only time-invariant confounders are removed.** A confounder that changes over time, such as a firm's growth opportunities, is untouched.
- **Feedback violates strict exogeneity.** If a bad year leads a firm to change its leverage, current leverage depends on past errors. With a lagged dependent variable and fixed effects, the within estimator is biased by roughly $-(1+\phi)/(T-1)$ in panels with T periods (Nickell, 1981). With ten periods and $\phi = 0.5$ the bias is about -0.17 , a third of the true coefficient. GMM estimators that use lagged levels as instruments address it (Arellano & Bond, 1991), and become weak-instrument problems when ϕ is near one.
- **Measurement error is amplified,** as §6.4 showed, because demeaning removes signal and keeps noise.
- **Identification from a few switchers.** If the regressor rarely changes within units (a state law, a CEO's characteristics), the estimate comes from the small and possibly unusual set of units where it does.
- **Fixed effects can absorb the variation of interest.** With firm fixed effects, the effect of a slow-moving characteristic like governance quality is estimated from its noise.
- **Substitutes for fixed effects are not fixed effects.** [Fact] Gormley & Matsa (2014) show that two common shortcuts in finance, subtracting the industry mean from the dependent variable ("industry-adjusting") and adding the group mean of the dependent variable as a control, give inconsistent estimates. Use the fixed-effects estimator.

In finance. Firm fixed effects are the workhorse of empirical corporate finance: capital structure, investment, and payout policy. Fund and manager fixed effects separate skill from style. Stock fixed effects in return-predictability regressions deserve caution, since demeaning a persistent predictor over a short window creates the finite-sample biases of §9.2.

When preferred. When the main worry is a stable unobserved characteristic of each unit, and the regressor changes within units for reasons unrelated to the outcome's recent history.

7.5 Difference-in-differences

Intuition. Compare the change over time in a group that was treated with the change in a group that was not. The untreated group's change stands in for what would have happened to the treated group.

Estimand. The ATT, for the treated group, in the post-treatment period.

The two-by-two case. With a treated and a control group, observed before and after,

$$\hat{\tau}_{\text{DiD}} = (\bar{Y}_{\text{treated, post}} - \bar{Y}_{\text{treated, pre}}) - (\bar{Y}_{\text{control, post}} - \bar{Y}_{\text{control, pre}}).$$

It equals the coefficient on the interaction in a regression of Y on a treated-group indicator, a post-period indicator, and their product. First-differencing removes each group's fixed level, and the second difference removes the common time shock.

Identifying assumption. Parallel trends: without treatment, the treated group's average outcome would have changed by the same amount as the control group's. Add no anticipation, meaning outcomes do not respond before treatment begins, and SUTVA. Note that parallel trends is a statement about a particular *scale*. If the groups start at different levels, trends cannot be parallel in both levels and logs, so choosing the scale is part of the assumption.

Estimation. The **two-way fixed effects** (TWFE) regression $Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + u_{it}$, where D_{it} switches on when unit i is treated. The **event-study** version replaces τD_{it} with separate coefficients for each period relative to treatment, including periods before it.

The staggered-adoption problem. With a single treatment date, TWFE equals the 2×2 estimator. When units are treated at *different* dates and effects vary over time, it does not, and the reason is worth working through by hand. Take two groups over three periods. The early group is treated from period 2, and the late group from period 3. Untreated outcomes are flat, so parallel trends holds exactly. Write τ_{E2} and τ_{E3} for the early group's treatment effects in periods 2 and 3, and τ_{L3} for the late group's effect in period 3.

By Frisch–Waugh–Lovell (§3.4), the TWFE coefficient is the regression of Y on the treatment indicator after removing unit and period means from it. For a balanced panel, that residual is $\tilde{D}_{it} = D_{it} - \bar{D}_i - \bar{D}_t + \bar{D}$:

	Period 1	Period 2	Period 3
Treatment D_{it} , early group	0	1	1
Treatment D_{it} , late group	0	0	1
Residual $\tilde{D}_{it}$, early group	−1/6	+1/3	−1/6
Residual $\tilde{D}_{it}$, late group	+1/6	−1/3	+1/6

The sum of squared residuals is $1/3$. The unit and period effects in the outcome are orthogonal to $\tilde{D}$, so only the treatment effects contribute to $\sum \tilde{D}_{it} Y_{it}$, and

$$\hat{\tau}_{\text{TWFE}} = 3 \left(\frac{1}{3} \tau_{E2} - \frac{1}{6} \tau_{E3} + \frac{1}{6} \tau_{L3} \right) = \tau_{E2} - \frac{1}{2} \tau_{E3} + \frac{1}{2} \tau_{L3}.$$

The early group's second-period effect enters with a negative weight. In period 3 the early group, already treated, is serving as the *control* for the late group, so growth in its treatment effect is subtracted as if it were a trend. If every effect equals 1, TWFE gives 1, correctly. If effects grow with exposure, $(\tau_{E2}, \tau_{E3}, \tau_{L3}) = (1, 2, 1)$, TWFE gives 0.5 while the average effect on the treated is $4/3$. If they grow faster, $(1, 4, 1)$, TWFE gives -0.5 : **a negative estimate when every treatment effect is positive**. This is the mechanism documented by Goodman-Bacon (2021) and de Chaisemartin & D'Haultfœuille (2020). **[Fact]**

The remedy is to compare treated units only with units that are not yet treated, or never treated. Several estimators do this: group-by-period effects that are then aggregated (Callaway & Sant'Anna, 2021), interaction-weighted event studies (Sun & Abraham, 2021), and imputation of untreated outcomes from untreated observations (Borusyak, Jaravel & Spiess, 2024). Roth, Sant'Anna, Bilinski & Poe (2023) is the guide to choosing among them. Baker, Larcker & Wang (2022) show that the problem affects published work in finance and accounting, where staggered designs are common.

Inference. Cluster at the level at which treatment is assigned. Bertrand, Duflo & Mullainathan (2004) is the warning (§4.4), and the few-treated-clusters problem of §4.4 applies with full force when a policy changes in only a handful of states.

Diagnostics.

- **The event-study plot.** Estimated effects in the periods before treatment should be near zero.
- **Placebo dates and placebo outcomes.** Pretend treatment happened earlier, or look at outcomes it could not have affected.
- **The scale.** Check whether conclusions survive switching between levels and logs.

Pre-trend tests deserve a caution of their own. They have low power, so passing them is weak evidence, and reporting only designs that pass them biases the resulting estimates (Roth, 2022). A more honest approach bounds how far post-treatment trends could depart from parallel, in proportion to the departures seen before treatment, and reports the range of effects consistent with that bound (Rambachan & Roth, 2023).

Failure modes.

- **Non-parallel trends from selection into timing.** Units adopt a treatment *because* of how their outcomes are evolving. **[Fact]** Workers who enter training programmes experience an earnings decline just before entering, which reverses whether or not the training works, so a naive DiD overstates the effect (Ashenfelter & Card, 1985). Firms adopt policies when they are in trouble, and mean reversion then masquerades as a treatment effect.
- **Anticipation.** In financial markets, prices respond at announcement, not at the effective date, so a DiD keyed to the effective date misses most of the effect.
- **Staggered timing with heterogeneous effects,** above.
- **Spillovers** to the control group, and changes in the composition of either group over time.
- **Policy endogeneity and institutional context.** **[Contested]** Karpoff & Wittry (2018) show that the effect of state anti-takeover laws on a firm's takeover protection depends on other state laws, the firm's existing defences, and court decisions, and that tests which ignore this context can support misleading inferences from the laws used as natural experiments.

In finance. Staggered state antitakeover laws (Bertrand & Mullainathan, 2003), bank deregulation, and regulatory changes that affect some firms or markets before others.

When preferred. When a policy or shock hits some units at some times, a comparison group plausibly shares the treated group's trends, and there are enough pre-treatment periods to check.

7.6 Regression discontinuity

Intuition. When treatment is assigned by whether a continuous **running variable** crosses a cutoff, units just above and just below the cutoff are nearly identical except for treatment. The jump in the average outcome at the cutoff is the treatment effect.

Estimand. The effect *at the cutoff*: $\tau(c) = \mathbb{E}[Y_i(1) - Y_i(0) \mid R_i = c]$, where R_i is the running variable and c the cutoff. In a **sharp** design, treatment switches on exactly at the cutoff. In a **fuzzy** design only the probability of treatment jumps, and the ratio of the outcome jump to the treatment jump is an IV estimate, a LATE for compliers at the cutoff (Hahn, Todd & van der Klaauw, 2001).

Identifying assumption. The average potential outcomes are continuous in the running variable at the cutoff. Nothing else jumps there, and units cannot precisely sort themselves to one side.

Estimation. Fit separate local linear regressions on each side of the cutoff, within a bandwidth, typically weighting nearer observations more heavily, and take the difference of the two fitted values at the cutoff. Choose the bandwidth by a data-driven mean-squared-error rule. Do **not** fit high-order polynomials to the whole range: they put erratic weight on observations far from the cutoff and give misleading confidence intervals (Gelman & Imbens, 2019).

Inference. A bandwidth chosen to minimise mean squared error deliberately leaves some bias, which invalidates conventional confidence intervals. Use **robust bias-corrected** intervals (Calonico, Cattaneo & Titiunik, 2014). A discrete running variable, such as an integer rank or a rounded score, needs special treatment (Kolesár & Rothe, 2018).

Diagnostics.

- **Manipulation.** A jump in the *density* of the running variable at the cutoff suggests units sorting themselves (McCrary, 2008; Cattaneo, Jansson & Ma, 2020).
- **Covariate balance.** Pre-determined characteristics should not jump at the cutoff.
- **Placebo cutoffs and bandwidth sensitivity.** There should be no jumps at fake cutoffs, and the estimate should not hinge on one bandwidth. A “donut” design that drops the observations closest to the cutoff checks for sorting right at the threshold.

Failure modes.

- **Manipulation of the running variable,** which recreates selection at exactly the point where the design needs none.

- **Compound treatments.** Other rules use the same cutoff, so the jump bundles several effects.
- **A coarse running variable,** which leaves no observations genuinely close to the cutoff.
- **Extrapolation.** The effect at the cutoff can differ from the effect anywhere else.
- **Specification search** over bandwidths, kernels, and polynomial orders.

In finance. The canonical application is the annual reconstitution of the Russell 1000 and Russell 2000 indices. Stocks are ranked by market capitalisation in May, and a stock just below the 1,000th rank lands at the top of the Russell 2000 with a large index weight, while one just above lands at the bottom of the Russell 1000 with a small weight. The design has been used to estimate the effect of index weight on prices (Chang, Hong & Liskovich, 2015) and of passive ownership on governance (Appel, Gormley & Keim, 2016). The complications illustrate the failure modes above. Russell ranks on its own proprietary capitalisation measure, which researchers can only approximate, and since 2007 a “banding” rule has meant that a stock’s assignment depends on its previous index as well as its rank. Both features blur the discontinuity and change which comparisons are valid, and the literature has argued at length about the right specification (Appel, Gormley & Keim, 2024). **[Contested]** Other thresholds used in finance include debt covenant limits (Chava & Roberts, 2008), close shareholder votes, and credit-rating boundaries.

When preferred. Whenever a rule assigns treatment by a threshold on a variable that units cannot precisely control. It is the most credible observational design, for the effect it identifies.

7.7 Synthetic control

Intuition. When a single aggregate unit is treated, such as a country, a state, or a market, build a weighted average of untreated units that tracked the treated unit closely before treatment. Its path after treatment is the counterfactual.

Estimand. The effect on the treated unit, period by period after treatment.

Identifying assumption. The weighted donor combination continues to track the treated unit’s untreated outcome after treatment. A sufficient condition is that outcomes follow a factor structure, with the treated unit’s factor loadings matched by the donors’. Also required: no spillovers onto the donors, and no other shock to the treated unit at the same time.

Estimation. Choose non-negative donor weights that sum to one and minimise the distance between the treated unit and the weighted donors in pre-treatment outcomes and predictors (Abadie & Gardeazabal, 2003; Abadie, Diamond & Hainmueller, 2010). The constraints keep the counterfactual an interpolation of real units rather than an extrapolation.

Inference. By placebo. Apply the method to every donor as if it had been treated, and see where the treated unit’s post-treatment divergence, relative to its pre-treatment fit, ranks among the placebos. With 19 donors, being the most extreme of 20 units corresponds to a p-value of 0.05.

Diagnostics. Pre-treatment fit, first of all: if the synthetic unit does not track the treated unit before treatment, do not use it. Also check the weights, which should be sparse and interpretable, and whether results survive dropping each donor in turn.

Failure modes.

- **Poor pre-treatment fit,** often because the treated unit is extreme and lies outside what any combination of donors can reproduce.
- **Short pre-periods,** which let the weights fit noise rather than structure.
- **Specification search** over predictors and donor pools, which can manufacture results (Ferman, Pinto & Possebom, 2020).
- **Contaminated donors** that were themselves affected by the treatment.

Synthetic difference-in-differences combines synthetic-control unit weights with time weights and DiD’s fixed effects, and is more robust than either parent when the pre-period fit is imperfect (Arkhangelsky, Athey, Hirshberg, Imbens & Wager, 2021). Abadie (2021) sets out the data requirements.

In finance. A regulation or market-structure change in one country’s market, or a shock to a single exchange, with other markets as donors.

When preferred. One or a few treated aggregate units, a long pre-treatment series, and a pool of comparable untreated units.

7.8 Structural estimation

Intuition. Write down a model of the decisions that generated the data, with parameters for preferences, technology, constraints, and beliefs. Estimate the parameters, then use the model to answer counterfactual questions, including questions about interventions that have never happened.

Estimand. Deep parameters, and the counterfactuals they imply.

Identifying assumption. The model itself: its functional forms, distributional assumptions, and equilibrium concept, together with whatever variation in the data pins down each parameter. **The model is the identification.** Good structural work shows which features of the data identify which parameters, so that a reader can judge whether identification comes from variation or from functional form.

Estimation. Maximum likelihood, GMM, or the **simulated method of moments**, which simulates the model and chooses parameters to match selected moments of the data.

Inference. The sandwich, or the bootstrap. Andrews, Gentzkow & Shapiro (2017) show how to report which moments drive which estimates, which makes the identification argument inspectable.

Diagnostics. Fit to moments that were *not* used in estimation, and plausibility of counterfactuals.

Failure modes.

- **Identification by functional form**, where the estimate is determined by an assumption no data could check.
- **Misspecification that propagates to counterfactuals**, often invisibly, because the counterfactual is by definition outside the data.
- **Computational fragility**: local optima and sensitivity to starting values.
- **Many modelling choices**, which create their own garden of forking paths.
- **Parameters that are not really invariant.** The Lucas critique (§1.2) motivates structural models, but it applies to them too if the “deep” parameters would change with policy.

In finance. Consumption-based asset pricing estimated by GMM (§5.3); dynamic corporate finance models estimated by simulated moments (Hennessy & Whited, 2007); demand-system asset pricing, which borrows the structural demand estimation of industrial organisation (Berry, Levinsohn & Pakes, 1995; Koijen & Yogo, 2019); and term-structure models estimated by the Kalman filter.

The debate. How much weight to put on design-based versus structural evidence is the longest-running argument in modern empirical economics. The design-based side argues that credible identification of a well-defined effect beats an ambitious model whose assumptions cannot be checked (Angrist & Pischke, 2010). The structural side argues that design-based estimates are local, often answer questions nobody asked, and cannot say anything about interventions outside the data without a model anyway (Keane, 2010; Heckman, 2010; Deaton, 2010). **[Contested]** My view is that the two are complements, and that the best current work uses quasi-experimental variation to discipline a structural model, so that the model's extrapolations are anchored by effects that were credibly identified.

When preferred. When the question is a counterfactual outside the range of the data, a welfare calculation, or a mechanism, and when the relevant policy would change the environment that generated the data.

7.9 Comparison

Design	Estimand	Identifying assumption	What the data can check	Characteristic failure
Randomised experiment	ATE; intention-to-treat	Random assignment	Balance, attrition, compliance	Spillovers; external validity
Regression discontinuity	Effect at the cutoff	Continuity at the cutoff	Density, covariate balance, placebo cutoffs	Manipulation; extrapolation
Instrumental variables	LATE for compliers	Exclusion, independence, monotonicity	First stage, reduced form, placebos	Weak or invalid instruments
Difference-in-differences	ATT	Parallel trends	Pre-trends (weakly), placebos	Selection into timing; staggered bias
Synthetic control	Effect on one treated unit	Donors track the untreated path	Pre-treatment fit, placebos	Poor fit; specification search
Panel fixed effects	Within-unit effect	Strict exogeneity	Within variation; switchers	Time-varying confounders; feedback
Selection on observables	ATE or ATT	Unconfoundedness	Overlap only	Unobserved confounding
Structural	Deep parameters; counterfactuals	The model	Untargeted moments	Misspecification

[Practice] My ranking of credibility, for the effect each design identifies and with competent execution: randomised experiments, then regression discontinuity, then instrumental variables with a genuinely random instrument and difference-in-differences with a well-understood policy, then synthetic control and panel fixed effects, then selection on observables. A structural model is as credible as its weakest assumption. The ranking is about *internal* validity. An experiment on the wrong population can be less useful than a careful observational study of the right one, and a regression discontinuity says nothing about units far from its cutoff.

§7 Key takeaways

1. Every research design buys an orthogonality condition at a price. Ask which condition, and whether the price is worth paying, before asking about estimators.
2. IV estimates the effect for compliers, not for everyone, and weak instruments bias it toward OLS while making its t -test lie. A true 5% test needs a first-stage F near 105, not 10, and screening on $F > 10$ makes reported results worse, not better.
3. Weak instruments turn small exclusion violations into large biases. Always report the reduced form.
4. Fixed effects remove only time-invariant confounders, fail under feedback from outcomes to regressors, and amplify measurement error. Industry-adjusting the dependent variable is not a substitute.
5. With staggered adoption and effects that change over time, two-way fixed effects can report a negative effect when every true effect is positive. Use estimators that compare treated units only with not-yet-treated or never-treated ones.
6. Pre-trend tests are weak evidence. Report sensitivity to plausible departures from parallel trends.
7. Regression discontinuity is the most credible observational design for the effect at its cutoff, and says nothing beyond it. Test for manipulation and use robust bias-corrected intervals.
8. Synthetic control is for one treated aggregate unit with a long pre-period; poor pre-treatment fit disqualifies it.
9. Structural models answer questions designs cannot, at the cost of assumptions designs do not need. The strongest evidence combines them.

Part IV — Time series

8. Time-series econometrics

A cross-section is many units observed once. A time series is one unit observed many times, and that single structural difference changes almost everything. There is only one history, so “repeated sampling” has to be reinterpreted. Observations are dependent, so the meat of every standard error changes. And variables can wander without limit, which breaks the asymptotic approximations of §2. This section covers the ideas a practitioner needs. Simple and Log Returns §6 applies several of them to returns specifically, and Market Regimes and Machine Learning takes regime-switching and change-point models much further.

8.1 What changes when observations are ordered

Stationarity replaces random sampling. A process is **covariance stationary** if its mean and variance are constant over time and the covariance between two observations depends only on how far apart they are. Combined with **ergodicity**, the requirement that dependence fade fast enough for time averages to converge to population averages, it lets a single long history stand in for many independent draws. Without it, the average of one path need not estimate anything.

The autocorrelation function $\rho_j = \text{Corr}(y_t, y_{t-j})$ summarises linear dependence. The **partial** autocorrelation at lag j is the correlation between y_t and y_{t-j} after removing the effect of the intermediate lags, which is the coefficient on y_{t-j} in a regression on j lags.

Three strengths of “unpredictable” mirror the three strengths of “unrelated” in §2.2. **White noise** is uncorrelated with its past. A **martingale difference sequence** has a conditional mean of zero given its past. An **i.i.d.** sequence is independent of its past. Returns come close to the middle one and are far from the last, because their variance is predictable.

Testing for autocorrelation illustrates why the distinction matters. The Ljung–Box statistic (Ljung & Box, 1978) sums squared sample autocorrelations and compares the total with a chi-squared distribution, which is valid when the series is i.i.d. On returns with volatility clustering it rejects too often, because the squared-autocorrelation variance it assumes is too small. The **variance ratio** compares the variance of q -period returns with q times the variance of one-period returns,

$$\text{VR}(q) = \frac{\text{Var}(r_t + \dots + r_{t-q+1})}{q \text{Var}(r_t)} = 1 + 2 \sum_{j=1}^{q-1} \left(1 - \frac{j}{q}\right) \rho_j,$$

which equals 1 for uncorrelated returns and exceeds 1 under positive autocorrelation. Lo & MacKinlay (1988) built a version whose standard errors are robust to heteroskedasticity, and rejected the random walk for weekly US stock index returns. The variance ratio as a trading statistic is developed in section 4.5.1 of Momentum in Financial Markets.

8.2 Autoregressions and the Wold decomposition

The AR(1) model, $y_t = c + \phi y_{t-1} + \varepsilon_t$ with $|\phi| < 1$, carries most of the intuition of time-series dynamics in one parameter.

Property	Formula	With $\phi = 0.9$
Mean	$\mu = c/(1 - \phi)$	
Variance	$\sigma^2/(1 - \phi^2)$	5.3 times the innovation variance
Autocorrelation at lag j	ϕ^j	0.59 at lag 5
Response at horizon h to a unit shock	ϕ^h	0.35 after 10 periods
Half-life of a shock	$\ln(0.5)/\ln \phi$	6.6 periods
Forecast	$\mathbb{E}[y_{t+h} \mathcal{F}_t] = \mu + \phi^h(y_t - \mu)$	Reverts to the mean
Effective sample size for the mean	$T(1 - \phi)/(1 + \phi)$	5.3% of T (§2.5)

Higher-order **autoregressions** add lags. **Moving-average** models, $y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots$, model a shock that echoes for a fixed number of periods and then stops, and **ARMA** models combine the two.

The Wold decomposition explains why these simple models are so general. Every covariance-stationary process can be written as a deterministic component plus an infinite moving average of its own one-step-ahead linear forecast errors. The consequence cuts both ways. **Linear ARMA models can reproduce the autocovariances of any stationary process, and they**

are blind to everything else. Volatility clustering, asymmetric responses, and any predictability that is not linear leave the autocovariances untouched, and so are invisible to them (§8.6).

The Box–Jenkins workflow, from Box and Jenkins’s 1970 book, is still the right order of operations: identify a candidate model from the sample autocorrelations and partial autocorrelations, estimate it, check that its residuals look like white noise, and only then forecast. Choosing lag lengths by information criteria involves a real trade-off. The **AIC** (Akaike, 1974) targets forecast accuracy and tends to choose too many lags. The **BIC** (Schwarz, 1978) penalises parameters more heavily and picks the true order with probability approaching one when the true model is among the candidates. **[Practice]** For forecasting use AIC or cross-validation; for describing the dynamics, BIC. The humbling historical footnote is that simple ARIMA models forecast US macroeconomic aggregates about as well as the large structural models of the 1960s (§13).

8.3 Unit roots and spurious regression

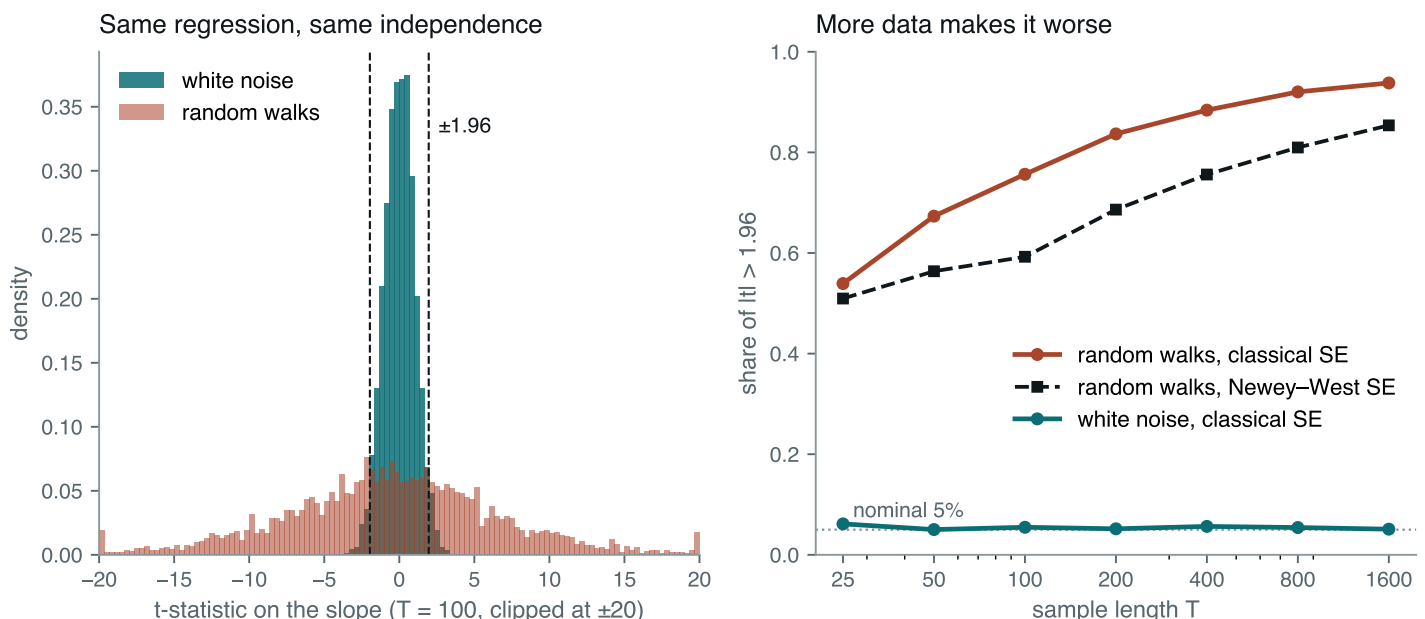
The random walk $y_t = y_{t-1} + \varepsilon_t$ is the AR(1) with $\phi = 1$, and at that value every property in the table above breaks. The variance grows linearly with time, so the process has no fixed variance to estimate. Shocks never decay, so the half-life is infinite. There is no mean for the forecast to revert to. A series like this is **integrated of order one**, written $I(1)$: its first difference is stationary, but its level is not.

Testing for a unit root. The **Dickey–Fuller** test regresses the change on the lagged level, $\Delta y_t = \alpha + \gamma y_{t-1} + \sum_j \delta_j \Delta y_{t-j} + \varepsilon_t$, and tests $\gamma = 0$ (Dickey & Fuller, 1979). Under the null the regressor is itself a random walk, so the t -statistic does not have a t distribution. Its 5% critical value with a constant is about -2.86 , far below the -1.65 of a standard one-sided test. The test also has low power: with samples of typical length it cannot reliably tell $\phi = 1$ from $\phi = 0.97$, which is a very different process, with a half-life of 23 periods. The **KPSS** test reverses the null to stationarity (Kwiatkowski, Phillips, Schmidt & Shin, 1992), and running both is common.

Whether macroeconomic series have unit roots was a major debate. Nelson & Plosser (1982) could not reject a unit root in most US macroeconomic series, and Perron (1989) showed that allowing for a few structural breaks reversed many of those conclusions. **[Contested]** The practical lesson is that unit-root tests are rarely decisive, and **[Practice]** the decision to difference a series should rest on the economics and on the question as much as on a test.

Spurious regression. Regress one random walk on another, independent one, and the regression will usually report a strong, significant relationship. The phenomenon was described by Yule (1926), quantified by Granger & Newbold (1974), and explained by Phillips (1986). Under the null, the estimated slope does not converge to zero but to a random variable, and the t -statistic grows without bound at rate $\sqrt{T}$.

The figure shows how severe this is. With 100 observations of two independent random walks, the conventional t -test rejects “no relationship” 76% of the time at a nominal 5%. With 1,600 observations it rejects 94% of the time: **more data makes it worse**. Newey–West standard errors lower the rates to 59% and 85%, and do not fix the problem. The median R^2 is about 0.17 at every sample length, and a third of the regressions report an R^2 above 0.3. **[Simulated]**



The remedies are to difference both series, which answers a different question (how *changes* relate); to include lagged levels

of both variables, which restores standard inference for some coefficients; or, if the levels genuinely move together, to model the relationship as cointegration. In finance, price levels, dividend–price ratios, interest rates, and volatility are all at or near unit roots, and a regression of one price level on another is spurious until shown otherwise.

8.4 Cointegration and error correction

Two $I(1)$ series are **cointegrated** if some linear combination of them, $y_t - \beta x_t$, is stationary. They share a common stochastic trend, wander together, and the gap between them keeps reverting. The combination is the **cointegrating relationship**.

Estimation. The Engle & Granger (1987) procedure has two steps. First, regress y_t on x_t in levels. When the series are cointegrated, this regression is not spurious. The estimate of β is **superconsistent**, converging at rate T rather than $\sqrt{T}$ (Stock, 1987), because any wrong β leaves a non-stationary residual that the least-squares criterion punishes heavily. Second, test the residual for a unit root, using Engle–Granger critical values, which are more negative than Dickey–Fuller ones because the residual was chosen to look stationary. The Johansen (1991) procedure estimates the whole system by maximum likelihood and allows for several cointegrating relationships.

The error-correction model. Cointegration implies, and is implied by, a model in which changes respond to the previous period’s deviation from equilibrium:

$$\Delta y_t = \alpha (y_{t-1} - \beta x_{t-1}) + \text{lagged changes} + \varepsilon_t,$$

with $\alpha < 0$ measuring the speed at which a gap closes. This equivalence is the Granger representation theorem.

In finance. Pairs trading is cointegration in practice: the spread between two related prices is traded on the expectation that it reverts (Gatev, Goetzmann & Rouwenhorst, 2006). Spot and futures prices, yields of different maturities, and log prices and log dividends are other examples. Whether a pair should be modelled as a stationary spread or a stationary ratio is its own decision (Simple and Log Returns §6.4).

Failure modes.

- **Multiple testing.** Screening 1,000 candidate pairs with a 5% cointegration test finds about 50 “cointegrated” pairs even if none is.
- **Low power and breaks.** Cointegration tests struggle to separate slow mean reversion from none, and a relationship that held in the sample can break precisely when a position is on.
- **Inference on β .** Superconsistency does not make the levels regression’s t -statistics standard, and inference needs estimators designed for it, such as dynamic OLS with leads and lags.

8.5 Vector autoregressions, impulse responses, and identification

The reduced-form VAR. Stack several series in a vector $\mathbf{y}_t$ and regress each on lags of all of them:

$$\mathbf{y}_t = \mathbf{c} + A_1 \mathbf{y}_{t-1} + \cdots + A_p \mathbf{y}_{t-p} + \mathbf{e}_t, \quad \text{Var}(\mathbf{e}_t) = \Sigma_e.$$

Each equation is estimated by OLS. Sims (1980) proposed VARs as an alternative to large structural models whose identifying restrictions he called “incredible.” With k variables and p lags there are $k^2 p$ coefficients, which is why shrinkage priors are standard for forecasting (§5.6).

Granger causality asks whether the past of one series improves forecasts of another, given the second series’s own past (Granger, 1969). It is a test of predictive content and nothing more. **[Fact]** Stock prices Granger-cause GDP because prices anticipate economic news, not because the stock market drives output. Forecasts of rain Granger-cause rain.

Structural VARs and the identification problem. The reduced-form errors $\mathbf{e}_t$ are mixtures of economically meaningful **structural shocks** ε_t , such as a monetary policy shock or an oil supply shock: $\mathbf{e}_t = B\varepsilon_t$. With structural shocks normalised to unit variance, $\Sigma_e = BB'$. That gives $k(k+1)/2$ equations for the k^2 unknowns in B , so $k(k-1)/2$ restrictions are needed. This is exactly the simultaneity problem of §6.4, in dynamic form. The standard solutions:

Approach	The restriction	Main weakness
Recursive (Cholesky) ordering	Variables are ordered so that each responds to earlier ones within the period, and not to later ones	Results depend on an ordering that is often arbitrary

Approach	The restriction	Main weakness
Long-run restrictions	Some shocks have no permanent effect on some variables (Blanchard & Quah, 1989)	Sensitive to how low-frequency dynamics are modelled
Sign restrictions	Shocks move variables in theoretically signed directions	Identify a set of models, not a single one
External instruments	An outside measure of the shock serves as an instrument (Mertens & Ravn, 2013)	The instrument's relevance and exclusion

High-frequency identification is the external-instrument approach with the most influence on finance. The change in federal funds futures prices in a narrow window around a monetary policy announcement measures the *surprise* component of the decision (Kuttner, 2001; Gürkaynak, Sack & Swanson, 2005). Within thirty minutes nothing else systematically moves interest rates, so the surprise is plausibly exogenous. It is an event study (§9.3) used as an instrument. One complication is that a surprise can also reveal the central bank's private information about the economy, so a rate rise can signal strength as well as tightening (Nakamura & Steinsson, 2018). **[Contested]**

Impulse responses and local projections. An **impulse response** traces the effect of a one-time structural shock on each variable over subsequent horizons. A VAR computes it by iterating the estimated dynamics forward. **Local projections** estimate it directly, with one regression per horizon of y_{t+h} on the shock and controls (Jordà, 2005). Local projections are more robust to misspecified dynamics and noisier. **[Fact]** In population, with unrestricted lag structures, the two estimate the same impulse responses, so the choice between them is a finite-sample bias–variance trade-off rather than a difference in what is estimated (Plagborg-Møller & Wolf, 2021). Because the dependent variable at horizon h overlaps with its neighbours, local-projection inference faces the overlap problem of §4.5, and augmenting the regression with lags of the outcome makes it much simpler (Montiel Olea & Plagborg-Møller, 2021).

8.6 Volatility: ARCH, GARCH, and realised variance

The stylised fact. **[Fact]** Asset returns are nearly uncorrelated, but their squares and absolute values are strongly and persistently autocorrelated. Calm periods and turbulent periods cluster (Cont, 2001). Volatility is predictable even when returns are not.

ARCH and GARCH. Engle (1982) modelled the conditional variance as a function of past squared shocks. The GARCH(1,1) model (Bollerslev, 1986) adds the past conditional variance:

$$\sigma_t^2 = \omega + a \varepsilon_{t-1}^2 + b \sigma_{t-1}^2,$$

where $\varepsilon_t = \sigma_t z_t$ and z_t is an i.i.d. shock with unit variance. Its properties are easy to read off.

- **The long-run variance** is $\omega/(1 - a - b)$, provided $a + b < 1$.
- **Persistence** is $a + b$. A variance shock has a half-life of $\ln 0.5 / \ln(a + b)$ periods: 34 days at 0.98, and 69 days at 0.99. **[Fact]** Estimates for daily equity index returns typically put $a + b$ in the high 0.9s.
- **GARCH(1,1) is an ARMA(1,1) in squared shocks.** Writing $v_t = \varepsilon_t^2 - \sigma_t^2$, which is a martingale difference, gives $\varepsilon_t^2 = \omega + (a + b)\varepsilon_{t-1}^2 + v_t - b v_{t-1}$. The machinery of §8.2 applies directly.

GARCH models are estimated by maximum likelihood, and Gaussian quasi-maximum likelihood with sandwich standard errors is consistent even though returns are not normal (§5.4). **Asymmetric** versions let volatility respond more to negative returns than to positive ones, which fits equity data better (Nelson, 1991; Glosten, Jagannathan & Runkle, 1993).

Realised variance. With intraday data, variance can be *measured* rather than modelled. Summing squared high-frequency returns over a day estimates that day's variance, with an error that shrinks as the sampling interval shrinks, up to the point where market microstructure noise takes over (Andersen, Bollerslev, Diebold & Labys, 2003). This is Merton's asymmetry (§2.5) put to work: finer sampling does nothing for the precision of a mean return and a great deal for the precision of a variance. The **HAR** model, which regresses tomorrow's realised variance on its daily, weekly, and monthly averages, is simple and hard to beat (Corsi, 2009).

Evaluating volatility forecasts has a trap of its own: the target is never observed, so forecasts are compared with a noisy proxy such as a squared return. Patton (2011) showed that only certain loss functions, notably mean squared error and the "QLIKE" loss, rank forecasts consistently when the proxy is noisy but unbiased. Others can prefer the wrong forecast. **[Fact]** In the most thorough horse race, Hansen & Lunde (2005) compared 330 ARCH-type models. For exchange rates they found no evidence that anything beats a GARCH(1,1). For IBM stock returns, models with a leverage effect were clearly better.

8.7 Structural breaks and parameter instability

A relationship estimated over a long sample assumes the parameters were constant throughout. Tests for **structural breaks** check that assumption.

- **A known break date.** The Chow (1960) test compares coefficients estimated before and after the date.
- **An unknown break date.** Computing the Chow statistic at every candidate date and taking the largest is a multiple-testing problem, so the maximum has its own, larger critical values (Andrews, 1993). A Chow test at a date chosen *after looking at a plot* is this sup-test with the wrong critical values.
- **Several breaks.** Bai & Perron (1998) estimate the number and location of multiple breaks.

Breaks and forecasting. When parameters change, a rolling estimation window adapts and an expanding window is more precise, so the choice is a bias–variance trade-off. Pesaran & Timmermann (2007) show that it can pay to include some pre-break data even after a break, trading a little bias for a large reduction in variance. **[Practice]** Averaging forecasts across several window lengths is a common robust alternative to choosing one. Regime-switching and change-point models, which treat instability as a process rather than an event, are the subject of Market Regimes and Machine Learning §5.

8.8 Forecasting and forecast evaluation

Choose the loss function first. The mean squared error is minimised by the conditional mean, the absolute error by the conditional median, and the asymmetric “tick” loss of quantile regression (§5.5) by a conditional quantile. A forecast is good or bad *for a loss*, and evaluating with a different loss from the one the forecast targets gives misleading rankings.

Evaluate out of sample, honestly. A **pseudo-out-of-sample** exercise re-estimates the model at each forecast date using only data available at that date, with either a rolling or an expanding window. “Only data available at that date” is stricter than it sounds. It includes the choice of model, the choice of predictors, the tuning of hyperparameters, and the data themselves. Macroeconomic data are revised after release, and a forecast evaluated with final revised data uses information nobody had (Croushore & Stark, 2001). Accounting data are available only months after the period they describe. The out-of-sample R^2 of §3.8 is the standard summary.

Comparing two forecasts. The **Diebold–Mariano** test takes the difference between two forecasts’ losses at each date and tests whether its mean is zero, with a HAC standard error (Diebold & Mariano, 1995). It was designed to compare *forecasts*. It is not valid for comparing a model with a larger model that nests it when both are estimated. Under the null that the extra predictors are useless, the larger model still pays for estimating them, which inflates its loss and makes the test too conservative, so real predictive content is missed. Clark & West (2007) adjust for exactly that noise. Giacomini & White (2006) reframe the question as a comparison of forecasting *methods*, estimation windows included.

The equity premium debate is the best-known application, and it is unresolved. **[Contested]**

- Goyal & Welch (2008) examined the predictors proposed in the literature (dividend yields, earnings yields, interest rates, and others) and found that they predicted poorly both in and out of sample over the preceding 30 years, were unstable, and would not have helped an investor.
- Campbell & Thompson (2008) showed that imposing weak economic restrictions, such as a non-negative forecast of the premium, lets many predictors beat the historical mean out of sample, with small R^2 values that are economically meaningful (§3.8).
- Cochrane (2008) argued that the right test is joint. If dividend growth is unpredictable, as it appears to be, then the variation in dividend yields has to come from predictable returns.
- Rapach, Strauss & Zhou (2010) found that simple averages of the individual forecasts beat the historical mean out of sample, even though most individual predictors did not. Goyal, Welch & Zafirov (2024) updated the original evaluation with newer predictors and found the record still weak.

My reading is that equity premium predictability is real, small, and unstable, and that the econometric problems of §9.2 make its in-sample strength look larger than it is.

Combine forecasts. Averaging forecasts from different models is one of the most reliable improvements in forecasting (Bates & Granger, 1969). **[Fact]** Simple equal weights usually beat weights estimated to be optimal. This “forecast combination puzzle” is explained by the error in estimating the weights, which outweighs the gain from optimising them (Smith & Wallis, 2009), the same logic as the shrinkage of §5.6.

§8 Key takeaways

1. Stationarity and ergodicity are what let one history stand in for many samples. Without them, time averages need not estimate anything.
2. Returns are close to a martingale difference sequence and far from i.i.d. Tests that assume independence are mis-sized on them; use heteroskedasticity-robust versions.
3. Linear time-series models describe autocovariances completely and are blind to predictable volatility and nonlinear dependence.
4. Unit-root tests have nonstandard critical values and low power. Decide whether to difference on economics as much as on tests.
5. Regressing one random walk on an independent one rejects “no relationship” most of the time, and more data make it worse. Newey–West does not fix it.
6. Cointegration makes levels regressions meaningful, and screening many pairs for it is a multiple-testing exercise.
7. Granger causality is predictive precedence, not causation. Structural VARs need identifying restrictions, and high-frequency event windows are the most credible source of them for monetary shocks.
8. Volatility is highly persistent and predictable. GARCH(1,1) is an ARMA(1,1) in squared shocks, and realised variance turns high-frequency data into precise variance measurement.
9. Choose the loss first, evaluate out of sample with only information available at the time, use Clark–West rather than Diebold–Mariano for nested models, and average forecasts.

Part V — Practice

9. Financial econometrics

Everything in Parts I–IV applies to financial data. This section is about where it bites hardest. Finance combines the problems of the earlier sections in an unusually unfavourable way: the signals are small, the noise is large and fat-tailed, the dependence runs across both time and assets, the predictors are persistent, the data have been searched by thousands of researchers for decades, and the act of publishing a finding changes the process that generated it.

9.1 Why financial data are statistically hostile

Property	Consequence	Where
Low signal-to-noise. Monthly return predictability rarely exceeds an R^2 of a few percent	The t -statistic of a strategy's mean is roughly its Sharpe ratio times the square root of the number of years, so a Sharpe ratio of 0.5 needs about 16 years to reach $t = 2$ (18 once the estimation-uncertainty correction of §2.6 is included)	§2.5, §4.7
Fat tails and volatility clustering	Classical and i.i.d. standard errors are wrong; higher moments are estimated imprecisely	§2.4, §8.6
Common factors across assets	Thousands of stocks at one date are far fewer independent observations; inference clusters by date	§4.4, §9.5
Persistent predictors. Valuation ratios and interest rates are near unit roots	Biased slopes and oversized tests in predictive regressions	§8.3, §9.2
Overlapping horizons	A 40-year sample of annual returns has about 40 observations, whatever the sampling frequency	§4.5
Non-stationarity and adaptation	Relationships break, and published anomalies decay	§8.7, §9.6
Survivorship and look-ahead	Samples that condition on the future, and datasets revised after the fact	§6.4, §9.7
Collective data mining	Decades of research on the same databases turn every new result into the best of an unknown number of tries	§4.8, §9.6
Microcaps	They are about 60% of listed US stocks and about 3% of market value (Fama & French, 2008), so equal-weighted results describe them	§9.6

The effect of publication deserves a number. **[Fact]** McLean & Pontiff (2016) studied 97 return predictors from the academic literature and found portfolio returns 26% lower in the period between the end of the original sample and publication, and 58% lower after publication. The first number is an upper bound on how much of the original result was statistical overfitting. The difference between the two, about a third, is consistent with investors trading on published results.

9.2 Predictive regressions

The canonical return-forecasting regression pairs a return equation with an equation for a persistent predictor such as the dividend yield:

$$r_{t+1} = \alpha + \beta x_t + u_{t+1}, \quad x_{t+1} = \theta + \phi x_t + v_{t+1}.$$

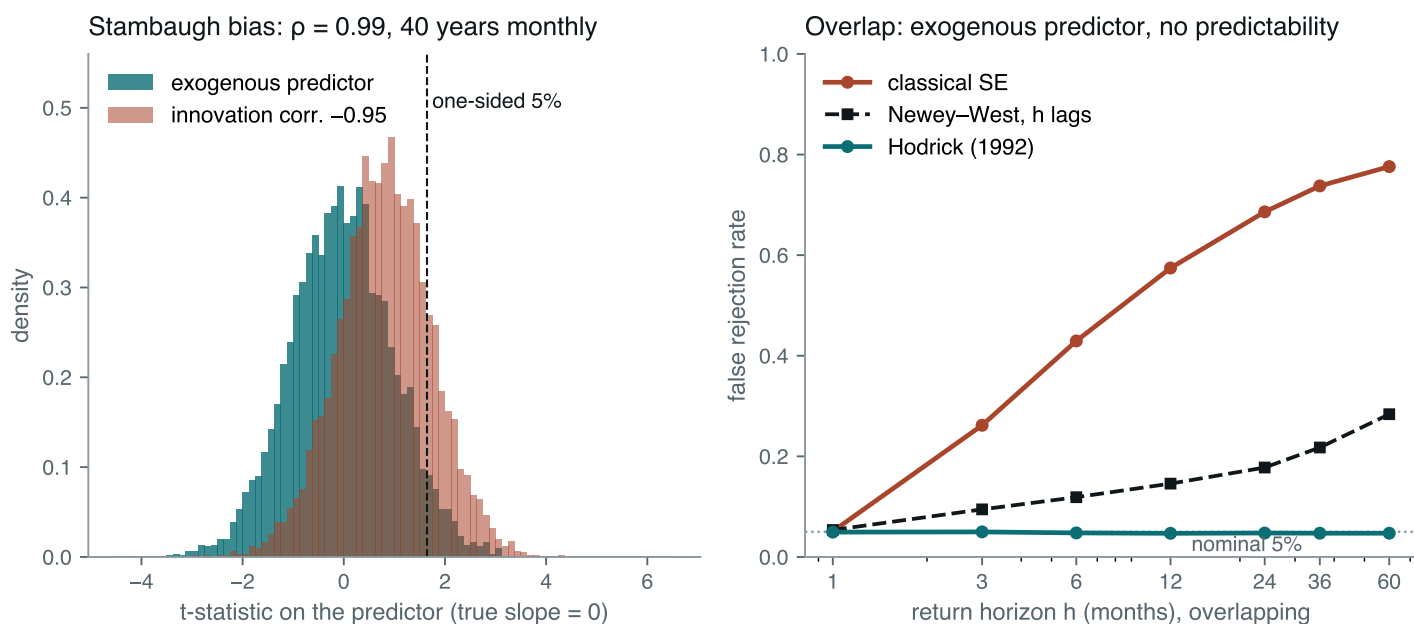
For valuation ratios, the two innovations are strongly negatively correlated. A positive return shock raises the price, and the higher price mechanically lowers the dividend yield in the same period.

Stambaugh bias. Because x_t is a lagged dependent variable in its own equation, $\hat{\phi}$ is biased downward, by roughly $-(1 + 3\phi)/T$ (§2.3). The return equation's slope inherits that bias through the correlation of the innovations (Stambaugh, 1999):

$$\mathbb{E}[\hat{\beta} - \beta] \approx \frac{\sigma_{uv}}{\sigma_v^2} \mathbb{E}[\hat{\phi} - \phi],$$

where σ_{uv} is the covariance of the innovations and σ_v^2 the variance of the predictor's innovation. With $\sigma_{uv} < 0$ and $\hat{\phi}$ biased down, $\hat{\beta}$ is biased *up*, toward finding that a high dividend yield predicts high returns.

The left panel of the figure simulates 40 years of monthly data with no predictability at all, a predictor with $\phi = 0.99$, and an innovation correlation of -0.95 . The predictor's autoregressive coefficient is underestimated by 0.009 on average, close to the approximation's 0.008. The average estimated slope is about 1.4 standard errors above zero (the approximation gives 1.2), the average t -statistic is 0.82 instead of zero, and a one-sided 5% test for positive predictability rejects 18% of the time. **[Simulated]**



Long-horizon regressions. Regressing returns over h months on today's predictor was once thought to reveal predictability that short horizons missed, because R^2 values grow with the horizon. Much of that growth is mechanical. The right panel removes the Stambaugh problem by making the predictor exogenous, and keeps only the overlap of \$4.5. With no predictability, the median R^2 at a 60-month horizon is 5.5%, and one simulated sample in ten reports an R^2 above 27%. Classical standard errors reject a true null 57% of the time at 12 months and 78% at 60. Newey-West with h lags rejects 15% and 28%. Hodrick (1992) standard errors, which impose the null of no predictability when estimating the meat, stay at 5%. **[Simulated]** Valkanov (2003) derives the non-standard behaviour of long-horizon regressions formally. Boudoukh, Richardson & Whitelaw (2008) show that estimates at different horizons are so highly correlated that the long-horizon evidence adds little to the short-horizon evidence it is built from.

What to do.

- **Estimate at the shortest sensible horizon**, and treat long-horizon results as a presentation of short-horizon evidence, not as additional evidence.
- **Use inference designed for persistent predictors.** Bias-corrected slopes, Hodrick standard errors, or the IVX approach, which remains valid whether the predictor is stationary, near a unit root, or exactly at one (Kostakis, Magdalinos & Stamatogiannis, 2015).
- **Evaluate out of sample** (§8.8), with the caveat that out-of-sample tests have low power and can miss real but small predictability.
- **Test the joint implications.** If a valuation ratio does not predict cash flows, it must predict returns (§8.8).

9.3 Event studies

The design. An event study measures how security prices respond to news, such as an earnings announcement, a merger, an index inclusion, or a regulatory decision, by comparing returns around the event with what they would normally have been

(Fama, Fisher, Jensen & Roll, 1969; MacKinlay, 1997).

1. Fix the event date and an **event window**, such as the day before to the day after.
2. In an earlier **estimation window**, fit a model of normal returns, typically the market model $r_{it} = a_i + b_i r_{mt} + e_{it}$.
3. Compute each day's **abnormal return**, $AR_{it} = r_{it} - (\hat{a}_i + \hat{b}_i r_{mt})$, sum it over the event window into a **cumulative abnormal return** (CAR), and average across events.

The identifying assumption. Prices incorporate the news within the window, nothing else moves the stock during the window, and the normal-return model is adequate. Over short windows the last condition barely matters: expected daily returns are tiny compared with event-day volatility, so any reasonable model gives nearly the same answer (Brown & Warner, 1985).

The joint-hypothesis problem. Any test of whether prices react “correctly” is jointly a test of a model of expected returns (Fama, 1970). Over a three-day window the model is nearly irrelevant. Over three years it dominates everything.

Failure modes.

- **Event clustering.** When many firms share an event date, as with an industry-wide regulation, their abnormal returns are cross-sectionally correlated and conventional t -statistics are overstated. Form a portfolio of the affected firms and treat it as one observation, or adjust for the correlation (Kolari & Pynnönen, 2010).
- **Event-induced variance.** Volatility rises on event days, which invalidates tests that standardise by estimation-window variance.
- **Leakage and anticipation.** If news leaks before the window, the measured response is too small.
- **Confounding news** inside the window.
- **Long horizons.** Over months or years, small errors in the expected-return model compound, long-horizon returns are skewed, and overlapping events are correlated. [Contested] Barber & Lyon (1997) show that common long-horizon test statistics are misspecified. Fama (1998) argues that long-term return anomalies tend to disappear under reasonable changes in method, and Mitchell & Stafford (2000) show that correcting for the correlation between overlapping events removes much of the evidence for long-run abnormal performance after corporate events. Behavioural-finance researchers dispute how general that conclusion is.

“Event study” means something related but different in the difference-in-differences literature, where it is a DiD with separate coefficients for each period around treatment (§7.5, §12.3).

9.4 Testing asset pricing models

A **factor model** says that the differences in expected returns across assets are explained by their exposure to a few common sources of risk, the **factors**, collected in a vector f_t of factor returns. An asset's **loadings** (its betas) b_i measure how strongly its return moves with each factor, and the model's prediction is that expected excess returns line up with the loadings: assets with more exposure to a priced factor earn more, and nothing else earns anything. Testing a factor model means testing that prediction.

Time-series tests. When the factors are themselves traded portfolio returns, regress each test asset's excess return on the factor returns, $r_{it}^e = \alpha_i + b_i' f_t + e_{it}$. The model predicts that every α_i is zero. The Gibbons, Ross & Shanken (1989) test checks them jointly, and has a clean economic reading: it asks whether adding the test assets to the factors would significantly raise the highest attainable Sharpe ratio.

Cross-sectional tests: Fama–MacBeth. The two-pass procedure of Fama & MacBeth (1973) first estimates each asset's factor loadings from time-series regressions. Then, at every date t , it regresses the cross-section of returns on the estimated loadings (or on firm characteristics) to get a slope $\hat{\lambda}_t$, the period's reward per unit of exposure. The estimate is the time-series average of $\hat{\lambda}_t$, and its standard error is the standard deviation of $\hat{\lambda}_t$ divided by $\sqrt{T}$.

The standard error's logic is worth understanding, because it defines what the procedure does and does not fix. **Each date contributes one observation, so correlation across assets within a date is handled automatically. Correlation of $\hat{\lambda}_t$ over time is not.** With persistent characteristics, the slopes are serially correlated, and the time series of $\hat{\lambda}_t$ needs a Newey–West standard error. The procedure also does nothing about firm-level persistence in the residuals (§9.5).

Errors in variables. The loadings used in the second pass are estimates, so the second-pass slopes suffer from the attenuation of §6.4 and their standard errors are too small. Shanken (1992) derived the correction to the standard errors. Grouping stocks into portfolios reduces the noise in the loadings, but it also discards cross-sectional variation, which is why Ang, Liu & Schwarz (2020) argue for individual stocks with an appropriate correction, and Jegadeesh, Noh, Pukthuanthong, Roll & Wang (2019) use loadings estimated in separate subsamples as instruments.

Two traps specific to these tests.

- **Useless factors look priced.** [Fact] Kan & Zhang (1999) show that a factor unrelated to returns can appear significantly priced in two-pass regressions, and that the problem gets *worse* with longer samples. It is the weak-instrument problem of §7.3 in asset-pricing form: a factor with near-zero loadings leaves the second pass dividing by nearly nothing. Tests robust to this exist (Kleibergen, 2009).
- **The test assets set a low bar.** Portfolios sorted on size and book-to-market have such a strong factor structure that almost any candidate factor correlated with the size and value factors fits their average returns with a high cross-sectional R^2 . Lewellen, Nagel & Shanken (2010) recommend adding test portfolios unrelated to the sorting characteristics, such as industries, and reporting a GLS R^2 , which is tied to the efficiency of the model's factor-mimicking portfolio and is a stricter hurdle.

The SDF approach. Any linear factor model can be written as a stochastic discount factor that is linear in the factors and estimated by GMM from $\mathbb{E}[M_{t+1}R_{t+1}^e] = 0$ (§5.3). The Hansen & Jagannathan (1997) distance then measures how far a misspecified model is from pricing the assets correctly, in units an investor can interpret.

9.5 Panel standard errors in finance

Finance panels have two dependence dimensions. A **firm effect** is persistence within a firm over time, for example in leverage or in a firm's residual returns. A **time effect** is correlation across firms at a date, as with market-wide shocks. The two call for different corrections, and the literature long used the wrong one for its setting. [Fact] Surveying 207 panel-data papers in the three leading finance journals, Petersen (2009) found that 42% did not adjust standard errors for dependence in the residuals. He also found that the common practice in asset pricing, Fama–MacBeth, corrects for time effects and not firm effects, while White standard errors correct for neither.

Dependence in the residuals	Use
Time effect only (cross-sectional correlation at each date)	Fama–MacBeth, or cluster by date, given many dates
Firm effect only (persistence within firms)	Cluster by firm
Both	Two-way clustering; or date fixed effects plus firm clustering, if the time effect is common to all firms
Time effects that persist across dates	Driscoll–Kraay standard errors (Driscoll & Kraay, 1998), or Newey–West on the Fama–MacBeth slopes

The limits of §4.4 still apply. Two-way clustering with ten years of annual data has ten time clusters, which is not enough.

9.6 Multiple testing, the factor zoo, and replication

Finance has run the multiple-testing experiment of §4.8 at an industrial scale. Several hundred variables have been published as predictors of the cross-section of stock returns, most of them tested on the same US databases. How many are real is one of the field's live disputes, and the positions are worth setting side by side. [Contested]

- **Raise the bar.** Harvey, Liu & Zhu (2016) argue that, given the search, a new factor should clear $t > 3.0$.
- **Most anomalies fail.** Hou, Xue & Zhang (2020) re-examine 452 anomalies with NYSE breakpoints and value-weighted returns, which limit the influence of microcaps. 65% fail to reach $|t| \geq 1.96$, and 82% fail a multiple-testing hurdle of 2.78.
- **Most anomalies replicate.** Jensen, Kelly & Pedersen (2023) build 153 factors in 93 countries and, using a Bayesian model that accounts for multiple testing, find that the large majority replicate, against the 35% implied by Hou, Xue & Zhang's criteria. The factors cluster into 13 themes and work out of sample and across countries. Chen & Zimmermann (2022) reproduce nearly all of several hundred published predictors using the original papers' own methods, and publish the code.
- **The case for higher hurdles is hard to identify.** Chen (2025) argues that because results that failed the old hurdle were never published, the data cannot pin down how much higher the hurdle should be, whereas false-discovery rates for published findings can be estimated well, and are modest.

My reading is that these conflict less than their titles suggest. Hou, Xue & Zhang ask whether an anomaly survives an implementation that downweights microcaps, which is closer to asking whether it is economically important. Jensen, Kelly & Pedersen, and Chen & Zimmermann, ask whether the original statistical finding reproduces. Both camps find that many anomalies are concentrated in small, illiquid stocks and have weakened since publication. The practical implication is the same from either side: a new predictor should be tested value-weighted, net of costs, out of sample, and against the existing factor themes before it is believed.

Data snooping in trading rules is the same problem one level up. Sullivan, Timmermann & White (1999) applied White's reality check (§4.8) to a universe of several thousand technical trading rules on a century of daily Dow Jones data, adjusting for the full search that produced the best rule. In the ten years that followed their original sample, the best rules were not profitable. Earlier, Lo & MacKinlay (1990) had shown how forming test portfolios on characteristics that were chosen *because* they are known to be related to returns biases asset-pricing tests toward rejection.

9.7 Backtests are econometrics

A backtest estimates the mean return of a strategy. The mean is the hardest parameter in finance to estimate (§2.5). The estimate comes from dependent data (§4.5), after a search over configurations (§4.8), frequently with information that was not available at the time (§6.4), and net of costs that were assumed rather than measured. Every failure in this document can appear in a backtest, and none of them announces itself in the output.

Failure	Mechanism	Fix
Look-ahead in the data	Restated financials, revised macroeconomic data, today's index constituents	Point-in-time data with realistic reporting lags
Look-ahead in the method	Full-sample normalisation, z-scores, or hyperparameters tuned on the test period	Strict walk-forward estimation of <i>everything</i>
Survivorship	Delisted securities, or their final returns, missing from the data	Include delistings and delisting returns
Best of many configurations	The reported strategy is the maximum of a noisy search	Holdout data; deflated Sharpe ratio; reality check (§4.8)
Overlapping positions	Holding periods longer than the rebalancing interval	Hodrick or HAC standard errors; the T/h rule (§4.5)
Microcap dominance	Equal weights on stocks that cannot be traded in size	Value weights, NYSE breakpoints, liquidity filters
Assumed costs	Spreads and impact omitted or understated	Cost models calibrated to trade size and capacity
Too short a sample	A Sharpe ratio of 0.5 needs about 16 years to reach $t = 2$	Compute the power first (§4.7)

Two numbers from earlier sections belong next to every backtest. If a strategy's true Sharpe ratio is 0.3 and it is tested for ten years, a backtest that clears $t = 2$ reports a Sharpe ratio of about 0.8 on average (§4.7). And the average published anomaly lost more than half its return after publication (§9.1). The expected out-of-sample performance of a strategy selected from a search is materially below its backtest, even if nothing was done wrong. Portfolio Construction, Appendix A.45–A.46, gives the standard error of a Sharpe ratio and the deflated Sharpe ratio. Momentum in Financial Markets and Market Regimes and Machine Learning cover walk-forward and purged validation in detail.

§9 Key takeaways

1. Financial data combine small signals, fat tails, cross-sectional and serial dependence, persistent predictors, and decades of collective data mining. Assume every problem in Parts I–IV is present until shown otherwise.
2. Published return predictors lose about a quarter of their return out of sample and more than half after publication.
3. Persistent predictors whose innovations are correlated with returns bias predictive slopes upward. Long-horizon regressions add little independent evidence and inflate R^2 mechanically.
4. Short-window event studies are robust to the choice of expected-return model; long-window ones are dominated by it. Correct for clustering of events in time.
5. Fama–MacBeth standard errors handle correlation across assets within a date, not persistence over time or within firms. First-pass betas are noisy regressors.
6. Useless factors can look priced in two-pass tests, and size/value-sorted test portfolios set a low bar. Add unrelated test assets.
7. Identify whether residuals have firm effects, time effects, or both before choosing a standard error; many published papers did not.

8. The replication debate turns partly on definitions. Either way, test a new predictor value-weighted, net of costs, out of sample, and against existing factor themes.
9. A backtest is an econometric estimate of the hardest parameter in finance. Expect its out-of-sample performance to be materially lower, even when nothing was done wrong.

10. Prediction, model selection, and machine learning

Machine learning has changed empirical finance more than any other development of the past fifteen years, and its relationship to econometrics is often misunderstood in both directions. Econometricians sometimes treat it as curve-fitting without inference. Machine-learning practitioners sometimes treat econometrics as a collection of outdated linear models. The accurate picture is that they answer different questions, share most of their mathematics, and are most powerful in combination.

10.1 Two questions: $\hat{Y}$ and $\hat{\beta}$

Breiman (2001) described “two cultures” of statistical modelling: one that assumes a data-generating model and estimates its parameters, and one that treats the mechanism as unknown and optimises predictive accuracy. Shmueli (2010) put the same point as *explaining* versus *predicting*. For applied work, Mullainathan & Spiess (2017) give the sharpest form: some problems are about $\hat{Y}$, a good prediction of the outcome, and some are about $\hat{\beta}$, a good estimate of a parameter.

Machine learning is built for $\hat{Y}$ problems, and it is very good at them. Its parameters are not built to be interpreted. When predictors are correlated, a LASSO fitted on two random halves of the same data can select quite different variables while producing nearly identical predictions. The predictions are stable, and the coefficients that generate them are not. **Feature importances from a predictive model are statements about the model, not about the world.**

Some decisions genuinely need only a prediction. Kleinberg, Ludwig, Mullainathan & Obermeyer (2015) call them “prediction policy problems”: deciding whether to carry an umbrella requires a forecast of rain, not the causal effect of umbrellas on rain. In finance, return forecasting, risk models, and default prediction are $\hat{Y}$ problems. Evaluating whether a hedging programme reduced a firm’s cost of capital, or whether a regulation reduced liquidity, is a $\hat{\beta}$ problem.

The two errors mirror each other. Reading a predictive model’s coefficients causally is the mistake of §1.4. Using a causal estimate to predict outcomes in a different population or period assumes an external validity the design never established.

10.2 Overfitting, measured

For linear regression with k estimated coefficients, n observations, and error variance σ^2 , the expected mean squared error in the estimation sample is $\sigma^2(1 - k/n)$, while the expected error on new observations with the same regressors is $\sigma^2(1 + k/n)$. **[Fact]** The gap, about $2\sigma^2 k/n$, is the **optimism** of in-sample fit, and it is the basis of Mallows’s C_p and of the AIC. With 20 coefficients and 100 observations, in-sample error understates the noise variance by 20% and out-of-sample error exceeds it by 20%.

This arithmetic is decisive for return prediction, and it is worth a short derivation. Suppose the true predictors explain a share ρ^2 of the variance of returns, and they are estimated by OLS with k coefficients. The forecast’s out-of-sample mean squared error is roughly $(1 - \rho^2)(1 + k/n)$ in units of the return variance, while the historical mean’s is roughly $1 + 1/n$. Out-of-sample R^2 compares the two, $R_{OS}^2 = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{null}}$ (§3.8), so it is approximately

$$R_{OS}^2 \approx \rho^2 - \frac{k}{n}(1 - \rho^2) \approx \rho^2 - \frac{k}{n},$$

which is positive only if $n > k/\rho^2$. With a true monthly R^2 of 1% and ten predictors, that is 1,000 months, or 83 years. **An unrestricted regression of monthly returns on ten individually sensible predictors is expected to lose to the historical mean out of sample, even if every predictor is genuinely informative.** This is my calculation, and it is the reason shrinkage, sign restrictions, and forecast combination (§8.8) matter so much in this setting: each one reduces the effective k .

The same logic is at the centre of a live debate. Kelly, Malamud & Zhou (2024) argue that heavily over-parameterised models, with far more parameters than observations and strong ridge shrinkage, can predict returns *better* than parsimonious ones, a “virtue of complexity.” Nagel (2025) shows that the method, trained in short rolling windows, reduces to a volatility-timed momentum strategy. That strategy happened to perform well historically, and the method constructs the same strategy on artificial data in which returns reverse, where it performs poorly. **[Contested]**

10.3 Model selection: AIC, BIC, and cross-validation

Choosing among models trades fit against complexity, and the three standard tools target different things.

Criterion	Formula or procedure	What it targets	Tendency
AIC	$-2 \log L + 2k$	Expected out-of-sample log loss	Picks larger models; good for prediction
BIC	$-2 \log L + k \log n$	The true model, if it is among the candidates	Picks smaller models; consistent for the true order
Cross-validation	Fit on part of the data, score on the held-out part, rotate	Out-of-sample loss, directly	Only as honest as its splits (§10.6)

L is the maximised likelihood, and k the number of estimated parameters.

Inference after selection is invalid. Choosing a model with the data and then reporting that model's standard errors as if it had been fixed in advance gives confidence intervals that are too narrow. The distribution of an estimator after data-driven selection is not normal and cannot be estimated uniformly well (Leeb & Pötscher, 2005). **[Fact]** The most common instance in applied work is choosing control variables by their significance. A control that is insignificant in the outcome equation but strongly correlated with the treatment gets dropped, and the omitted-variable bias of §3.5 returns. §10.5 gives the fix.

10.4 Regularisation

Shrinkage (§5.6) is the main defence against overfitting when the number of candidate predictors is large.

- **Ridge** shrinks all coefficients proportionally. It suits many small, correlated effects, which is the usual shape of return predictability.
- **LASSO** shrinks and sets some coefficients exactly to zero (Tibshirani, 1996). Its selections are unstable when predictors are correlated, and the retained coefficients are biased toward zero.
- **Elastic net** mixes the two penalties.
- **Trees, boosting, and neural networks** allow nonlinearity and interactions, and are regularised by depth, learning rate, early stopping, and dropout rather than by an explicit coefficient penalty.

Every penalty strength or architecture choice is a hyperparameter, tuned by cross-validation. The benchmark evidence for return prediction is Gu, Kelly & Xiu (2020). With more than 900 candidate predictors, ordinary least squares on all of them had a negative out-of-sample R^2 . Penalised and dimension-reduction linear models recovered small positive values, and tree ensembles and neural networks did best, with the best neural network reaching a monthly out-of-sample R^2 of about 0.4% for individual stocks. By the arithmetic of §3.8 those are economically meaningful numbers, and by the arithmetic of §10.2 they are about what careful regularisation should be expected to deliver.

10.5 Machine learning for causal parameters

The most important recent development at the boundary is the use of machine learning *inside* causal estimators. The problem it solves is common. You want the effect θ of a treatment D on Y , and there are many potential confounders X , more than a regression can handle without overfitting, or entering in unknown nonlinear ways.

Why naive approaches fail. Fit a LASSO of Y on D and X , and the penalty drops controls that are weakly related to Y even when they are strongly related to D . That is omitted-variable bias created by the regularisation itself.

Post-double selection. Belloni, Chernozhukov & Hansen (2014) select controls that predict Y , separately select controls that predict D , and regress Y on D and the *union*. A confounder strongly related to either side is kept, and inference on θ is valid under approximate sparsity.

Double/debiased machine learning (DML). Chernozhukov et al. (2018) generalise the idea. Take the partially linear model $Y = \theta D + g(X) + u$ and $D = \pi(X) + v$, where $\pi(X)$ is a flexible first stage, the nonlinear counterpart of the coefficient π of §7.3. Write $\ell(X) \equiv \mathbb{E}[Y \mid X] = \theta \pi(X) + g(X)$ for the best prediction of Y from X alone. It is not the same function as g , because X predicts Y partly through D .

1. Predict Y from X with any machine-learning method to get $\hat{\ell}(X)$, and predict D from X the same way to get $\hat{\pi}(X)$.
2. Regress the residual $Y - \hat{\ell}(X)$ on the residual $D - \hat{\pi}(X)$. The slope is $\hat{\theta}$.

This is Frisch–Waugh–Lovell (§3.4) with flexible first stages. Two ingredients make it work:

- **Neyman orthogonality.** The moment condition $\mathbb{E}[(Y - \ell(X) - \theta(D - \pi(X)))(D - \pi(X))] = 0$ is insensitive, to first order, to small errors in the two prediction functions. The regularisation bias of the machine learning step therefore does not pass through to $\hat{\theta}$.

- **Cross-fitting.** The prediction functions are estimated on folds of the data that are not used to compute the residuals, so their overfitting does not contaminate the second step.

The result is a $\sqrt{n}$ -consistent, approximately normal $\hat{\theta}$ with an ordinary sandwich standard error. It is the master form of §1.5 with a carefully chosen ψ . For effects that vary with characteristics, **causal forests** estimate the conditional average treatment effect with valid confidence intervals (Wager & Athey, 2018). Athey & Imbens (2019) survey the toolkit.

What machine learning does not do: it does not create identification. DML still requires unconfoundedness given X (§6.6). It makes adjusting for many controls feasible without bias; it does not make the controls sufficient. Machine learning is an estimator, not a research design.

10.6 Validation with dependent data

Cross-validation estimates out-of-sample loss honestly only if the held-out data are genuinely unseen. Random K -fold cross-validation assumes the observations are exchangeable. With time series, overlapping labels, or assets that share shocks, that assumption fails, and information leaks from the training folds into the test folds.

- **Serial dependence.** A test observation's neighbours in time sit in the training set and carry information about it.
- **Overlapping labels.** A 20-day forward return on Monday and one on Tuesday share 19 days. With one in training and one in testing, the model is partly scored on the data it learned from.
- **Cross-sectional dependence.** Stocks on the same date share market and industry shocks, so splitting assets rather than dates leaks the date's shock.

The remedies follow from the leak:

- **Walk-forward (forward-chaining) validation.** Train only on the past, and test on the future.
- **Purging and embargo.** Remove training observations whose label windows overlap the test period, and a buffer after it (López de Prado, 2018; Market Regimes and Machine Learning §10.3 and Appendix A.33).
- **Group by date** when splitting panels.
- **Nested validation.** Tune hyperparameters inside the training folds only, and keep a final holdout that is used once.

One more trap sits on top of all of these. **Hyperparameter search is multiple testing.** The cross-validated loss of the best of 200 configurations is an optimistic estimate of that configuration's out-of-sample loss, for exactly the reason in §4.8. The holdout exists to measure how optimistic.

§10 Key takeaways

1. Machine learning answers $\hat{Y}$ questions and econometric designs answer $\hat{\beta}$ questions. Feature importances describe a model, not the world.
2. In-sample fit overstates out-of-sample fit by about $2\sigma^2 k/n$. For return prediction, an unrestricted regression needs roughly k/ρ^2 observations just to beat the historical mean.
3. AIC targets prediction, BIC targets the true model, and cross-validation targets out-of-sample loss directly. Standard errors after data-driven model selection are invalid.
4. Shrinkage is the main defence against overfitting. Realistic out-of-sample monthly R^2 values for individual stocks are fractions of a percent, and are economically meaningful.
5. Selecting controls by significance recreates omitted-variable bias. Use double selection or double machine learning, whose orthogonal moments and cross-fitting deliver valid inference on a causal parameter.
6. Machine learning is an estimator, not a research design. DML still needs unconfoundedness.
7. Random K -fold cross-validation leaks on time series, overlapping labels, and panels. Walk forward, purge, embargo, group by date, and treat hyperparameter search as multiple testing.

11. The practice of applied econometrics

The previous sections describe methods. This one describes the habits that decide whether methods produce knowledge. Almost none of it is mathematically difficult, and almost all of it is routinely skipped.

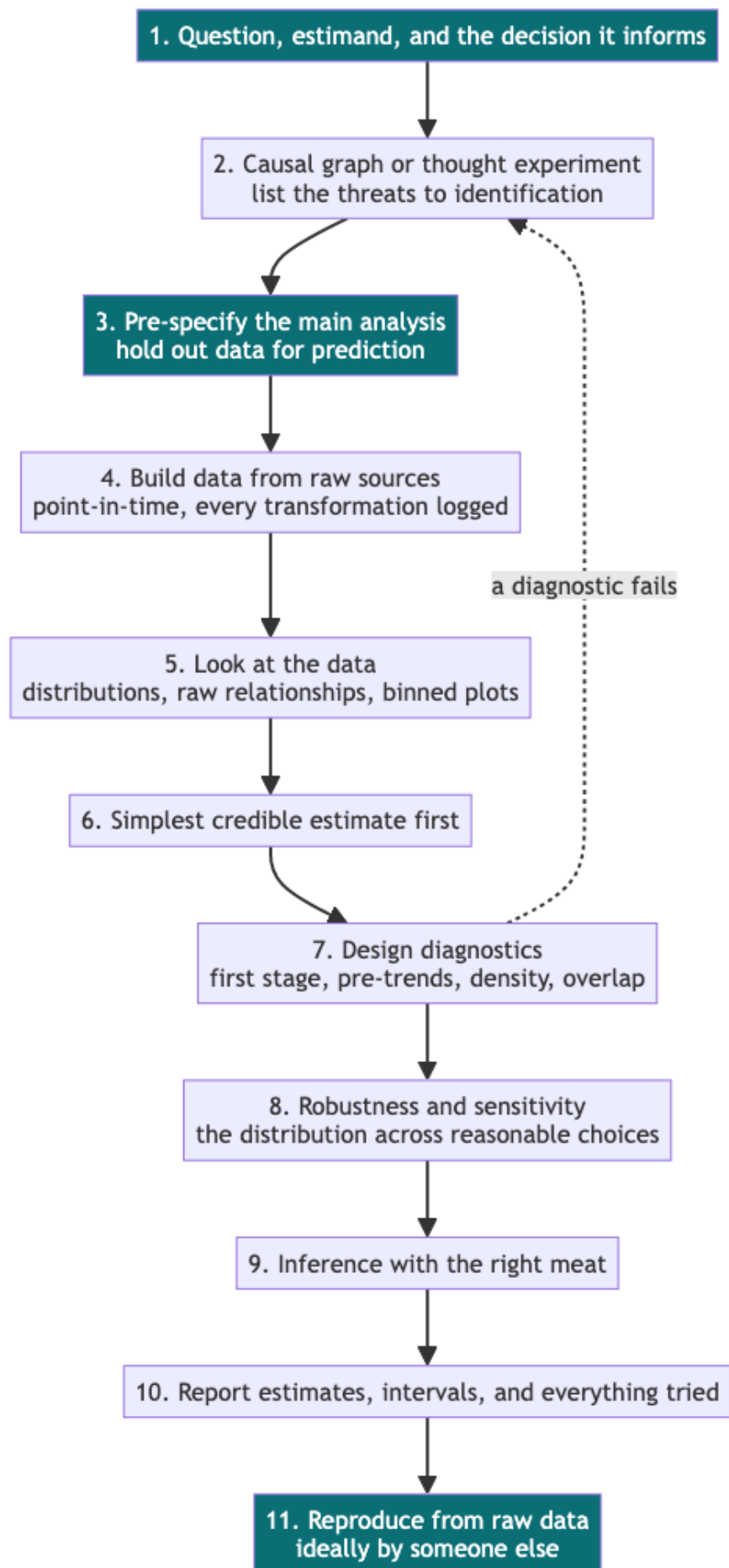
11.1 Where the effort should go

[Practice] Ranked by how much they typically determine whether a result is right:

1. **The question and the estimand.** What decision will the answer inform, and what exact population quantity would inform it? A wrong estimand makes everything downstream irrelevant.
2. **The data.** How each variable was measured, when it was known, and who is missing. Data errors are the most common cause of wrong results, and the least discussed.
3. **The identification strategy**, or for a prediction problem, the validation design. This is where the orthogonality condition is won or lost.
4. **The specification:** functional form, controls, sample definition.
5. **Inference:** the meat.
6. **The estimator.**
7. **Software and computation.**

Effort is usually allocated in close to the reverse order, because the bottom of the list is where the interesting tools are and the top is where the tedious work is. The allocation that works is the one on the list, and early stages are not preliminaries to the analysis: they are most of it.

11.2 A workflow



Two steps deserve emphasis because they are the ones most often skipped. **Step 3** separates choices from the data that judge them (§4.8). A dated note written before the outcome data are examined is enough for most internal research. **Step 5** catches problems no estimator can: Anscombe (1973) constructed four datasets with identical means, variances, correlations, and regression lines that look completely different when plotted. Binned scatter plots are the standard way to look at a conditional mean with many observations, and they have their own pitfalls, notably residualising the controls in a way that distorts the picture (Cattaneo, Crump, Farrell & Feng, 2024).

The dotted arrow matters as much as the solid ones. When a design diagnostic fails, the right response is to return to the identification argument, not to search for a specification in which the diagnostic passes.

11.3 Data work

[Practice] Most wrong results in empirical finance are data errors, not econometric errors. The recurring ones:

- **Joins.** Merges that silently duplicate or drop rows. Count rows before and after every join, and check key uniqueness on both sides.
- **Units and scales.** Returns in percent in one table and decimals in another; thousands versus millions; currencies; share counts before and after splits.
- **Timing.** When was each value *knowable*? Accounting data are released months after the fiscal period they describe, which is why the standard academic convention matches fiscal-year accounting data with returns only from the middle of the following calendar year. Macroeconomic series are revised after release (§8.8). Timestamps need time zones, and closing prices from different exchanges are not simultaneous.
- **Corporate actions and delistings.** Splits, spin-offs, ticker changes, and the final returns of delisted securities.
- **Outliers.** Winsorising at the 1st and 99th percentiles is a choice of estimand as well as a cleaning step: a winsorised mean is a different quantity. In finance the extreme observations are often the economically important ones, so every outlier rule should be justified by the question and reported.
- **Missing data.** Dropping incomplete observations is harmless only if missingness is unrelated to the outcome. In finance it rarely is. Firms stop reporting on the way to distress, and funds stop reporting after bad performance.

11.4 Specification and robustness

Functional form. A linear specification estimates the best linear approximation to the conditional mean (§3.2), which is fine when it is close and misleading when it is not. Binned plots, splines, and tests for omitted nonlinearity, such as the RESET test (Ramsey, 1969), show which case you are in.

Diagnostics are also forks. Running a diagnostic, respecifying when it fails, and reporting the specification that passes is a specification search with extra steps. Decide in advance which diagnostic results would change the specification, and report them whatever they show.

Robustness tables versus specification curves. A conventional robustness table shows a handful of alternatives chosen by the author, and so shows the tail of a distribution the author selected. A **specification curve** estimates the model under every combination of reasonable analytical choices, sorts the estimates, and displays which choices produced which estimates (Simonsohn, Simmons & Nelson, 2020). The idea goes back to Leamer's "extreme bounds" (Leamer, 1983), which asked whether a coefficient keeps its sign across all specifications, and to Sala-i-Martin (1997), who argued for looking at the whole distribution of estimates across millions of control sets rather than the extremes. The claim a specification curve supports is honest: "across the choices a reasonable analyst might make, the estimate lies in this range."

Sensitivity to what was not observed belongs in the same section of a report as sensitivity to what was (§6.7).

11.5 Reporting

- **Report magnitudes with intervals**, in units a decision-maker understands (§4.10).
- **Show the design, not just the estimate.** For IV, the first stage and the reduced form. For difference-in-differences, the event-study plot. For regression discontinuity, the binned outcome around the cutoff and the density test. For synthetic control, the pre-treatment fit and the placebo distribution. For selection on observables, the overlap of propensity scores.
- **Report what was tried.** State the number of specifications estimated, the choices made after seeing the data, and which results were confirmatory and which exploratory.
- **Make it reproducible.** Leading economics journals now require data and code sufficient to reproduce the results, and internal research should meet the same standard.

The reason for these norms is measurable. **[Fact]** Brodeur, Lé, Sangnier & Zylberberg (2016) found that the distribution of test statistics published in top economics journals has a shortage just below conventional significance thresholds and an excess just above them. Brodeur, Cook & Heyes (2020) examined thousands of hypothesis tests in 25 leading journals and found that the problem varies by method: it is substantial in IV and difference-in-differences studies, and much smaller in randomised trials and regression discontinuity designs. The designs that leave more discretion to the researcher show more signs of it.

[Practice] In a quantitative research group, the equivalent of a pre-registration is a **research log** that records every backtest run, not just the ones that were kept. It turns “how many things did we try?” from a guess into a number, which is what the deflated Sharpe ratio and every multiple-testing correction need as an input (§4.8).

11.6 A catalogue of failure modes

The table collects the failure modes of the whole document in one place, with a pointer to where each is explained.

Failure	Mechanism	How it shows up, and how to check	Fix
Confounding	A common cause of regressor and outcome	Coefficient moves when controls are added; check stability relative to R^2	Research design; sensitivity analysis (§3.5, §6.7)
Bad control	Conditioning on a mediator or collider	Estimate changes sharply, even in sign, when a control is added; check timing	Drop post-treatment and collider controls (§6.5)
Survivorship and selection	The sample depends on the outcome	Results differ between the survivor sample and the full universe	Point-in-time universe with delistings (§6.4)
Measurement error	Noise in the regressor	Attenuation; within estimates far below pooled	A second measure as an instrument (§6.4)
Simultaneity	The outcome moves the regressor	Implausible signs; check timing and economics	IV; high-frequency identification (§6.4, §8.5)
Weak instruments	A tiny first stage	Small robust first-stage F ; IV close to OLS	Anderson–Rubin inference; better instruments (§7.3)
Invalid exclusion	The instrument affects the outcome directly	Placebo outcomes respond	Map every channel; bound the violation (§7.3)
Staggered two-way fixed effects	Already-treated units used as controls	Negative weights; odd event-study patterns	Clean-comparison DiD estimators (§7.5)
Non-parallel trends	Selection into treatment timing	Pre-trends; dips before treatment	Sensitivity bounds; better comparison groups (§7.5)
Ignored clustering	Correlation within groups	Classical standard errors far below clustered	Cluster at the assignment level (§4.4)
Few treated clusters	The meat comes from a handful of sums	Huge t -statistics from one or two treated groups	Wild cluster bootstrap; randomisation inference (§4.4)
Overlapping observations	Shared returns across rows	t -statistics and R^2 grow with horizon	Hodrick standard errors; count T/h (§4.5, §9.2)
Persistent predictors	Stambaugh bias	High persistence and correlated innovations	Bias correction; IVX (§9.2)
Spurious regression	Independent trending series	High R^2 ; non-stationary residuals	Difference, or test for cointegration (§8.3)
Structural breaks	Parameters change	Drifting rolling estimates; out-of-sample collapse	Break tests; window averaging (§8.7)
Overfitting	Too many parameters for the signal	In-sample far above out-of-sample	Shrinkage; fewer parameters (§10.2)
Validation leakage	Dependence across folds	Cross-validated score far above live performance	Walk-forward, purging, grouping by date (§10.6)
Specification search	The garden of forking paths	Test statistics bunched just above thresholds	Pre-specification; holdout; specification curves (§4.8, §11.4)
Winner's curse	Low power plus a significance filter	Effects shrink when replicated	Compute power first; shrink estimates (§4.7)
Look-ahead	Information unavailable at the time	Backtest far above live	Point-in-time data (§9.7)
Manual two-stage estimation	Second-stage residuals use fitted values	Wrong standard errors	Estimate 2SLS as one step (§5.2)
Generated regressors	A regressor estimated in a first step	Standard errors too small	Bootstrap both steps together (Pagan, 1984)
Mechanical correlation	Regressor and outcome share a component	Significance with no economic mechanism	Rebuild the variables (§7.3)
Regression to the mean	Selecting units for extreme values	Extremes “revert” with or without treatment	A comparison group selected the same way
Aggregation reversal	Pooling groups of different composition	Pooled and within-group slopes differ in sign	Within-group estimates; fixed effects (§3.4)

Extrapolation	Estimates where the data are sparse	Poor overlap; estimates driven by functional form	Trim; state the local estimand (§6.6, §7.6)
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§11 Key takeaways

1. Effort should go to the question, the data, and identification before specification, inference, and estimators, which is close to the reverse of how it is usually spent.
2. Most wrong results in empirical finance are data errors: joins, units, timing, corporate actions, delistings.
3. Plot the data before estimating anything. Summary statistics can be identical for data that look nothing alike.
4. Diagnostics that trigger respecification are forks in the path. Decide in advance what they would change.
5. Report a specification curve rather than a curated robustness table: the claim is the range of estimates across reasonable choices.
6. Show the design (first stage, event-study plot, density test, pre-period fit, overlap), not just the estimate.
7. Published test statistics bunch just above significance thresholds, most of all in the designs with the most researcher discretion. Keep a log of everything tried.

Part VI — Synthesis

12. Taxonomy and equivalences

12.1 The design space as a product

A flat list of methods (“OLS, IV, GMM, fixed effects, DiD, clustered errors, Newey–West, LASSO...”) hides the fact that these are answers to different questions. Every empirical study is a *coordinate* in a product space of four independent choices, and most methods in the literature are names for a choice on one axis only.

Axis	What it decides	Choices
1. Estimand	Which question is answered	A projection (description); a forecast under a loss; ATE, ATT, LATE, or CATE; a structural parameter
2. Identifying variation	What is assumed uncorrelated with the error (the orthogonality condition)	Randomisation; controls; an instrument; within-unit changes; treatment timing; a cutoff; donor weights; a model
3. Estimator	How the sample moment condition is formed and solved	OLS; IV and GMM; maximum and quasi-maximum likelihood; quantile and robust M-estimators; shrinkage; machine-learning-assisted (DML)
4. Meat	What is independent of what	Independent; heteroskedastic; clustered one-way or two-way; HAC; Driscoll–Kraay; Fama–MacBeth; bootstrap by the independent unit

A prediction problem replaces axis 2 with a **validation design**: in-sample fit, random cross-validation, walk-forward, or purged and grouped splits (§10.6).

Seen this way, “difference-in-differences with clustered standard errors” is a coordinate (ATT, timing, OLS with fixed effects, clustered by state), and so is “a Fama–MacBeth regression” (a projection or risk premium, characteristics as regressors, period-by-period OLS, a meat that treats each date as one observation). The empty cells are informative too. There is no reason, for instance, not to pair a regression-discontinuity design with a wild cluster bootstrap when the running variable is assigned at the cluster level.

Axis 2: where the identifying variation comes from

By construction:
randomised experiment

By a rule or by nature:
instrument; cutoff

By timing: difference-in-differences;
synthetic control; event study

By comparison within units:
fixed effects

By assumption: selection on
observables; structural model

Axis 4: the meat, which every design needs

independent

heteroskedastic

clustered, one-way or two-way

serially correlated (HAC)

bootstrap by independent unit

The shading runs from variation that is random by construction to variation that is random only by assumption. The meat is drawn separately because it is not a design: every design needs one, and choosing it is a question about the data rather than about identification.

12.2 Equivalences

Things that look different and are provably the same. The status column says whether the identity is exact, holds only asymptotically, or holds under a stated condition.

Relationship	Status	Where
Estimators		
OLS is the method of moments with $\psi_i = X_i(Y_i - X_i'\beta)$	Exact	§3.3
OLS is the Gaussian maximum likelihood estimator of β	Exact	§5.4
Just-identified IV, 2SLS, and GMM with any weighting matrix coincide	Exact	§5.2–§5.3
2SLS is GMM with $W = (\mathbf{Z}'\mathbf{Z}/n)^{-1}$, and efficient GMM under homoskedasticity	Exact; efficiency conditional	§5.3
Simple IV is the reduced-form slope divided by the first-stage slope	Exact	§5.2
A multiple-regression coefficient is the slope on the residualised regressor (Frisch–Waugh–Lovell)	Exact	§3.4
Unit dummies and within-unit demeaning give the same coefficients	Exact	§7.4
First differences equal the within estimator when $T = 2$	Exact	§7.4
Random effects plus unit means of the regressors reproduce fixed effects (Mundlak)	Exact	§7.4
The 2×2 difference-in-differences equals the interaction coefficient in two-way fixed effects	Exact	§7.5
Staggered two-way fixed effects is a weighted average of 2×2 comparisons, with weights that can be negative	Exact decomposition	§7.5
Fuzzy regression discontinuity is IV with the cutoff indicator as the instrument	Exact, locally	§7.6
Ridge is a posterior mean under a normal prior; LASSO a posterior mode under a Laplace prior	Exact	§5.6
Double machine learning in the partially linear model is Frisch–Waugh–Lovell with flexible first stages	Exact in form	§10.5
Interpretation		
Omitted variable bias is a special case of the endogeneity term $\mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i u_i]$	Exact	§3.2, §3.5
Selection bias in potential outcomes is omitted variable bias in regression language	Conceptually identical	§6.2
Heckman's selection correction adds the omitted inverse-Mills-ratio term, $\varphi(\cdot)/\Phi(\cdot)$	Exact under normality	§6.4
OLS with a binary treatment and saturated controls is a variance-weighted average of cell effects	Exact	§3.4
IV with heterogeneous effects estimates the effect for compliers (LATE)	Exact under monotonicity	§7.3
Inference		

Relationship	Status	Where
Classical standard errors are the sandwich with a homoskedastic, independent meat	Exact	§4.2
Cluster-robust with one observation per cluster is heteroskedasticity-robust	Exact up to a finite-sample factor	§4.4
Two-way clustered variance is firm-clustered plus time-clustered minus heteroskedasticity-robust	Exact	§4.4
Newey–West is a Bartlett-kernel estimate of the long-run variance (the spectral density at frequency zero)	Exact	§4.5
HC3 is close to the jackknife variance	Approximate	§4.3
Wald, likelihood ratio, and Lagrange multiplier tests agree	Asymptotic, under correct specification	§4.6
The F -statistic for one restriction is the squared t -statistic	Exact	§4.6
R^2 is the squared correlation between Y and $\hat{Y}$ (with an intercept)	Exact	§3.8
Time series and asset pricing		
A stationary AR(1) is an MA(∞) with coefficients ϕ^j	Exact	§8.2
Every covariance-stationary process is deterministic plus MA(∞) (Wold)	Exact	§8.2
GARCH(1,1) is ARMA(1,1) in squared shocks	Exact	§8.6
Cointegration is equivalent to an error-correction representation	Exact (Granger representation)	§8.4
Local projections and VARs estimate the same impulse responses	Exact in population	§8.5
The variance ratio is one plus a weighted sum of autocorrelations	Exact	§8.1
The GRS test asks whether test assets raise the factors' maximum Sharpe ratio	Exact	§9.4
A linear factor model is a stochastic discount factor linear in the factors	Exact	§9.4

12.3 Same name, different thing

Term	One meaning	Another meaning
Fixed effects	Unit-specific intercepts, allowed to correlate with the regressors (econometrics)	Non-random effects, as opposed to random effects in a mixed model (statistics)
Random effects	Intercepts assumed uncorrelated with the regressors (panel econometrics)	Hierarchical coefficients with a distribution, estimated with shrinkage (mixed models)
Robust	Standard errors valid under heteroskedasticity or dependence	Estimators insensitive to outliers; or a robustness check across specifications
Event study	Abnormal returns around news (finance, §9.3)	A difference-in-differences with coefficients for each period around treatment (§7.5)
Causality	The effect of an intervention (§6.1)	Granger causality: the past of one series predicts another (§8.5)
Structural	A model of economic behaviour (§7.8)	A VAR with identified shocks (§8.5); a structural <i>break</i> is a parameter change (§8.7)

Term	One meaning	Another meaning
Exogenous	Uncorrelated with the error (§3.2)	Strictly exogenous: uncorrelated at every date (§7.4); determined outside the economic model
Bias	Finite-sample bias, $\mathbb{E}[\hat{\theta}] - \theta_0$	Inconsistency, a probability limit away from the truth; the two are routinely conflated
Efficient	Minimum-variance among a class of estimators	Informationally efficient markets
Significance	Statistical: signal relative to noise	Economic: size relative to what matters (§4.10)
R^2	In-sample fit (§3.8)	Out-of-sample R^2 against a benchmark (§8.8); GLS R^2 in asset-pricing tests (§9.4)
Heterogeneity	Treatment effects that differ across units (§6.1)	Unobserved unit characteristics, as absorbed by fixed effects (§7.4)
Identification	The estimand is a function of the observable distribution (§1.3)	Uniqueness of parameters in a statistical model; recovering structural shocks in a VAR
Factor	A priced risk factor	A statistical factor from principal components; a categorical variable in software
Instrument	A variable satisfying relevance and exclusion (§7.3)	A tradable security

12.4 Which choice matters most

The product space has four axes, and they do not matter equally.

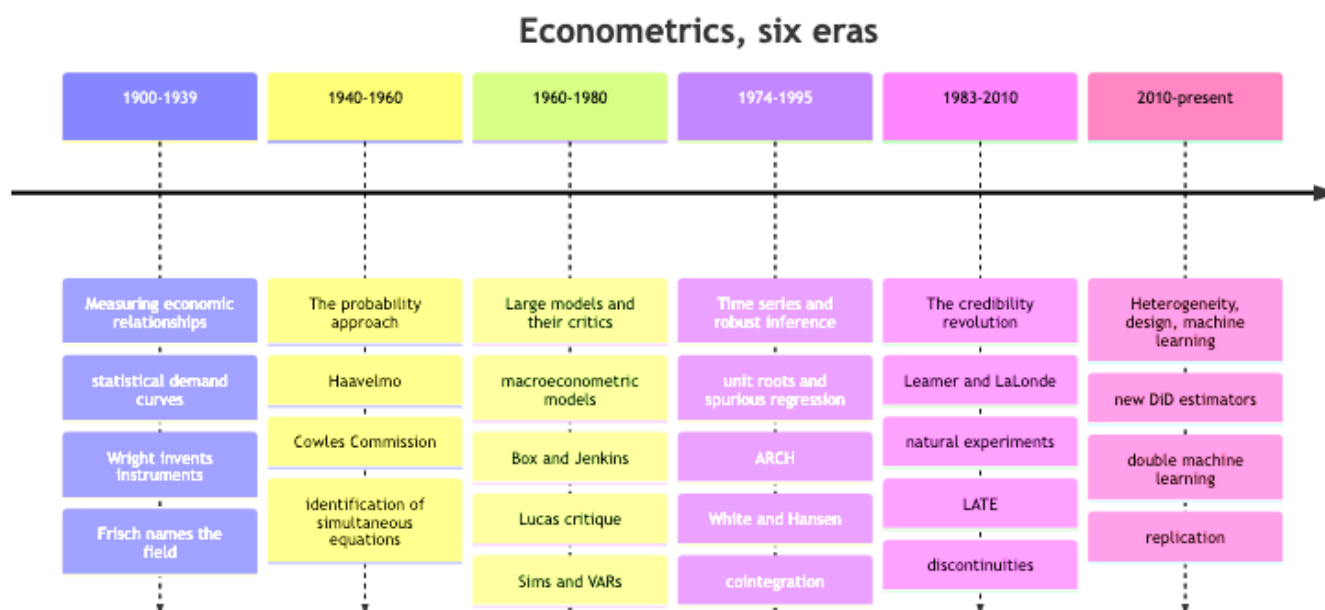
- **For causal questions, the identifying variation dominates.** LaLonde’s comparison (§1.1) and its successors show estimates on the same data varying by more than the effect itself when the source of variation changes. Under a *fixed* identifying assumption with good overlap, different estimators tend to agree with each other (Imbens & Xu, 2025), and whether they agree with the truth depends on the assumption, not on which estimator was used. **[Practice]**
- **For inference, the meat dominates.** Choosing the wrong dependence structure changes standard errors by factors of two to seven in realistic settings (§4.4, §4.5, §9.2). No estimator choice has an effect of that size on precision.
- **For prediction, the effective number of parameters relative to the signal dominates.** With low signal-to-noise, the k/n term of §10.2 decides whether a model beats a naive benchmark, which is why shrinkage and combination beat cleverness.
- **The estimator is rarely the binding choice**, except in the weak-identification cases (weak instruments, many moments, near-unit roots) where the default estimator’s approximations break down.

§12 Key takeaways

1. Every empirical study is a coordinate on four axes: estimand, identifying variation, estimator, and meat. Most method names describe one axis only.
2. The meat is not a design. Every design needs one, and choosing it is a question about the dependence in the data.
3. Many apparently different methods are algebraically identical: IV is a ratio of regressions, fixed effects are demeaning, DML is Frisch–Waugh–Lovell, fuzzy RD is IV, ridge is a Bayesian posterior.
4. Many shared names hide different objects: “event study,” “robust,” “fixed effects,” “structural,” and “causality” each mean at least two things.
5. Identifying variation dominates causal conclusions, the meat dominates inference, and the parameter count relative to signal dominates prediction. The estimator is rarely the binding choice.

13. How the field evolved

The history is worth a few thousand words because it explains why the field's concerns are what they are: why “identification” is the first word an econometrician reaches for, why standard errors are clustered by reflex, and why a finance seminar asks about data mining before it asks about the model. Each era is summarised in four parts: its contribution, what changed as a result, its limitations, and what survives.



13.1 Era I — Measuring economic relationships (1900–1939)

Thesis: statistics met economic theory, and immediately ran into the identification problem.

Contribution. Economists began estimating demand curves from market data. Working (1927) showed that a scatter of prices and quantities traces a demand curve only if supply shifts while demand stays put, which stated the identification problem two decades before it was named. The following year, Philip Wright's book on tariffs on animal and vegetable oils included an appendix that derived instrumental variables as a solution (Stock & Trebbi, 2003). Yule (1926) documented nonsense correlations between time series, and Frisch & Waugh (1933) proved the partialling-out theorem. Ragnar Frisch coined “econometrics” and helped found the Econometric Society and its journal. The first volume of *Econometrica* carried Cowles (1933), which examined the recommendations of 45 professional forecasting agencies and found no evidence that they did better than chance: the first test of market efficiency, and the first finance paper in the field's flagship journal.

What changed. Economic relationships became quantities to be estimated rather than asserted.

Limitations. There was no probability model for the errors, identification was handled case by case, and the most ambitious project, Tinbergen's statistical business-cycle models, drew a critique from Keynes (1939) that the era could not answer: the method needs a complete list of causes, relationships that stay stable, and a way to tell causation from correlation.

Lasting influence. Instrumental variables, partialling out, spurious correlation, and the identification problem are all still core, which makes this the era with the highest ratio of lasting ideas to papers.

13.2 Era II — The probability approach (1940–1960)

Thesis: economic data are draws from a probability model, and identification is a property of that model.

Contribution. Haavelmo (1943) showed that least squares applied to one equation of a simultaneous system is inconsistent. Haavelmo (1944) argued that economic data should be treated as realisations from a joint distribution generated by a system of behavioural equations, the framing of §1.2. The Cowles Commission turned this into a programme. Koopmans (1949) formalised identification, with the rank and order conditions for simultaneous equations, and Anderson and Rubin developed estimation and inference for them (Anderson & Rubin, 1949).

What changed. Econometrics became a probabilistic discipline with its own theory, and “identification” acquired a precise meaning.

Limitations. The identifying restrictions came from economic theory and were taken on trust. Computing power and data were too limited to test much. Koopmans (1947) attacked the empirical business-cycle measurement of the NBER as “measurement without theory,” opening an argument between theory-led and data-led empirical work that is still running (§7.8).

Lasting influence. The structural tradition and the language of identification. Frisch and Tinbergen shared the first economics Nobel prize in 1969, and Haavelmo received his in 1989.

13.3 Era III — Large models and their critics (1960–1980)

Thesis: large structural models promised policy evaluation, and lost on forecasting to simple time-series methods and on theory to rational expectations.

Contribution. Macroeconometric models with hundreds of equations were built for forecasting and policy analysis. Box and Jenkins’s 1970 book systematised ARIMA modelling. Nelson (1972) found that simple ARIMA forecasts contained information that the forecasts of the large FRB–MIT–Penn model missed, and were competitive with them. Granger (1969) gave a testable definition of predictive causality. Lucas (1976) argued that the models’ estimated relationships would shift whenever policy changed, because they summarised decisions made under the old policy. Sims (1980) called the models’ identifying restrictions “incredible” and proposed vector autoregressions instead.

What changed. Confidence in large structural models collapsed. Forecasting and policy analysis came apart, and time-series methods rose.

Limitations. VARs bought credibility by giving up economic content, and the Lucas critique applies with equal force to reduced-form relationships.

Lasting influence. Structural VARs remain the workhorse of empirical macroeconomics (§8.5), and the Lucas critique is the standard warning about relationships that break when deployed. Sargent and Sims shared the 2011 Nobel prize.

13.4 Era IV — Time series and robust inference (1974–1995)

Thesis: the classical assumptions were systematically wrong for economic data, and the field built tools that did not need them.

Contribution. Nonstationarity: Granger & Newbold (1974) on spurious regression, Dickey & Fuller (1979) on unit-root testing, Engle & Granger (1987) on cointegration. Volatility: Engle (1982) on ARCH. Robust inference: White (1980) on heteroskedasticity, White (1982) on misspecified likelihoods, Hansen (1982) on GMM, and Newey & West (1987) on autocorrelation. Microeconometrics: Heckman (1979) on selection, discrete-choice models, and panel methods. Finance built its own econometrics in parallel: event studies (Fama, Fisher, Jensen & Roll, 1969), two-pass tests (Fama & MacBeth, 1973), Euler-equation GMM (Hansen & Singleton, 1982), and portfolio efficiency tests (Gibbons, Ross & Shanken, 1989), gathered in Campbell, Lo and MacKinlay’s 1997 textbook.

What changed. Standard errors robust to heteroskedasticity and dependence became the default, and nonstationarity and time-varying volatility were modelled rather than ignored.

Limitations. Robust inference did nothing about identification, and the specification searches that Leamer was about to attack went on unchanged.

Lasting influence. Close to all of the standard-error practice of §4, GARCH, GMM in asset pricing, and cointegration. Heckman and McFadden shared the 2000 Nobel prize, Engle and Granger the 2003 prize, and Hansen shared the 2013 prize with Fama and Shiller.

13.5 Era V — The credibility revolution (1983–2010)

Thesis: credible causal claims come from research design, not from modelling.

Contribution. Leamer (1983) argued that specification searches made most econometric conclusions fragile, and LaLonde (1986) showed non-experimental estimators failing against an experimental benchmark (§1.1). The response was to look for variation that was random for reasons outside the model: a draft lottery (Angrist, 1990), quarter of birth (Angrist & Krueger, 1991), a minimum-wage increase in one state but not its neighbour (Card & Krueger, 1994). The theory followed: LATE (Imbens & Angrist, 1994), weak instruments (Bound, Jaeger & Baker, 1995; Staiger & Stock, 1997), regression discontinuity

(Hahn, Todd & van der Klaauw, 2001), synthetic control (Abadie & Gardeazabal, 2003), and clustering in difference-in-differences (Bertrand, Duflo & Mullainathan, 2004). Angrist and Pischke's *Mostly Harmless Econometrics* (2009) and their 2010 essay named the movement.

What changed. “What is your identification strategy?” became the first question in applied microeconomics and much of empirical corporate finance, and randomised field experiments spread through development economics. Banerjee, Duflo, and Kremer shared the 2019 Nobel prize for experiments, and Card, Angrist, and Imbens the 2021 prize for natural experiments.

Limitations. Local estimands and limited external validity, argued forcefully by the structural side (Keane, 2010; Deaton, 2010; Heckman, 2010). And, as the later evidence on p-hacking shows (§11.5), designs constrain researcher discretion without removing it.

Lasting influence. It is the dominant paradigm of applied microeconomics and of causal work in corporate finance.

13.6 Era VI — Heterogeneity, design, and machine learning (2010–present)

Thesis: the credibility revolution's tools were re-examined under heterogeneous effects, and machine learning was absorbed as an estimator.

Contribution. Difference-in-differences with staggered timing was shown to fail under heterogeneous effects, and new estimators followed within a few years (§7.5). Weak-instrument practice was reassessed (Andrews, Stock & Sun, 2019; Lee, McCrary, Moreira & Porter, 2022), and the foundations of clustering were reworked (Abadie, Athey, Imbens & Wooldridge, 2023). Machine learning entered causal inference with valid inference (Belloni, Chernozhukov & Hansen, 2014; Chernozhukov et al., 2018; Wager & Athey, 2018). Meta-research documented low power and selective reporting (Ioannidis, Stanley & Doucouliagos, 2017; Brodeur, Cook & Heyes, 2020), and journals began requiring replication packages. In finance, the factor zoo and its replication debate (§9.6), machine-learning asset pricing (Gu, Kelly & Xiu, 2020), and demand-system asset pricing (Kojien & Yogo, 2019) reshaped the empirical agenda.

What changed. Two-way fixed effects stopped being the default for staggered designs, weak-instrument-robust inference became expected, machine-learning estimators gained valid confidence intervals, and replication became a condition of publication.

Limitations. Methods now turn over faster than practice can absorb them, each new estimator carries assumptions of its own, and none of them substitutes for a credible source of variation.

Lasting influence. It is too early to judge. My expectation is that heterogeneity-robust difference-in-differences and double machine learning will be routine within a decade, and that the argument over whether complexity is a virtue in return prediction will still be open.

§13 Key takeaways

1. The identification problem was stated in 1927 and instrumental variables were invented in 1928, to estimate supply and demand. The field's central concern is as old as the field.
2. Haavelmo's probability approach made identification a property of a model, and the structural tradition followed from it.
3. Large structural models lost on forecasting to simple time-series models and on theory to the Lucas critique; VARs replaced them in macroeconomics.
4. The 1980s made inference robust to heteroskedasticity, autocorrelation, and misspecification, and did nothing for identification.
5. The credibility revolution moved the source of credibility from the model to the research design, at the cost of local estimands.
6. The current era is re-examining those designs under heterogeneous effects, absorbing machine learning as an estimator, and confronting evidence of low power and selective reporting.

14. Synthesis

14.1 The framework on one page

Everything in this document is three questions, one equation, and two ways to fail.

The three questions (§1.3). What is the estimand? Is it identified? How precisely is it estimated? They are answered in that order, errors flow downward, and prediction problems replace the second question with an honest validation design.

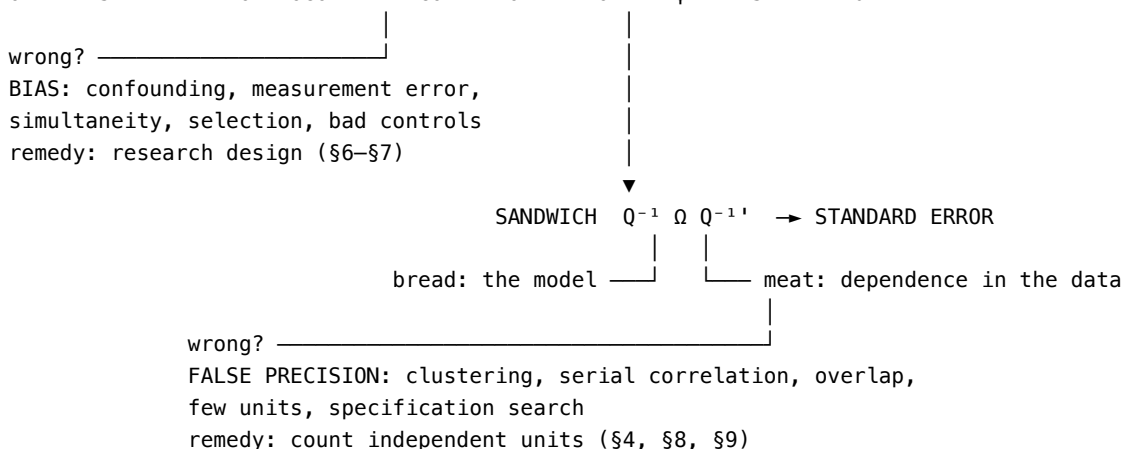
The one equation (§1.5–§1.6). Every estimator solves a sample moment condition, and every standard error is the sandwich around it:

$$\frac{1}{n} \sum_{i=1}^n \psi_i(\hat{\theta}) = 0, \quad \text{Avar}(\sqrt{n}(\hat{\theta} - \theta_0)) = Q^{-1} \Omega Q^{-1'}.$$

The choice of ψ states the orthogonality condition, which is the identifying assumption. The bread Q comes from the model. The meat Ω comes from the dependence in the data.

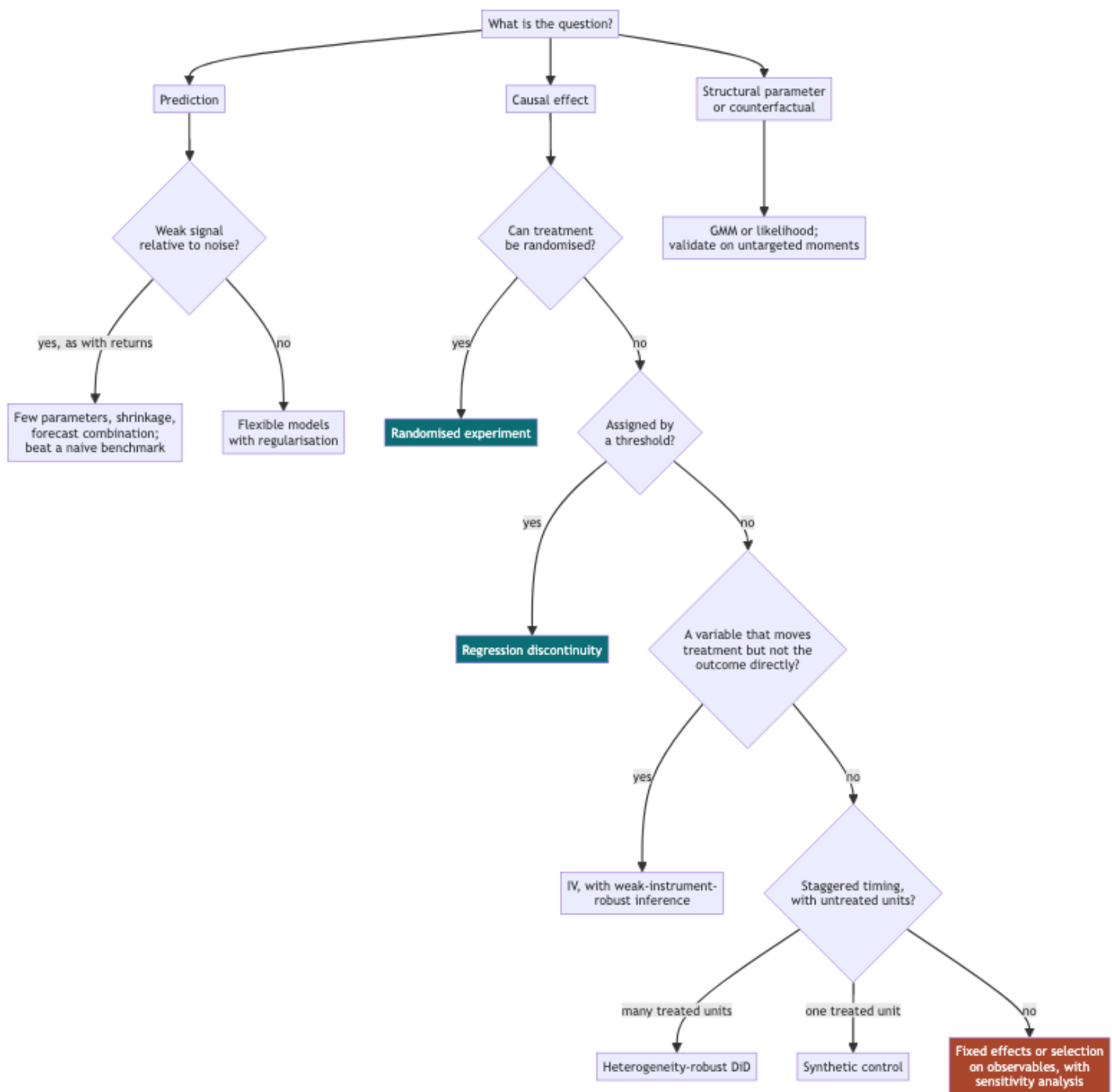
The two ways to fail. Bias comes from a false orthogonality condition: confounding, measurement error, simultaneity, selection, or a bad control (§6). False precision comes from a wrong meat: ignored clustering, serial correlation, overlap, too few independent units, or a specification search that the standard error never saw (§4). The research designs of §7 exist to make the first failure less likely. The standard-error machinery of §4, §8, and §9 exists to prevent the second. Shrinkage and validation (§5.6, §10) trade a little of the first for a large reduction in variance when the goal is prediction.

QUESTION → ESTIMAND → ORTHOGONALITY CONDITION → MOMENT ψ → ESTIMATE $\hat{\theta}$



14.2 A decision tree

The tree branches on the questions that actually change what you should do, and ends in named methods. The steps listed after it apply whichever branch you take.



Whatever the branch: choose the meat by the dependence in the data, count the independent units, check when every variable was known, and report intervals together with everything that was tried.

The shaded warning on the selection-on-observables leaf is deliberate. It is where analyses end up when no better source of variation exists, and it is where a sensitivity analysis (§6.7) is not optional.

14.3 A roadmap for building an empirical research capability

For a team or an individual building the capability to produce empirical results that survive contact with new data, in stages, each with a gate that has to be passed before moving on. The first two stages are infrastructure, not the interesting part, and skipping them is the usual reason later stages fail.

Stage	Build	Gate before moving on
0. Data	Point-in-time data with reporting lags; delistings and corporate actions; tests on every join	Reproduce a published series, such as a standard factor's returns, to within a small tolerance
1. Inference core	Regression with heteroskedasticity-robust, clustered, two-way, HAC, Driscoll-Kraay, and Fama-MacBeth errors; a simulation harness	Recover known answers in simulations with known dependence, including this document's clustering and overlap simulations
2. Designs	IV with Anderson-Rubin inference; heterogeneity-robust DiD; RD with robust bias-corrected intervals	Replicate one published design end to end
3. Time series and forecasting	Unit-root and cointegration tools; GARCH and HAR; walk-forward evaluation; Diebold-Mariano and Clark-West	Reproduce a known <i>negative</i> out-of-sample result
4. Prediction	Shrinkage and tree models; purged, grouped validation; a research log with trial counts	Every reported result has a computable number of trials behind it
5. Causal machine learning and structure	Double machine learning; causal forests; a small structural model estimated by simulated moments	DML matches OLS in simple simulated designs and removes regularisation bias in hard ones

A reading order that follows the same arc: *Mastering 'Metrics* for intuition; *Mostly Harmless Econometrics*, *The Mixtape*, or *The Effect* for designs; Wooldridge for panels, or Hansen for theory; Hamilton for time series; Campbell, Lo and MacKinlay, and Cochrane for finance; and the *Applied Causal Inference* book for machine learning. §15.1 has the details.

14.4 Things I would tell someone starting today

1. **Write down the estimand before touching the data.** If you cannot say which population quantity would answer the question, no estimate will.
2. **Ask what experiment the study approximates**, and where the approximation breaks.
3. **Count the independent units** before trusting a standard error: clusters, treated clusters, non-overlapping periods. The rows do not matter.
4. **Plot the raw data and the reduced form** before estimating anything more elaborate.
5. **Controls are not free.** Draw the causal graph, and never control for anything the treatment could have caused.
6. **Treat any t -statistic below about 3 that came out of a search as a hypothesis**, not a finding.
7. **Compute the minimum detectable effect first.** If the plausible effect is smaller, a significant result is more likely an exaggeration than a discovery.
8. **A backtest estimates a mean from dependent data after a search.** Expect it to shrink out of sample, even when nothing was done wrong.
9. **Use the simplest estimator the design allows**, and spend the effort on data and diagnostics.
10. **Simulate your method on data where you know the answer**, including data where the answer is zero. Every figure in this document was produced that way, and each one found a failure that the textbook formula hides.
11. **Keep a log of everything you run.** It is the only way to know how many things you tried.
12. **For prediction, the only sample that counts is the one the model never saw. For causation, the nearest equivalent is a placebo test.**

14.5 What is and is not known

Settled. The mathematics of estimation and inference, meaning moment conditions, the sandwich, and the asymptotic theory behind both, is settled, and so are the mechanisms of failure: omitted variables, measurement error, simultaneity, selection, colliders, spurious regression, and specification search. So is the need for inference that respects the dependence in the data. For causal questions it is also settled that the source of variation matters far more than the choice of estimator.

Contested. How much weight local, design-based estimates deserve relative to structural models (§7.8). How many published return predictors reflect real, exploitable effects (§9.6). Whether heavily over-parameterised models genuinely improve return prediction (§10.2). How far pre-analysis plans and multiple-testing hurdles should go, given their cost in missed discoveries (§4.8, §9.6).

Open. Three problems seem to me both important and unsolved. The first is **inference with few independent episodes**. Finance has one history, a handful of crises, and a few regimes, and no standard-error formula creates independence the data lack. The second is **external validity**: when an effect estimated in one market, period, or population transfers to another. Most of the field's tools say nothing about it. The third is **interference and reflexivity**. In markets, one participant's treatment changes the prices faced by the others, violating the no-interference assumption at the heart of causal inference, and publishing a finding changes the process that produced it.

Econometrics cannot make data answer a question the data do not contain the answer to. What it can do, and what this document has tried to make usable, is say precisely which questions a given dataset can answer, under which assumptions, and with how much confidence.

15. References

Grouped by kind, because the kinds are read differently. Each entry says why it matters. Where a free copy exists it is the link, usually a working-paper or author version, which can differ from the published version in detail. Entries linked to a publisher page may be paywalled.

15.1 Textbooks and guides

- **Angrist & Pischke (2015)**. *Mastering 'Metrics: The Path from Cause to Effect*. Princeton University Press. — The best first book: five research designs taught through examples, with almost no matrix algebra.
- **Angrist & Pischke (2009)**. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press. — The design-based toolkit at graduate level, and the source of the regression-as-projection presentation in §3.
- **Huntington-Klein (2021)**. *The Effect: An Introduction to Research Design and Causality*. CRC Press; free online. — The clearest book on *thinking* about identification.
- **Cunningham (2021)**. *Causal Inference: The Mixtape*. Yale University Press; free online. — Designs with code in several languages; strong on causal graphs and difference-in-differences.
- **Hernán & Robins (2020)**. *Causal Inference: What If*. Chapman & Hall/CRC; free online. — The epidemiology tradition's treatment of potential outcomes and graphs, excellent on time-varying treatments.
- **Imbens & Rubin (2015)**. *Causal Inference for Statistics, Social, and Biomedical Sciences*. Cambridge University Press. — The potential-outcomes framework in full, especially experiments and selection on observables.
- **Pearl (2009)**. *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge University Press. — The graphical approach to causation, including the back-door criterion of §6.5.
- **Wooldridge (2010)**. *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. MIT Press. — The reference for microeconometrics and panels.
- **Hansen (2022)**. *Econometrics*. Princeton University Press. — A modern, rigorous graduate text built on projection and the sandwich; the textbook closest to this document's organisation.
- **Hayashi (2000)**. *Econometrics*. Princeton University Press. — GMM as the organising principle of the whole field, with finance applications.
- **Hamilton (1994)**. *Time Series Analysis*. Princeton University Press. — Still the standard graduate time-series reference.
- **Kilian & Lütkepohl (2017)**. *Structural Vector Autoregressive Analysis*. Cambridge University Press. — Identification in VARs, comprehensively.
- **Elliott & Timmermann (2016)**. *Economic Forecasting*. Princeton University Press. — Loss functions, evaluation, combination, and breaks, rigorously.
- **Hyndman & Athanasopoulos (2021)**. *Forecasting: Principles and Practice*, 3rd ed. OTexts; free online. — Practical forecasting with code.
- **Campbell, Lo & MacKinlay (1997)**. *The Econometrics of Financial Markets*. Princeton University Press. — The classic financial econometrics text: event studies, predictability, asset-pricing tests.
- **Cochrane (2005)**. *Asset Pricing*, revised ed. Princeton University Press. — Asset pricing as GMM on the stochastic discount factor.
- **Cattaneo, Idrobo & Titiunik (2020)**. *A Practical Introduction to Regression Discontinuity Designs: Foundations*. Cambridge University Press. — The implementation guide to regression discontinuity, by the authors of its standard tools.
- **Chernozhukov, Hansen, Kallus, Spindler & Syrgkanis (2024)**. *Applied Causal Inference Powered by ML and AI*. Free online. — Double machine learning and its relatives, from the people who developed them.
- **Hastie, Tibshirani & Friedman (2009)**. *The Elements of Statistical Learning*, 2nd ed. Springer; free online. — The machine-learning reference an econometrician can read.
- **Gelman, Hill & Vehtari (2020)**. *Regression and Other Stories*. Cambridge University Press. — Applied regression with an emphasis on simulation and model checking.
- **López de Prado (2018)**. *Advances in Financial Machine Learning*. Wiley. — Purged and embargoed validation and backtest overfitting; opinionated, and useful for the validation chapters alone.

15.2 Classic books

No stable free copies are linked for these.

- **Wright, P. G. (1928)**. *The Tariff on Animal and Vegetable Oils*. Macmillan. — Its Appendix B derives instrumental variables; see Stock & Trebbi (2003) for the story.
- **Tinbergen, J. (1939)**. *Statistical Testing of Business-Cycle Theories*. League of Nations. — The first macroeconomic models, and the target of Keynes (1939).

- **Box, G. E. P. & Jenkins, G. M. (1970).** *Time Series Analysis: Forecasting and Control*. Holden-Day. — The ARIMA methodology of §8.2.
- **Goldberger, A. S. (1991).** *A Course in Econometrics*. Harvard University Press. — Econometrics built from the conditional expectation function, and the origin of “micronumerosity.”

15.3 Foundations and history

- **Yule (1926).** “Why Do We Sometimes Get Nonsense-Correlations between Time-Series?” *Journal of the Royal Statistical Society* 89(1), 1–63. — Spurious correlation between trending series, sixty years before the theory.
- **Working (1927).** “What Do Statistical ‘Demand Curves’ Show?” *Quarterly Journal of Economics* 41(2), 212–235. — The identification problem, stated before it had a name.
- **Frisch & Waugh (1933).** “Partial Time Regressions as Compared with Individual Trends.” *Econometrica* 1(4), 387–401. — The partialling-out theorem of §3.4.
- **Cowles (1933).** “Can Stock Market Forecasters Forecast?” *Econometrica* 1(3), 309–324. — 45 forecasting services, no evidence of skill: the first test of market efficiency.
- **Keynes (1939).** “Professor Tinbergen’s Method.” *Economic Journal* 49(195), 558–577. — The first great critique of econometric modelling; most of its objections are still live.
- **Haavelmo (1943).** “The Statistical Implications of a System of Simultaneous Equations.” *Econometrica* 11(1), 1–12. — Least squares is inconsistent in a simultaneous system.
- **Haavelmo (1944).** “The Probability Approach in Econometrics.” *Econometrica* 12 (Supplement). — The founding statement of econometrics as inference about a probability model (§1.2).
- **Koopmans (1947).** “Measurement Without Theory.” *Review of Economics and Statistics* 29(3), 161–172. — The theory-versus-data argument, in its first form.
- **Koopmans (1949).** “Identification Problems in Economic Model Construction.” *Econometrica* 17(2), 125–144. — Identification made formal.
- **Anderson & Rubin (1949).** “Estimation of the Parameters of a Single Equation in a Complete System of Stochastic Equations.” *Annals of Mathematical Statistics* 20(1), 46–63. — LIML, and the test that remains valid with weak instruments (§7.3).
- **Nelson (1972).** “The Prediction Performance of the FRB-MIT-PENN Model of the U.S. Economy.” *American Economic Review* 62(5), 902–917. — Simple ARIMA forecasts carried information a large structural model missed.
- **Lucas (1976).** “Econometric Policy Evaluation: A Critique.” *Carnegie-Rochester Conference Series on Public Policy* 1, 19–46. — Estimated relationships shift when policy changes.
- **Sims (1980).** “Macroeconomics and Reality.” *Econometrica* 48(1), 1–48. — “Incredible” identifying restrictions, and the VAR alternative.
- **Stock & Trebbi (2003).** “Retrospectives: Who Invented Instrumental Variable Regression?” *Journal of Economic Perspectives* 17(3), 177–194. — Stylometric detective work on the origin of IV, and a clear introduction to it.
- **Angrist & Pischke (2010).** “The Credibility Revolution in Empirical Economics: How Better Research Design Is Taking the Con out of Econometrics.” *Journal of Economic Perspectives* 24(2), 3–30. — The manifesto of design-based empirical work.
- **Imbens & Xu (2025).** “Comparing Experimental and Nonexperimental Methods: What Lessons Have We Learned Four Decades after LaLonde (1986)?” *Journal of Economic Perspectives* 39(4), 173–201. — What modern methods do and do not fix, on LaLonde’s own data.

15.4 Regression, inference, and standard errors

- **Kendall (1954).** “Note on Bias in the Estimation of Autocorrelation.” *Biometrika* 41(3/4), 403–404. — The finite-sample bias of the autoregressive coefficient behind Stambaugh bias.
- **Anscombe (1973).** “Graphs in Statistical Analysis.” *The American Statistician* 27(1), 17–21. — Four datasets, one regression line: plot the data.
- **Ramsey (1969).** “Tests for Specification Errors in Classical Linear Least-Squares Regression Analysis.” *Journal of the Royal Statistical Society, Series B* 31(2), 350–371. — The RESET test for omitted nonlinearity.
- **White (1980).** “A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity.” *Econometrica* 48(4), 817–838. — Robust standard errors; retired the homoskedasticity assumption.
- **Pagan (1984).** “Econometric Issues in the Analysis of Regressions with Generated Regressors.” *International Economic Review* 25(1), 221–247. — Why regressors estimated in a first step need corrected standard errors.
- **Mackinnon & White (1985).** “Some Heteroskedasticity-Consistent Covariance Matrix Estimators with Improved Finite Sample Properties.” *Journal of Econometrics* 29(3), 305–325. — HC2 and HC3.
- **Kiefer & Vogelsang (2005).** “A New Asymptotic Theory for Heteroskedasticity-Autocorrelation Robust Tests.” *Econometric Theory* 21(6). [[paywalled]] — Fixed- b asymptotics for HAC tests.

- **Newey & West (1987).** “A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix.” *Econometrica* 55(3), 703–708. — The HAC estimator of §4.5.
- **Moulton (1990).** “An Illustration of a Pitfall in Estimating the Effects of Aggregate Variables on Micro Units.” *Review of Economics and Statistics* 72(2), 334–338. — The Moulton factor.
- **Angrist (1998).** “Estimating the Labor Market Impact of Voluntary Military Service Using Social Security Data on Military Applicants.” *Econometrica* 66(2), 249–288. — Regression with a binary treatment as a variance-weighted average of cell effects.
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- **Cameron, Gelbach & Miller (2011).** “Robust Inference with Multiway Clustering.” *Journal of Business and Economic Statistics* 29(2), 238–249. — Two-way clustering.
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15.5 Estimation: moments, likelihood, and shrinkage

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- **Bollerslev & Wooldridge (1992)**. “Quasi-Maximum Likelihood Estimation and Inference in Dynamic Models with Time-Varying Covariances.” *Econometric Reviews* 11(2), 143–172. — Why Gaussian GARCH estimates survive fat tails.
- **Altonji & Segal (1996)**. “Small-Sample Bias in GMM Estimation of Covariance Structures.” *Journal of Business and Economic Statistics* 14(3), 353–366. — Efficient GMM’s finite-sample problem.
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15.6 Causal inference: potential outcomes, graphs, and selection

- **Rubin (1974)**. “Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies.” *Journal of Educational Psychology* 66(5), 688–701. — Potential outcomes.
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- **Rosenbaum & Rubin (1983)**. “The Central Role of the Propensity Score in Observational Studies for Causal Effects.” *Biometrika* 70(1), 41–55. — The propensity score.
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- **Imbens (2020)**. “Potential Outcome and Directed Acyclic Graph Approaches to Causality: Relevance for Empirical Practice in Economics.” *Journal of Economic Literature* 58(4), 1129–1179. — Graphs and potential outcomes, com-

pared.

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15.7 Research designs

Instrumental variables

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15.10 Prediction and machine learning

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15.11 Critiques, debates, and meta-research

A bibliography that lists only a field’s methods is advertising. These are the papers that challenge how the methods are used.

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- **Sala-i-Martin (1997)**. “I Just Ran Two Million Regressions.” *American Economic Review* 87(2), 178–183. — The distribution of estimates across specifications; the link is the working-paper version, “four million.”
- **Gelman & Carlin (2014)**. “Beyond Power Calculations: Assessing Type S (Sign) and Type M (Magnitude) Errors.” *Perspectives on Psychological Science* 9(6), 641–651. — The winner’s curse, quantified.
- **Gelman & Loken (2014)**. “The Statistical Crisis in Science.” *American Scientist* 102(6), 460–465. — The garden of forking paths.
- **Brodeur, Lé, Sangnier & Zylberberg (2016)**. “Star Wars: The Empirics Strike Back.” *American Economic Journal: Applied Economics* 8(1), 1–32. — Test statistics bunch above significance thresholds.
- **Ioannidis, Stanley & Doucouliagos (2017)**. “The Power of Bias in Economics Research.” *Economic Journal* 127(605), F236–F265. — Median power of 18% or less.
- **Karpoff & Wittry (2018)**. “Institutional and Legal Context in Natural Experiments: The Case of State Antitakeover Laws.” *Journal of Finance* 73(2), 657–714. — Context changes the inferences of nine natural-experiment studies.
- **Brodeur, Cook & Heyes (2020)**. “Methods Matter: p-Hacking and Publication Bias in Causal Analysis in Economics.” *American Economic Review* 110(11), 3634–3660. — More in IV and DiD, less in RCTs and RD.
- **Simonsohn, Simmons & Nelson (2020)**. “Specification Curve Analysis.” *Nature Human Behaviour* 4(11), 1208–1214. — Reporting the whole distribution of specifications.
- **Wardlaw (2020)**. “Measuring Mutual Fund Flow Pressure as Shock to Stock Returns.” *Journal of Finance* 75(6), 3221–3243. — An instrument that contained the outcome.
- **Young (2022)**. “Consistency without Inference: Instrumental Variables in Practical Application.” *European Economic Review* 147, 104112. — Published IV inference is fragile.
- **Heath, Ringgenberg, Samadi & Werner (2023)**. “Reusing Natural Experiments.” *Journal of Finance* 78(4), 2329–2364. — Many tests on one experiment need multiple-testing corrections.
- **Mellon (2025)**. “Rain, Rain, Go Away: 194 Potential Exclusion-Restriction Violations for Studies Using Weather as an Instrumental Variable.” *American Journal of Political Science* 69(3), 881–898. — One instrument, hundreds of

channels.

15.12 If you only read ten things

In this order:

1. **Angrist & Pischke (2015)**, *Mastering 'Metrics*, for the intuition behind research designs.
2. **Huntington-Klein (2021)**, *The Effect*, for thinking about identification before estimation.
3. **Leamer (1983)** and **Angrist & Pischke (2010)**, read together, for why research design became the centre of the field.
4. **Cameron & Miller (2015)**, for what standard errors assume and how to cluster.
5. **Cinelli, Forney & Pearl (2024)**, for which controls help and which hurt.
6. **Roth, Sant'Anna, Bilinski & Poe (2023)**, for difference-in-differences as it is now done.
7. **Andrews, Stock & Sun (2019)**, for weak instruments and what to do about them.
8. **Stambaugh (1999)** and **Goyal & Welch (2008)**, for what goes wrong in return-prediction regressions.
9. **Harvey, Liu & Zhu (2016)** and **Jensen, Kelly & Pedersen (2023)**, read together, for the replication debate in finance.
10. **Chernozhukov et al. (2018)**, or the *Applied Causal Inference* book, for machine learning in the service of causal questions.

Appendix A: Concepts and prerequisites

Everything the main text leans on without stopping to explain. The reader this is written for is mathematically comfortable but does not do econometrics for a living: they meet “fixed- b critical values” or “Neyman orthogonality” somewhere in the middle of an argument, want the idea rather than a textbook chapter, and would rather not leave the document to get it.

Entries are ordered by **dependency**, not alphabetically, and grouped into seven parts that are themselves in dependency order, so the appendix reads as a build-up. Each entry gives the idea in words first, then the formal definition, then why it appears in this document, then where to go deeper. Notation follows the main text’s Notation block. Where a standard formula from another literature would collide with it, the entry says so and keeps the other literature’s letters local to that entry.

Nothing here is needed to follow the *argument* of the document. A good deal of it is needed to implement it.

Index. Where each concept first appears:

Concept	First used	Concept	First used
Information sets and martingale differences	§1.4	Local polynomial regression	§7.6
Stationarity and ergodicity	§2.4	Approximate sparsity	§10.5
Convergence, Slutsky, continuous mapping	§1.6	Neyman orthogonality, cross-fitting	§10.5
The chi-squared distribution	§5.3	The Kalman filter	§7.8
Long-run variance, spectral density	§4.5	Granger representation theorem	§8.4
Fixed- b asymptotics	§4.5	Market microstructure noise	§8.6
Kullback–Leibler, pseudo-true values	§5.4	MSE and QLIKE for volatility	§8.6
Fisher information, Cramér–Rao	§5.4	IVX	§9.2
Priors as penalties	§5.6	Excess returns, Sharpe ratio	§2.4
Stein’s paradox, empirical Bayes	§5.6	Risk aversion and the SDF	§1.5
Truncated normal, inverse Mills ratio	§6.4	Portfolio sorts and breakpoints	§1.4
Jackknife and block bootstraps	§4.3	Tangency portfolio, GLS R^2	§9.4
Wild cluster bootstrap	§4.4	The Shanken correction	§9.4
Randomisation inference	§4.4	Driscoll–Kraay standard errors	§1.6
Family-wise error and FDR	§4.8	The deflated Sharpe ratio	§4.8
Reality check, SPA, stepdown	§4.8	Trees, forests, boosting	§10.4
The Hausman test	§7.4	Mallows’s C_p	§10.2
d-separation, back-door criterion	§6.5		
Oster’s δ , robustness value	§6.7		
Weak-instrument tools	§7.3		
Dynamic panel GMM	§7.4		

Part I — Probability and asymptotics. The approximations that every standard error in the document relies on, and the time-series conditions under which they hold.

A.1 Information sets, filtrations, and martingale differences

The idea. A forecast can only use what is known when it is made. An **information set** is the formal name for “everything known at time t ”: past prices, past data releases, the forecaster’s own past forecasts. Information accumulates, so each period’s set contains the previous one. A sequence is a **martingale difference** if nothing in the information set helps predict

its next *value on average*, although it may still help predict its size. That gap between unpredictable levels and predictable magnitudes is the statistical signature of asset returns.

Formally. A **filtration** is an increasing sequence of information sets (σ -algebras) $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \dots$; a process x_t is **adapted** if x_t is known at t . A sequence ε_t with $\mathbb{E}|\varepsilon_t| < \infty$ is a martingale difference sequence if

$$\mathbb{E}[\varepsilon_{t+1} \mid \mathcal{F}_t] = 0.$$

If second moments exist, this implies $\mathbb{E}[\varepsilon_{t+1} h] = 0$ for any h known at t , so the sequence is uncorrelated with every function of the past. It does not imply independence: $\mathbb{E}[\varepsilon_{t+1}^2 \mid \mathcal{F}_t]$ can vary, which is exactly what GARCH models (§8.6).

Why it appears here. The prediction estimand in §1.4 is a conditional expectation given $\mathcal{F}_t$. §2.2 and §8.1 place returns between “uncorrelated” and “independent” using this definition. The Hodrick standard errors of §4.5 and §9.2 impose the null that returns are a martingale difference sequence, and the Euler equations of §5.3 are martingale-difference conditions on pricing errors.

Deeper. Hamilton (1994), chapter 7; section 8 of Stochastic Processes builds filtrations from σ -algebras.

A.2 Stationarity and ergodicity

The idea. With one history, the only way to learn about a process is to average over time. That works only if two things hold. The process must be **stationary**: its statistical behaviour does not change with the calendar, so early and late observations are draws from the same rules. And it must be **ergodic**: a single long path eventually explores the full range of the process’s behaviour, so the time average converges to the average over all possible paths. A process can be stationary without being ergodic. Flip a coin once, then report its outcome every day forever: the series is stationary, but its time average is the one flip, not one-half.

Formally. A process is **strictly stationary** if the joint distribution of $(y_{t_1+h}, \dots, y_{t_k+h})$ does not depend on h . It is **covariance stationary** if $\mathbb{E}[y_t] = \mu$ and $\text{Cov}(y_t, y_{t-j}) = \gamma_j$ do not depend on t . A covariance-stationary process is **ergodic for the mean** if $\bar{y}_T \xrightarrow{p} \mu$; a sufficient condition is absolutely summable autocovariances, $\sum_j |\gamma_j| < \infty$, which is also what makes the long-run variance of A.5 finite.

Why it appears here. §2.4 states that the law of large numbers needs “independence or dependence that fades with distance”, and ergodicity is the precise version. §8.1 makes stationarity the time-series substitute for random sampling, and §8.3 is about what happens when it fails.

Deeper. Hamilton (1994), chapter 3; Hansen (2022), the time-series chapters.

A.3 Convergence in probability and in distribution; Slutsky and continuous mapping

The idea. Asymptotic theory uses two different senses of “settling down”. An estimate **converges in probability** to a number when, in large samples, it is almost certainly close to that number; that is consistency. A statistic **converges in distribution** when the *shape* of its sampling distribution approaches a limit, even though the statistic itself keeps varying from sample to sample; that is what licenses normal critical values. Two workhorse results let you combine the two kinds of limit the way you would combine ordinary numbers.

Formally. $X_n \xrightarrow{p} c$, written $\text{plim } X_n = c$, if $\Pr(|X_n - c| > \epsilon) \rightarrow 0$ for every $\epsilon > 0$. $X_n \xrightarrow{d} X$ if the distribution function of X_n converges to that of X at every point where the latter is continuous. The **continuous mapping theorem**: if g is continuous, $X_n \xrightarrow{p} c$ implies $g(X_n) \xrightarrow{p} g(c)$ and $X_n \xrightarrow{d} X$ implies $g(X_n) \xrightarrow{d} g(X)$. **Slutsky’s theorem**: if $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, then $X_n + Y_n \xrightarrow{d} X + c$, $X_n Y_n \xrightarrow{d} cX$, and $X_n/Y_n \xrightarrow{d} X/c$ for $c \neq 0$.

Why it appears here. The sandwich derivation of §1.6 and §4.2 uses both: the bread converges in probability, the scaled moment sum converges in distribution, and Slutsky combines them. A t -statistic with an *estimated* standard error is asymptotically normal by Slutsky (§2.6). The delta method of §2.6 is the continuous mapping theorem with a first-order expansion.

Deeper. Hansen (2022), the chapters on asymptotic theory; Hayashi (2000), chapter 2.

A.4 The chi-squared distribution and degrees of freedom

The idea. A test of several restrictions at once asks how far the estimates are from the null in several directions, and squared distances add up. If each direction contributes an independent standard normal discrepancy, the sum of the squares has a **chi-squared** distribution. Its **degrees of freedom** count the independent directions in which the null could be violated.

Formally. If $Z_1, \dots, Z_q$ are independent $N(0, 1)$, then $\sum_{j=1}^q Z_j^2 \sim \chi_q^2$, with mean q and variance $2q$. The 5% critical values are 3.84 for $q = 1$ (the square of 1.96), 5.99 for $q = 2$, and 18.31 for $q = 10$. A **Wald statistic** for q linear restrictions $R\theta = r$ is

$$(R\hat{\theta} - r)' [R \hat{V} R']^{-1} (R\hat{\theta} - r) \xrightarrow{d} \chi_q^2$$

under the null, where $\hat{V}$ is the estimated variance of $\hat{\theta}$; it is a squared t -statistic generalised to several dimensions. In the J -test the degrees of freedom are $\ell - k$ because estimating k parameters lets the estimator set k combinations of the ℓ sample moments to zero, leaving only $\ell - k$ that can reveal a violation.

Why it appears here. The J -test of §5.3, the Ljung–Box test of §8.1, the Hausman test (A.17), the Anderson–Rubin test (A.20), and every joint test in §4.6.

Deeper. Hansen (2022), the chapters on hypothesis testing and GMM.

A.5 The long-run variance as a spectral density at frequency zero

The idea. Any stationary series can be decomposed into cycles of different frequencies, fast wiggles and slow swells, and the **spectral density** says how much of the variance sits at each frequency. A long average washes out the fast wiggles and keeps the slow swells. The variance of a long average is therefore governed by the spectral density at the slowest frequency, zero. That is why the “meat” of a time-series standard error and the spectral density at zero are the same object.

Formally. For a covariance-stationary vector series ψ_t with autocovariances Γ_j , the spectral density is $f(\omega) = \frac{1}{2\pi} \sum_{j=-\infty}^{\infty} \Gamma_j e^{-i\omega j}$. At $\omega = 0$,

$$2\pi f(0) = \sum_{j=-\infty}^{\infty} \Gamma_j = \Omega,$$

the long-run variance of §4.5. Kernel estimators such as Newey–West estimate $f(0)$ by a weighted sum of sample autocovariances; the **bandwidth** L controls how many are included, which is equivalent to how wide a band of low frequencies is averaged.

Why it appears here. §4.5 defines the HAC meat as $\sum_j \Gamma_j$, and §12.2 lists Newey–West as a Bartlett-kernel estimate of the spectral density at zero.

Deeper. Hamilton (1994), chapters 6 and 10.

A.6 Fixed- b asymptotics

The idea. Conventional theory for HAC standard errors assumes the bandwidth is a vanishing fraction of the sample, so the estimated long-run variance can be treated as exact. In real samples the bandwidth is a noticeable fraction of T , the variance estimate is itself noisy, and a t -statistic that divides by it has fatter tails than a normal. **Fixed- b** theory holds the ratio $b = L/T$ fixed as the sample grows, derives the distribution the t -statistic actually has, and gives critical values that account for the noise in the denominator.

Formally. With b fixed, the HAC t -statistic converges to a nonstandard distribution that depends on the kernel and on b , and whose critical values rise with b . For the equal-weighted cosine estimator with ν terms, the limit is exactly Student- t with ν degrees of freedom, which makes the method easy to use. Larger bandwidths reduce the bias of the variance estimate, and fixed- b critical values pay for the extra noise, which is the trade Lazarus, Lewis, Stock & Watson (2018) optimise.

Why it appears here. The recommended procedure of §4.5: a Newey–West bandwidth of $1.3 T^{1/2}$ with fixed- b critical values, or the cosine estimator with $\nu = 0.4 T^{2/3}$ and t_ν critical values.

Deeper. Kiefer & Vogelsang (2005); Lazarus, Lewis, Stock & Watson (2018).

Part II — Likelihood, information, and priors. What maximum likelihood is estimating when the model is wrong, why it is efficient when the model is right, and how priors turn into penalties.

A.7 Kullback–Leibler divergence and pseudo-true parameters

The idea. A misspecified model cannot match the true distribution of the data, but some parameter values bring it closer than others. **Kullback–Leibler divergence** measures “closer” as the average shortfall in log-likelihood from using the model instead of the truth. Maximum likelihood on a wrong model converges to the parameter that minimises this divergence, called the **pseudo-true** value. Whether that value is the parameter you care about depends on which parts of the model are wrong.

Formally. For a true density p and a model density q_θ ,

$$\text{KL}(p \parallel q_\theta) = \mathbb{E}_p[\log p(Y) - \log q_\theta(Y)] \geq 0,$$

with equality only when the two coincide. Maximising the expected log-likelihood $\mathbb{E}_p[\log q_\theta(Y)]$ is the same as minimising the divergence, so the MLE satisfies $\hat{\theta} \xrightarrow{p} \theta^* = \arg \min_\theta \text{KL}(p \parallel q_\theta)$. For Gaussian quasi-likelihood in a regression or GARCH model, θ^* equals the true conditional-mean and conditional-variance parameters whenever those two equations are correctly specified, even if the shocks are not normal.

Why it appears here. §5.4’s quasi-maximum likelihood: which misspecified likelihoods still estimate the right thing, and why their standard errors need the sandwich.

Deeper. White (1982).

A.8 Fisher information, the information matrix equality, and the Cramér–Rao bound

The idea. If the log-likelihood falls away steeply on either side of its peak, the data pin the parameter down tightly; if it is flat, many values fit almost equally well. **Fisher information** measures that curvature at the true value. The **Cramér–Rao bound** turns it into a floor on precision: no unbiased estimator can have a smaller variance than the inverse of the information. Maximum likelihood attains the bound in large samples, when the model is correct.

Formally. With per-observation score $s_i(\theta) = \partial \log f(Y_i \mid X_i; \theta) / \partial \theta$, which has mean zero at θ_0 , the information is

$$\mathcal{I}(\theta_0) = \mathbb{E}[s_i(\theta_0)s_i(\theta_0)'] = -\mathbb{E}\left[\frac{\partial^2 \log f(Y_i \mid X_i; \theta_0)}{\partial \theta \partial \theta'}\right].$$

The second equality, the **information matrix equality**, requires correct specification. The Cramér–Rao bound says $\text{Var}(\tilde{\theta}) \geq \mathcal{I}(\theta_0)^{-1}/n$ for any unbiased $\tilde{\theta}$. For a normal mean with known variance σ^2 , $\mathcal{I} = 1/\sigma^2$ and the bound is σ^2/n , which the sample mean attains. Under misspecification the two expectations differ, and the MLE’s variance is the sandwich $H^{-1}JH^{-1}/n$, with H minus the expected Hessian and J the variance of the score.

Why it appears here. §5.4 explains MLE’s efficiency as the meat equalling minus the bread, which is this equality in sandwich language.

Deeper. Hansen (2022), the chapters on maximum likelihood.

A.9 Priors as penalties

The idea. A Bayesian estimate combines what the data say (the likelihood) with what was believed beforehand (the prior). A prior belief that coefficients are probably small pulls estimates toward zero, and the strength of the pull depends on how confident the prior is relative to how noisy the data are. Written out, the most probable coefficients under such a prior are exactly the solution of a penalised regression, so every penalty in machine learning corresponds to a prior, and every prior to a penalty.

Formally. With $\mathbf{Y} \mid \beta \sim N(\mathbf{X}\beta, \sigma^2 I)$ and prior $\beta \sim N(0, \tau^2 I)$, the posterior mean and mode are

$$(\mathbf{X}'\mathbf{X} + \kappa I)^{-1}\mathbf{X}'\mathbf{Y}, \quad \kappa = \sigma^2/\tau^2,$$

the ridge estimator of §5.6. A Laplace prior with density proportional to $e^{-|\beta_j|/c}$ gives a posterior mode that minimises $\|\mathbf{Y} - \mathbf{X}\beta\|^2 + (2\sigma^2/c) \sum_j |\beta_j|$, which is the LASSO. The **Minnesota prior** for vector autoregressions centres each equation on a random walk (a coefficient of one on the variable's own first lag, zero elsewhere), with prior variances that shrink as the lag length grows and that are tighter on other variables' lags than on a variable's own.

Why it appears here. §5.6 (“a penalty is a prior, and a prior is a penalty”), the Bayesian VARs mentioned there, and the regularised predictors of §10.4.

Deeper. Litterman (1986); Tibshirani (1996).

A.10 Stein's paradox and empirical Bayes

The idea. Estimate the skill of 2,000 fund managers from their track records, and the top of the ranking is populated partly by skill and partly by luck. Pulling every estimate toward the group average removes some of the luck and lowers the total error, even though it biases each individual estimate. **Stein's paradox** is the surprising theorem that this shrinkage beats the unshrunk estimates for any true values once three or more are estimated together. **Empirical Bayes** chooses how much to shrink from the data: the spread of the estimates, compared with their noise, reveals how much genuine variation there is.

Formally. Let $\hat{\theta}_i \sim N(\theta_i, s^2)$ independently for $i = 1, \dots, p$ with $p \geq 3$. The James–Stein estimator $\tilde{\theta}_i = (1 - (p - 2)s^2 / \sum_j \hat{\theta}_j^2) \hat{\theta}_i$ has lower total mean squared error than $\hat{\theta}_i$ for every value of the θ_i . In the empirical Bayes version, the true values are drawn from $N(\mu, A)$, and the posterior mean is

$$\mu + \frac{A}{A + s^2} (\hat{\theta}_i - \mu),$$

with μ and A estimated from the cross-section (A as the variance of the estimates minus s^2). The noisier the estimates, or the less true dispersion, the harder the shrinkage.

Why it appears here. §5.6 states Stein's paradox and recommends shrinking manager and analyst estimates. The same logic underlies the forecast-combination puzzle of §8.8 and the winner's curse of §4.7.

Deeper. Efron & Morris (1977).

A.11 The truncated normal and the inverse Mills ratio

The idea. If an outcome is observed only when some unobserved index is high enough (a wage only when someone chose to work, a fund's return only when it survived), the observed errors are not mean zero: they are the errors of units that were lucky enough to be selected. How far their average is shifted depends on how likely selection was. For normal errors the shift has a closed form, the **inverse Mills ratio**, and Heckman's selection correction adds it to the regression as the omitted variable.

Formally. For $v \sim N(0, 1)$, $\mathbb{E}[v \mid v > -a] = \varphi(a)/\Phi(a) \equiv \lambda(a)$, which is large when a is small (selection is rare and only large v get through) and near zero when a is large. In Heckman's model, $Y_i = X_i'\beta + u_i$ is observed only if $W_i'\gamma_s + v_i > 0$, with (u_i, v_i) jointly normal, $\text{Var}(v_i) = 1$, and correlation ρ_s . Then

$$\mathbb{E}[Y_i \mid X_i, \text{selected}] = X_i'\beta + \rho_s \sigma_u \lambda(W_i'\gamma_s),$$

so a probit for selection gives $\hat{\lambda}$, which is added as a regressor. Because λ is nearly linear over much of its range, credible identification needs a variable in W_i that affects selection but not Y_i . The letters W_i , γ_s , and ρ_s are local to this entry.

Why it appears here. §6.4 describes selection as an omitted variable, and the §12.2 equivalences table names this term.

Deeper. Heckman (1979); Wooldridge (2010), the chapter on sample selection.

Part III — Testing and resampling. Standard errors computed by resampling, tests that are exact by design, and the corrections for testing many hypotheses at once.

A.12 The jackknife, the percentile- t bootstrap, and block bootstraps

The idea. The **jackknife** recomputes an estimate n times, each time leaving out one observation, and uses how much the estimate moves to estimate its variance. The bootstrap (§4.9) resamples with replacement instead. Two refinements matter. Resampling the t -statistic rather than the estimate, the **percentile- t** method, gives more accurate intervals because a t -statistic's distribution depends less on unknown parameters. And for dependent data, resampling **blocks** of consecutive observations preserves the dependence inside each block.

Formally. The jackknife variance is $\frac{n-1}{n} \sum_i (\hat{\theta}_{(-i)} - \bar{\theta}_{(\cdot)})^2$, where $\hat{\theta}_{(-i)}$ omits observation i and $\bar{\theta}_{(\cdot)}$ averages them. A percentile- t interval bootstraps $t^* = (\hat{\theta}^* - \hat{\theta}) / \text{se}^*$ and reports $[\hat{\theta} - t_{0.975}^* \text{se}, \hat{\theta} - t_{0.025}^* \text{se}]$. The **moving block bootstrap** resamples overlapping blocks of fixed length L_b , which must grow with the sample (a common rate is $T^{1/3}$). The **stationary bootstrap** draws blocks of random, geometrically distributed length with mean L_b , so that the resampled series is itself stationary.

Why it appears here. §4.3 notes that HC3 is close to the jackknife, and §4.9's table and refinements rest on these constructions.

Deeper. Efron (1979); Künsch (1989); Politis & Romano (1994).

A.13 The wild and wild cluster bootstrap

The idea. Keep the regressors fixed, and create new artificial samples by multiplying each residual by a random sign. Each observation keeps its own error variance, so heteroskedasticity is preserved without modelling it. The **cluster** version flips the sign of a whole cluster's residuals at once, which also preserves any correlation inside the cluster. Because the procedure uses the actual cluster structure rather than an approximation that assumes many clusters, it remains accurate with far fewer clusters than the cluster-robust t -test.

Formally. Estimate the model imposing the null (the **restricted** version), with residuals $\tilde{u}_i$. For $b = 1, \dots, B$, draw a weight w_g for each cluster, either ± 1 with equal probability (Rademacher) or from the six-point Webb distribution $\{\pm\sqrt{1/2}, \pm 1, \pm\sqrt{3/2}\}$; form $Y_i^* = X_i' \tilde{\beta} + w_{g(i)} \tilde{u}_i$, re-estimate, and compute the cluster-robust t_b^* . The p-value is the share of $|t_b^*|$ at least as large as the observed $|t|$. With G clusters, Rademacher weights generate only 2^G distinct samples, which is why Webb weights are used when G is small. When only one or two clusters are treated, sign flips of their residual sums cannot reproduce the treatment's noise, and the procedure becomes very conservative (§4.4).

Why it appears here. The remedy for few clusters in §4.4, simulated in that section's figure.

Deeper. Cameron, Gelbach & Miller (2008); MacKinnon, Nielsen & Webb (2023).

A.14 Randomisation inference

The idea. In an experiment, the uncertainty comes from the coin flips that assigned treatment. Under the null that treatment did nothing to anyone, every unit's outcome would have been the same under any assignment, so the outcomes under every alternative assignment are known. Recompute the test statistic for all the assignments the design could have produced, and see how extreme the actual one is. The resulting p-value is exact in any sample size and needs no distributional assumption.

Formally. Test the **sharp null** $Y_i(1) = Y_i(0)$ for all i . For a statistic $T(\mathbf{d}, \mathbf{Y})$, such as the difference in means under assignment vector $\mathbf{d}$, the p-value is the share of assignments $\mathbf{d}$ allowed by the randomisation design with $|T(\mathbf{d}, \mathbf{Y})| \geq |T(\mathbf{D}, \mathbf{Y})|$, where $\mathbf{D}$ is the realised assignment. When there are too many assignments to enumerate, a large random sample of them suffices. The test concerns the sharp null of no effect for anyone, which is stronger than a zero *average* effect.

Why it appears here. §4.4 and §7.2 recommend it for small experiments and single treated units, and the placebo inference of synthetic control in §7.7 is a version of it.

Deeper. Imbens & Rubin (2015), the chapter on Fisher's exact p-values.

A.15 Family-wise error and false discovery rates

The idea. When many hypotheses are tested, "5%" can mean two different things. The **family-wise error rate** is the probability of making *any* false discovery; controlling it is strict and appropriate when a single false claim is costly. The **false discovery rate** is the expected *share* of discoveries that are false; controlling it allows many discoveries while bounding how contaminated they are, which suits screening.

Formally. With K tests and p-values sorted as $p_{(1)} \leq \dots \leq p_{(K)}$:

- **Bonferroni** rejects H_j when $p_j \leq \alpha/K$; the family-wise error rate is at most α .
- **Holm** rejects $H_{(1)}, H_{(2)}, \dots$ in order while $p_{(j)} \leq \alpha/(K - j + 1)$; it controls the same rate and is never less powerful.
- **Benjamini–Hochberg** finds the largest j with $p_{(j)} \leq j\alpha/K$ and rejects $H_{(1)}, \dots, H_{(j)}$; the false discovery rate is at most α for independent or positively dependent tests.

With $K = 100$ and $\alpha = 5\%$, Bonferroni requires $p \leq 0.0005$, a two-sided $|t|$ above about 3.48.

Why it appears here. The table of corrections in §4.8, and the $t > 3$ hurdle for new return predictors in §9.6.

Deeper. Benjamini & Hochberg (1995); Harvey, Liu & Zhu (2016).

A.16 Testing the best of many: reality check, SPA, and stepdown

The idea. After searching over many strategies or forecasting models, the question worth testing is whether *any* of them genuinely beats a benchmark, and the natural statistic is the performance of the best one. Its distribution under the null must reflect both the size of the search and the correlation among the candidates, which share the same data. White's reality check obtains that distribution by bootstrapping all candidates together over the same resampled dates.

Formally. For candidate k at date t , let $d_{k,t}$ be its performance relative to the benchmark, such as a loss difference or a return difference, with sample mean $\bar{d}_k$. The null is $\max_k \mathbb{E}[d_{k,t}] \leq 0$ and the statistic is $V = \max_k \sqrt{T} \bar{d}_k$. A stationary bootstrap (A.12) resamples the same dates for every candidate, and the p-value is the share of bootstrap values $\max_k \sqrt{T}(\bar{d}_k^* - \bar{d}_k)$ that exceed V . Hansen's **SPA** test studentises each candidate and stops very poor candidates from inflating the null distribution, which makes it more powerful. The **Romano–Wolf stepdown** procedure goes further and identifies *which* candidates beat the benchmark, while controlling the family-wise error rate.

Why it appears here. §4.8's table, and the trading-rule evidence in §9.6.

Deeper. White (2000); Hansen (2005); Romano & Wolf (2005).

A.17 The Hausman test

The idea. Compare two estimators: one that is consistent whether or not an assumption holds but is imprecise, and one that is more efficient but consistent only if the assumption holds. If the assumption is true, both converge to the same value and their difference is just noise. If the difference is too large to be noise, reject the assumption.

Formally. With $\hat{\theta}_C$ consistent under both hypotheses and $\hat{\theta}_E$ efficient under the null,

$$H = (\hat{\theta}_C - \hat{\theta}_E)' [\hat{V}_C - \hat{V}_E]^{-1} (\hat{\theta}_C - \hat{\theta}_E) \xrightarrow{d} \chi_q^2,$$

where q is the number of coefficients compared. The simple variance of the difference, $\hat{V}_C - \hat{V}_E$, is valid only because $\hat{\theta}_E$ is efficient under the null. In panels, fixed effects play $\hat{\theta}_C$ and random effects play $\hat{\theta}_E$, and the null is that the unit effects are uncorrelated with the regressors. With clustered or heteroskedastic errors random effects is no longer efficient, and the usual robust alternative is the Mundlak regression of §7.4 with a cluster-robust Wald test on the unit means. The same logic compares OLS with IV to test for endogeneity.

Why it appears here. §7.4's choice between fixed and random effects.

Deeper. Hausman (1978); Wooldridge (2010), the chapter on linear panel models.

Part IV — Identification machinery. The formal tools behind the research designs of §6, §7, and §10.5.

A.18 d-separation and the back-door criterion

The idea. A causal graph is a map of which variables can pass statistical association to which others. Association flows along any path between two variables, whatever the direction of its arrows, unless something blocks it. Conditioning on a variable in the middle of a chain or a fork blocks the flow; conditioning on a collider, where two arrows meet, *opens* a path that was blocked. The back-door criterion uses these rules to decide which controls identify a causal effect.

Formally. A path between D and Y is **blocked** by a set of variables S if it contains a chain $A \rightarrow B \rightarrow C$ or a fork $A \leftarrow B \rightarrow C$ with B in S , or a collider $A \rightarrow B \leftarrow C$ with neither B nor any of its descendants in S . If every path is blocked, D and Y are **d-separated** given S , and they are then independent conditional on S in every distribution the graph allows. A set S satisfies the **back-door criterion** for the effect of D on Y if no member of S is a descendant of D and S blocks every path from D to Y that begins with an arrow into D . The effect is then identified by the adjustment formula

$$\mathbb{E}[Y \mid \text{do}(D = d)] = \sum_s \mathbb{E}[Y \mid D = d, S = s] \Pr(S = s),$$

where $\text{do}(D = d)$ denotes setting D by intervention rather than observing it.

Why it appears here. §6.5's rules for confounders, mediators, and colliders, and its warning that some pre-treatment controls are colliders.

Deeper. Pearl (2009); Cinelli, Forney & Pearl (2024).

A.19 Sensitivity analysis: Oster's δ and the robustness value

The idea. Unobserved confounding cannot be tested, but its required strength can be measured: how strong would a confounder have to be to explain the estimate away? Both methods here calibrate that strength against something observed. Oster's δ compares it with the selection implied by the controls you have. The robustness value expresses it as the share of variance the confounder would need to explain.

Formally. *Oster.* Let $\hat{\beta}, \hat{R}^2$ come from the regression without controls and $\tilde{\beta}, \tilde{R}^2$ from the regression with them. Suppose selection on unobservables is δ times selection on observables, and that a regression including every confounder would reach $R_{\max}^2$. The bias-adjusted coefficient is approximately

$$\beta^* \approx \tilde{\beta} - \delta (\hat{\beta} - \tilde{\beta}) \frac{R_{\max}^2 - \tilde{R}^2}{\tilde{R}^2 - \hat{R}^2}.$$

One reports the δ that sets $\beta^* = 0$, or β^* at $\delta = 1$ with $R_{\max}^2 = 1.3 \tilde{R}^2$. *Cinelli and Hazlett.* With t_D the treatment's t -statistic and df the residual degrees of freedom, let $f = |t_D|/\sqrt{df}$. The **robustness value** $\frac{1}{2}(\sqrt{f^4 + 4f^2} - f^2)$ is the smallest partial R^2 , with both the treatment and the outcome, that a confounder would need to reduce the estimate to zero.

Why it appears here. §6.7 recommends reporting one of these whenever an estimate rests on selection on observables.

Deeper. Oster (2019); Cinelli & Hazlett (2020).

A.20 Weak-instrument tools: Anderson–Rubin, LIML, and the effective F

The idea. When instruments are weak, the 2SLS t -statistic is far from normal (§7.3). There are three responses. Use a test whose validity does not depend on instrument strength (Anderson–Rubin). Use an estimator that is less biased toward OLS (LIML). And measure strength with a statistic suited to the dependence in the data (the effective F).

Formally. *Anderson–Rubin.* To test $\beta = \beta_0$, regress $Y_i - X_i' \beta_0$ on the instruments and any exogenous controls, and test that the instrument coefficients are zero, with a robust Wald statistic compared to χ_{ℓ}^2 . Under the null the dependent variable is just the structural error, so the test is valid however weak the instruments are. Collecting the β_0 values it does not reject gives a confidence set, which can be very wide or unbounded when the instruments are weak, honestly reflecting how little they reveal. *LIML* belongs to the family of k -class estimators, which includes OLS ($k = 0$) and 2SLS ($k = 1$); it chooses k from the data, slightly above one. It equals 2SLS in just-identified models, and is approximately median-unbiased with many instruments, at the cost of fatter tails. *The effective F* of Montiel Olea and Pflueger rescales the first-stage F -statistic to account for heteroskedasticity, clustering, or serial correlation, and compares it with critical values for the bias of 2SLS; in just-identified models it equals the robust first-stage F .

Why it appears here. §7.3's inference and diagnostics for IV.

Deeper. Anderson & Rubin (1949); Montiel Olea & Pflueger (2013); Andrews, Stock & Sun (2019).

A.21 Dynamic panel GMM

The idea. With a lagged dependent variable and unit fixed effects, demeaning biases the estimate (Nickell bias, §7.4). First-differencing removes the fixed effect but creates a new problem: the differenced lag is correlated with the differenced error, because both contain last period's shock. Levels of the dependent variable from two or more periods back are correlated with the differenced lag but, if the shocks are serially uncorrelated, not with the differenced error, so they can serve as instruments.

Formally. In $Y_{it} = \phi Y_{i,t-1} + X'_{it}\beta + \alpha_i + u_{it}$, differencing gives $\Delta Y_{it} = \phi \Delta Y_{i,t-1} + \Delta X'_{it}\beta + \Delta u_{it}$, where $\Delta Y_{i,t-1}$ and Δu_{it} both contain $u_{i,t-1}$. If the u_{it} are serially uncorrelated, the moment conditions $\mathbb{E}[Y_{i,t-s} \Delta u_{it}] = 0$ for $s \geq 2$ identify ϕ by GMM (§5.3). Two diagnostics are standard: the differenced residuals should show first-order but no second-order autocorrelation, and the J -test should not reject. The number of instruments grows with the square of T , and too many instruments bias the estimate toward OLS. When ϕ is near one, lagged levels predict differences poorly and the instruments are weak; “system GMM”, which adds equations in levels, is the usual response.

Why it appears here. §7.4 names it as the fix for the Nickell bias.

Deeper. Arellano & Bond (1991); Wooldridge (2010), the chapter on dynamic panel models.

A.22 Local polynomial regression and bandwidth choice

The idea. To estimate a regression function at one point without assuming its shape, fit a straight line using only the data near that point, weighting the nearest observations most. The **bandwidth** defines “near”. A wide bandwidth uses more data, which cuts variance but lets curvature further away bias the fit; a narrow one does the reverse. Regression discontinuity needs these fits at the edge of the data, on each side of the cutoff, where global polynomials behave worst.

Formally. On the right of a cutoff c , with running variable R_i and kernel K , solve

$$\min_{a,b} \sum_{i: R_i \geq c} K\left(\frac{R_i - c}{h}\right) (Y_i - a - b(R_i - c))^2,$$

do the same on the left, and estimate the jump as $\hat{a}_+ - \hat{a}_-$. The triangular kernel $K(z) = \max(0, 1 - |z|)$ is standard. The bias of a local linear fit is of order h^2 , and the bandwidth that minimises mean squared error shrinks like $n^{-1/5}$. At that bandwidth the bias is of the same order as the standard error, so a conventional confidence interval is centred in the wrong place. **Robust bias correction** estimates the bias with a higher-order fit, subtracts it, and widens the standard error to account for the correction.

Why it appears here. §7.6's estimation and inference for regression discontinuity.

Deeper. Calonico, Cattaneo & Titiunik (2014); Cattaneo, Idrobo & Titiunik (2020).

A.23 Approximate sparsity

The idea. With hundreds of candidate controls, estimation is only feasible if most of them barely matter. **Sparsity** means that only a few have non-zero coefficients. **Approximate sparsity** relaxes this to realism: a few controls matter a lot, and the many small effects that remain add up to an error smaller than the sampling noise. Under that condition, a selection method that finds the important few loses little.

Formally. A conditional mean $\mathbb{E}[Y | X] = \sum_{j=1}^p \beta_j X_j$ with possibly $p > n$ is approximately sparse if it can be approximated by some s of the regressors with an approximation error whose root mean square is of order $\sqrt{s/n}$, where s is small relative to n (inference results require s to grow slowly relative to n , up to logarithmic factors in p). The LASSO then selects a model whose prediction error is close to what an oracle that knew the s important controls would achieve.

Why it appears here. §10.5 states that post-double-selection inference is valid under approximate sparsity.

Deeper. Belloni, Chernozhukov & Hansen (2014); Chernozhukov, Hansen, Kallus, Spindler & Syrgkanis (2024).

A.24 Neyman orthogonality and cross-fitting

The idea. Machine-learning predictions are deliberately biased by regularisation. If a causal estimate depends on them directly, that bias passes straight through. The fix is to build the estimating equation so that small errors in the auxiliary predictions do not move the causal estimate *to first order*. Their effect then enters only as the product of two small errors,

which is negligible. **Cross-fitting** adds a second protection: the predictions used for each observation are made by models that never saw that observation, so overfitting cannot contaminate the estimate.

Formally. A moment function $\psi(W; \theta, \eta)$ with nuisance functions η is **Neyman orthogonal** if its expectation has zero derivative in every direction of η at the truth:

$$\frac{\partial}{\partial r} \mathbb{E}[\psi(W; \theta_0, \eta_0 + r(\eta - \eta_0))] \Big|_{r=0} = 0.$$

For §10.5's partialling-out moment, $\psi = (Y - \ell(X) - \theta(D - \pi(X)))(D - \pi(X))$ with $\eta = (\ell, \pi)$. Perturbing ℓ changes the expectation by a multiple of $\mathbb{E}[D - \pi(X) \mid X] = 0$, and perturbing π by a multiple of $\mathbb{E}[u \mid X] = 0$, so both derivatives vanish. The remaining bias is of order $\|\hat{\ell} - \ell\| \cdot \|\hat{\pi} - \pi\|$, which is small enough for $\sqrt{n}$ -inference if each prediction error shrinks faster than $n^{-1/4}$. **Cross-fitting** splits the sample into folds, fits $\hat{\ell}$ and $\hat{\pi}$ on all folds but one, computes residuals on the held-out fold, rotates, and estimates θ from the pooled residuals. Here W collects the data, and r is a scalar.

Why it appears here. The two “ingredients” of double machine learning in §10.5.

Deeper. Chernozhukov et al. (2018).

Part V — Time series and volatility.

A.25 The Kalman filter

The idea. Many economic quantities are never observed directly, such as the true level of a trend, a factor that drives the yield curve, or an expected return. The **Kalman filter** tracks such a hidden state from noisy observations. Each period it predicts the state and the next observation, sees the actual observation, and corrects the prediction by an amount proportional to the surprise. The correction is larger when the prediction was uncertain and smaller when the observation is noisy. The prediction errors also give the likelihood of the model, so its parameters can be estimated by maximum likelihood.

Formally. With state $\xi_t = F\xi_{t-1} + \eta_t$ and observation $y_t = H\xi_t + w_t$, where $\text{Var}(\eta_t) = \Sigma_\eta$ and $\text{Var}(w_t) = \Sigma_w$:

$$\begin{aligned} \text{predict: } \hat{\xi}_{t|t-1} &= F\hat{\xi}_{t-1|t-1}, & P_{t|t-1} &= FP_{t-1|t-1}F' + \Sigma_\eta, \\ \text{update: } \mathcal{G}_t &= P_{t|t-1}H'(HP_{t|t-1}H' + \Sigma_w)^{-1}, \\ \hat{\xi}_{t|t} &= \hat{\xi}_{t|t-1} + \mathcal{G}_t(y_t - H\hat{\xi}_{t|t-1}), & P_{t|t} &= (I - \mathcal{G}_tH)P_{t|t-1}. \end{aligned}$$

P is the variance of the state estimate and $\mathcal{G}_t$ the **gain**. The letters in this entry are local to it.

Why it appears here. §7.8 lists term-structure models estimated by the Kalman filter among structural models in finance.

Deeper. Hamilton (1994), chapter 13; section 6.6 of Trend-Following in Financial Markets applies it to trend estimation.

A.26 The Granger representation theorem

The idea. If several non-stationary series are cointegrated, the gaps between them are stationary, so something must keep pulling them back together. The **Granger representation theorem** says the pull has to show up in the short-run dynamics: at least one of the series must respond to last period's deviation from the long-run relationship. Cointegration and error correction are two descriptions of the same system.

Formally. Let $\mathbf{y}_t$ be a vector of n_y integrated series. They are cointegrated with rank r if there are r linearly independent vectors, the columns of an $n_y \times r$ matrix B , such that $B'\mathbf{y}_t$ is stationary. Then the vector autoregression can be written as the **vector error-correction model**

$$\Delta \mathbf{y}_t = A B' \mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \mathbf{y}_{t-j} + \mathbf{e}_t,$$

where the $n_y \times r$ matrix A holds the speeds of adjustment. The rank of AB' is r . With $r = 0$ the model is a VAR in differences, and with $r = n_y$ the series are stationary in levels. The Johansen procedure estimates r by testing the rank of this matrix. The letters A , B , Γ_j , and n_y are local to this entry.

Why it appears here. §8.4's error-correction model and the equivalence listed in §12.2.

Deeper. Engle & Granger (1987); Johansen (1991).

A.27 Market microstructure noise

The idea. Transaction prices are not the “efficient” price. They bounce between the bid and the ask, move in discrete ticks, and carry temporary price impact. Each deviation is tiny, but unlike genuine price moves it does not shrink when you sample more often. At high enough frequency the noise dominates squared returns, and realised variance measures the noise rather than the volatility.

Formally. Write the observed log price as $p_t^{\text{obs}} = p_t^* + v_t$, with v_t independent noise of variance σ_v^2 . An observed return over a short interval is then the efficient return plus $v_t - v_{t-1}$. With M intraday returns, realised variance has expectation of about the integrated variance plus $2M\sigma_v^2$, so its bias grows linearly with the sampling frequency. The same differenced noise gives observed returns a negative first-order autocorrelation. The standard responses are sampling at moderate frequencies, such as every five minutes, or noise-robust estimators such as realised kernels and pre-averaging.

Why it appears here. §8.6 names noise as the limit on realised variance's precision.

Deeper. Andersen, Bollerslev, Diebold & Labys (2003).

A.28 Loss functions for volatility forecasts: MSE and QLIKE

The idea. Variance is never observed, so volatility forecasts are scored against a noisy stand-in such as a squared return. Some loss functions rank competing forecasts in the same order whether they are scored against the true variance or against an unbiased noisy stand-in; others can prefer the wrong forecast. Mean squared error and QLIKE are in the first group.

Formally. Let $\hat{h}_t$ be a variance forecast and $\hat{\sigma}_t^2$ a proxy with $\mathbb{E}[\hat{\sigma}_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2$. The two standard robust losses are

$$\text{MSE}_t = (\hat{\sigma}_t^2 - \hat{h}_t)^2, \quad \text{QLIKE}_t = \frac{\hat{\sigma}_t^2}{\hat{h}_t} - \log \frac{\hat{\sigma}_t^2}{\hat{h}_t} - 1.$$

Both have expected loss minimised by $\hat{h}_t = \sigma_t^2$. QLIKE depends only on the ratio of proxy to forecast, so it is scale-free and less dominated by a few turbulent days than MSE. Losses such as the mean squared error of volatilities or of log variances are not robust in this sense.

Why it appears here. §8.6's warning about evaluating volatility forecasts.

Deeper. Patton (2011); Hansen & Lunde (2005).

A.29 IVX

The idea. A predictor that is nearly a random walk gives the predictive regression a nonstandard sampling distribution, and the right correction depends on exactly how persistent the predictor is, which is itself poorly estimated (§9.2). **IVX** sidesteps the problem. It builds an instrument from the predictor's own past changes, filtered so that the instrument is deliberately less persistent than the predictor, and uses it in an IV regression. The resulting test statistic is approximately normal whatever the predictor's persistence.

Formally. For $r_{t+1} = \alpha + \beta x_t + u_{t+1}$, construct $\tilde{z}_t = \sum_{j=0}^{t-1} \varrho^j \Delta x_{t-j}$ with $\varrho = 1 - c_z/T^a$, for a constant $c_z > 0$ and $a \in (0, 1)$. This makes $\tilde{z}_t$ “mildly integrated”, less persistent than a unit root but more than a stationary series. Using $\tilde{z}_t$ as an instrument for x_t gives a Wald statistic with a χ^2 limit whether x_t is stationary, local to a unit root, or exactly integrated. The price is some loss of power when the predictor is in fact stationary and exogenous. The symbols ϱ , c_z , and a are local to this entry.

Why it appears here. §9.2's list of inference methods for persistent predictors.

Deeper. Kostakis, Magdalinos & Stamatogiannis (2015).

Part VI — Finance. The finance vocabulary and machinery used in §3.8, §4.7, and §9.

A.30 Excess returns and the Sharpe ratio

The idea. An investment's **excess return** is what it earns above a risk-free rate, the reward for bearing risk. The **Sharpe ratio** divides the average excess return by its volatility: reward per unit of risk. It does not change with leverage, since doubling a position doubles both the numerator and the denominator. It is also, almost exactly, the signal-to-noise ratio that determines how hard a strategy's mean return is to estimate.

Formally. With excess return r_t^e , $SR = \mathbb{E}[r_t^e] / \text{sd}(r_t^e)$. For independent returns, a per-period Sharpe ratio scales to annual units by $\sqrt{A}$, with A periods per year. The t -statistic of the mean excess return over T periods is approximately $SR_{\text{period}} \sqrt{T}$, which equals the annual Sharpe ratio times the square root of the number of years. For independent normal returns the standard error of an estimated per-period Sharpe ratio is approximately $\sqrt{(1 + SR^2/2)/T}$ (§2.6).

Why it appears here. Throughout: the precision of means (§2.5–§2.6), the value of predictability (§3.8), the winner's curse (§4.7), and backtests (§9.1, §9.7).

Deeper. Lo (2002); Appendix A.32 and A.45 of Portfolio Construction and the Covariance Matrix.

A.31 Risk aversion, rational expectations, and the stochastic discount factor

The idea. An investor values a payoff by how much an extra dollar is worth in the state of the world where it arrives. A dollar in a recession, when consumption is low, is worth more than a dollar in a boom. The **stochastic discount factor** is that state-by-state valuation, and the price of every asset is its expected payoff weighted by it. **Risk aversion** controls how steeply the valuation rises in bad states. **Rational expectations** is the assumption that investors' expectations are the true conditional expectations, which is what allows an econometrician to test the resulting pricing equations using realised data.

Formally. With power (constant relative risk aversion) utility $u(C) = C^{1-\eta}/(1-\eta)$, relative risk aversion $-Cu''(C)/u'(C)$ equals η . An investor who can buy an asset with gross return R_{t+1} is at an optimum only if $u'(C_t) = d \mathbb{E}_t[u'(C_{t+1})R_{t+1}]$, which rearranges to

$$\mathbb{E}_t[M_{t+1}R_{t+1}] = 1, \quad M_{t+1} = d(C_{t+1}/C_t)^{-\eta},$$

and for an excess return, $\mathbb{E}_t[M_{t+1}r_{t+1}^e] = 0$. Under rational expectations $\mathbb{E}_t$ is the mathematical expectation given $\mathcal{F}_t$ (A.1), so any variable known at t is a valid instrument. The mean-variance investor of §3.8, who maximises expected return minus $\gamma_r/2$ times variance, corresponds to quadratic utility, whose marginal utility is linear, so that investor's discount factor is linear in the return on their portfolio.

Why it appears here. The Euler-equation row of §1.5's table, GMM estimation in §5.3, the SDF approach to asset-pricing tests in §9.4, the rational-expectations critique in §13.3, and the optimal portfolio in §3.8.

Deeper. Cochrane (2005); Hansen & Singleton (1982).

A.32 Portfolio sorts, value weighting, and NYSE breakpoints

The idea. The simplest non-parametric test of whether a characteristic predicts returns is to rank stocks on it every period, group them into portfolios, and track each portfolio's subsequent return. The return of the top group minus the bottom group is the characteristic's "factor" return. Two implementation choices decide whether the result describes the economy or its smallest stocks: how stocks are weighted inside each portfolio, and where the group boundaries are drawn.

Formally. At each rebalancing date, compute percentile **breakpoints** of the characteristic, assign stocks to groups, and compute next-period portfolio returns, either equal-weighted, $\frac{1}{N_p} \sum_i r_i$, or **value-weighted**, $\sum_i \frac{ME_i}{\sum_j ME_j} r_i$, where ME_i is market capitalisation at formation and N_p the number of stocks in the portfolio. **NYSE breakpoints** compute the percentiles from New York Stock Exchange stocks only, which are larger on average, and then assign every stock. Without them, the extreme groups fill with microcaps, which are about 60% of stocks and about 3% of market value (§9.1). Evidence comes from the time series of the long-short return, with HAC standard errors, and from its alpha against existing factor models (§9.4).

Why it appears here. §1.4 lists portfolio sorts as a descriptive tool, and the replication debate in §9.6 turns partly on weighting and breakpoints.

Deeper. Fama & French (2008); Hou, Xue & Zhang (2020).

A.33 Mean-variance efficiency, the tangency portfolio, and the GLS R^2

The idea. Among all combinations of a set of risky assets, one has the highest Sharpe ratio: the **tangency portfolio**. A factor model is exactly right when some combination of its factors reaches that maximum, so that adding the test assets cannot improve on the factors. That is why the time-series test of a factor model can be read as a question about Sharpe ratios, and why a goodness-of-fit measure for a factor model should weight pricing errors by how much they matter for the maximum Sharpe ratio rather than equally.

Formally. For assets with mean excess returns μ and covariance matrix Σ , the maximum squared Sharpe ratio is $\mu' \Sigma^{-1} \mu$, attained by weights proportional to $\Sigma^{-1} \mu$. If α collects the test assets' intercepts in regressions on the factors, and Σ_e their residual covariance, then

$$\alpha' \Sigma_e^{-1} \alpha = \text{SR}^2(\text{factors and test assets}) - \text{SR}^2(\text{factors}),$$

which is what the GRS statistic scales and tests. A **GLS R^2** in a cross-sectional regression weights pricing errors by the inverse covariance matrix, so it measures errors on the portfolio combinations that matter for the maximum Sharpe ratio, and it is tied to how close the model's **factor-mimicking portfolio**, the combination of test assets most correlated with the factor, comes to the tangency portfolio. In this entry μ and α are vectors over test assets.

Why it appears here. §9.4's reading of the GRS test and Lewellen, Nagel and Shanken's recommendation.

Deeper. Gibbons, Ross & Shanken (1989); Lewellen, Nagel & Shanken (2010); Cochrane (2005).

A.34 Errors in betas: the Shanken correction

The idea. Fama–MacBeth standard errors treat the first-pass betas as if they were known. They are estimated, so the second-pass risk premia inherit extra uncertainty that the standard errors miss. Shanken derived the adjustment. It inflates the variance by a factor that grows with the squared Sharpe ratio of the factors, and adds a term for the sampling variation of the factors themselves.

Formally. For an OLS second pass with K factors, estimated premia $\hat{\lambda}$, factor covariance matrix Σ_f , and Fama–MacBeth variance $\hat{V}_{\text{FM}}$, the corrected variance is approximately

$$\hat{V}_{\text{Shanken}} = (1 + \hat{\lambda}' \Sigma_f^{-1} \hat{\lambda}) \hat{V}_{\text{FM}} + \frac{\Sigma_f}{T}.$$

For traded factors, $\lambda' \Sigma_f^{-1} \lambda$ is the factors' maximum squared Sharpe ratio per period, which is small in monthly data (a monthly Sharpe ratio of 0.15 gives a multiplier of about 1.02), so the correction matters most with several factors or high Sharpe ratios. Here K is the number of factors.

Why it appears here. §9.4's discussion of errors in variables in two-pass tests.

Deeper. Shanken (1992); Cochrane (2005), the chapter on regression-based tests.

A.35 Driscoll–Kraay standard errors

The idea. Sum each date's moment contributions across all firms, producing a single time series with one observation per date. Whatever correlation exists across firms at a date is inside that sum. Then apply Newey–West to the time series, which handles correlation across dates. The result is robust to arbitrary cross-sectional dependence and to serial dependence that fades, provided there are many dates.

Formally. With $h_t = \sum_i X_{it} \hat{e}_{it}$ and Bartlett weights $w_j = 1 - j/(L + 1)$,

$$\hat{V}_{\text{DK}} = (\mathbf{X}'\mathbf{X})^{-1} \left[\sum_t h_t h_t' + \sum_{j=1}^L w_j \sum_t (h_t h_{t-j}' + h_{t-j} h_t') \right] (\mathbf{X}'\mathbf{X})^{-1}.$$

Consistency requires the number of dates to grow; the number of firms can be fixed or large.

Why it appears here. The standard-error table in §1.6 and the panel table in §9.5.

Deeper. Driscoll & Kraay (1998); Petersen (2009).

A.36 The deflated Sharpe ratio

The idea. The best of many backtests reports a Sharpe ratio biased upward by the search that selected it. The **deflated Sharpe ratio** asks whether the best result beats what the best of that many unskilled trials would have produced by luck, after also accounting for the sample length and for non-normal returns. It converts “my best configuration scored 1.4” into the probability that the configuration has genuine skill.

Formally. The expected maximum Sharpe ratio across K independent unskilled trials is approximately

$$SR_0 = \sqrt{\text{Var}(\widehat{SR}_k)} \left[(1 - \gamma_E) \Phi^{-1}\left(1 - \frac{1}{K}\right) + \gamma_E \Phi^{-1}\left(1 - \frac{1}{Ke}\right) \right],$$

where $\gamma_E \approx 0.5772$ is the Euler–Mascheroni constant and $\text{Var}(\widehat{SR}_k)$ is the variance of the Sharpe ratios across trials. The deflated Sharpe ratio is then

$$\Phi \left(\frac{(\widehat{SR} - SR_0) \sqrt{T-1}}{\sqrt{1 - \hat{s}_3 \widehat{SR} + \frac{\hat{s}_4 - 1}{4} \widehat{SR}^2}} \right),$$

with $\hat{s}_3$ and $\hat{s}_4$ the skewness and kurtosis of returns. The number of trials K should count *effectively independent* trials, which is smaller than the raw count when configurations are correlated (§4.8). In this entry K is the number of trials.

Why it appears here. §4.8 and §9.7 list it among the adjustments for search, and §11.5’s research log exists to supply K .

Deeper. Bailey & López de Prado (2014); Appendix A.46 of Portfolio Construction and the Covariance Matrix.

Part VII — Machine learning.

A.37 Trees, forests, boosting, and neural-network regularisation

The idea. A **regression tree** splits the data into boxes using thresholds on the predictors (“book-to-market above 0.8 and size below the median”) and predicts the average outcome in each box, so it captures nonlinearities and interactions without being told about them. A single deep tree overfits badly. A **random forest** averages many trees, each grown on a bootstrap sample and allowed to split only on a random subset of predictors, which reduces variance. **Boosting** grows a sequence of small trees, each fitted to what the ensemble so far has failed to explain, and adds each with a small weight, which reduces bias gradually. **Neural networks** stack layers of nonlinear transformations and are kept from overfitting by weight penalties, **early stopping** (halting training when validation loss stops improving), **dropout** (randomly silencing units during training), and averaging several networks. A **causal forest** grows trees whose splits seek differences in treatment effects rather than in outcomes, and uses separate subsamples to choose the splits and to estimate the effects within them, which is what makes its confidence intervals valid.

Formally. A tree predicts $\hat{f}(x) = \sum_m c_m \mathbb{1}\{x \in \mathcal{R}_m\}$ for boxes $\mathcal{R}_m$ chosen greedily to reduce squared error, with c_m the mean outcome in box m . A forest averages B trees, $\bar{f}(x) = \frac{1}{B} \sum_b \hat{f}_b(x)$. Boosting builds $\hat{f}_M(x) = \sum_{m=1}^M \varsigma \hat{t}_m(x)$, where each small tree $\hat{t}_m$ is fitted to the current residuals (for squared loss) and ς is a small learning rate. The tuning parameters (depth, number of trees, learning rate, penalties) are chosen by validation (§10.6). The symbols in this entry are local to it.

Why it appears here. §10.4’s comparison of return-prediction methods, and the causal forests of §10.5.

Deeper. Hastie, Tibshirani & Friedman (2009), chapters 9, 10, and 15; Wager & Athey (2018); Gu, Kelly & Xiu (2020).

A.38 Mallows's C_p and the optimism of in-sample fit

The idea. A model's error on the data it was fitted to understates its error on new data, and the understatement grows with the number of parameters, because each parameter lets the fit absorb some of the noise. **Mallows's** C_p adds the expected understatement back to the in-sample error, so that models of different sizes can be compared on an estimate of their out-of-sample error without holding out data.

Formally. For linear regression with k coefficients, n observations, and error variance σ^2 , the expected in-sample mean squared error falls short of the expected out-of-sample error at the same regressor values by $2\sigma^2 k/n$ (§10.2). One common form of the criterion is

$$C_p = \frac{1}{n} \sum_{i=1}^n \hat{e}_i^2 + \frac{2k\hat{\sigma}^2}{n},$$

with $\hat{\sigma}^2$ estimated from a large model. Choosing the model with the smallest C_p approximately minimises expected prediction error. With normal errors and known σ^2 , ranking models by C_p is equivalent to ranking them by the AIC.

Why it appears here. §10.2 derives the optimism and names C_p as the criterion built on it.

Deeper. Hastie, Tibshirani & Friedman (2009), chapter 7.

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